

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	7
Bill of Material:	9
MDMT Summary:	10
Warnings and Errors:	12
Input Echo:	13
XY Coordinate Calculations:	24
Internal Pressure Calculations:	25
Element: Btm_head	26
Element: Shell-1	28
Element: Shell-2	30
Element: Shell-3	32
Element: Shell-4	34
Element: Shell-5	36
Element: Shell-6	38
Element: Shell-7	40
Element: Top_head	42
Shop Pressure Test Results:	44
Field Pressure Test Results:	46
External Pressure Calculations:	48
Element and Detail Weights:	49
Nozzle Flange MAWP:	52
Natural Frequency Calculation:	54
Wind Load Calculation:	55
Earthquake Load Calculation:	59
Wind/Earthquake Shear, Bending:	61
Wind/Earthquake Deflection:	62
Longitudinal Stress Constants:	63
Longitudinal Allowable Stresses:	64
Longitudinal Stresses due to:	65
Stress due to Combined Loads:	68
Skirt Stress Calculations:	74
Center of Gravity Calculation:	79
Basering Calculations:	80
Nozzle Calcs.: N3_6"	84
Nozzle Calcs.: M1B_24"	94
Nozzle Calcs.: N11A_2"	100
Nozzle Calcs.: M1A_24"	105
Nozzle Calcs.: N1_8"	111
Nozzle Calcs.: N11B_2"	124
Nozzle Calcs.: N7_2"	127
Nozzle Calcs.: N5_2"	130
Nozzle Calcs.: N9_8"	133
Nozzle Calcs.: N2_8"	146
Nozzle Calcs.: N4_6"	156
Nozzle Calcs.: N6_2"	170
Nozzle Calcs.: N8_6"	173
Nozzle Calcs.: N10_4"	187



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A165QFA10CQ002Instrument air receiver.pvdb

Date of Analysis : Sep 15, 2026 5:33pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 8200.000 mm ID	
Vessel Design Length, Tangent to Tangent	20500.00 mm
Distance of Bottom Tangent above Grade	3800.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	3500.00 mm
Internal Design Temperature	85 °C
Internal Design Pressure	1.100 MPa
External Design Temperature	85 °C
Maximum Allowable Working Pressure	1.11 MPa
Shop Test Pressure	1.444 MPa
Field Test Pressure	1.444 MPa
Required Minimum Design Metal Temperature	-10.0 °C
Warmest Computed Minimum Design Metal Temperature	-28.0 °C
Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N
Head	SA-516 70	...	...	K02700	+N
Skirt	SA-285 C	...	...	K02801	
Skirt	SA-516 70	...	...	K02700	+N
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 70	...	...	K02700	+N
Nozzle	SA-105	...	...	K03504	+N
Re-Pad	SA-516 70	...	...	K02700	+N
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-285 C	...	...	K02801	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	1.5000	N/A	No
Skirt-1	...	...	...	1.5000	N/A	No
Btm_head	1.100	0.00	1.130	1.5000	No	No
Shell-1	1.100	0.00	1.120	1.5000	N/A	No
Shell-2	1.100	0.00	1.120	1.5000	N/A	No
Shell-3	1.100	0.00	1.120	1.5000	N/A	No
Shell-4	1.100	0.00	1.120	1.5000	N/A	No
Shell-5	1.100	0.00	1.120	1.5000	N/A	No
Shell-6	1.100	0.00	1.120	1.5000	N/A	No
Shell-7	1.100	0.00	1.120	1.5000	N/A	No
Top_head	1.100	0.00	1.130	1.5000	No	No

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
---------	-----------	---------	---------	----------	----------	----------	------	------



Type	mm	Length mm	Thickness mm	Thickness mm	Internal mm	External mm	Eff	Eff
Skirt	-2000.0	1500.0	16.0	16.0	...	...	0.70	0.70
Skirt	0.0	2000.0	16.0	16.0	...	...	0.70	0.70
Ellipse	50.0	50.0	42.0	35.0	34.2	...	1.00	1.00
Cylinder	3050.0	3000.0	35.0	35.0	34.4	...	1.00	1.00
Cylinder	6050.0	3000.0	35.0	35.0	34.4	...	1.00	1.00
Cylinder	9050.0	3000.0	35.0	35.0	34.4	...	1.00	1.00
Cylinder	12050.0	3000.0	35.0	35.0	34.4	...	1.00	1.00
Cylinder	15050.0	3000.0	35.0	35.0	34.4	...	1.00	1.00
Cylinder	18050.0	3000.0	35.0	35.0	34.4	...	1.00	1.00
Cylinder	20450.0	2400.0	35.0	35.0	34.4	...	1.00	1.00
Ellipse	20500.0	50.0	42.0	35.0	34.2	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_6"	WRC-107/537	0.878	n/a	Passed
N1_8"	WRC-297	0.978	Shell	Passed
N9_8"	WRC-297	0.978	Shell	Passed
N2_8"	WRC-107/537	0.878	n/a	Passed
N4_6"	WRC-107/537	0.878	n/a	Passed
N8_6"	WRC-107/537	0.878	n/a	Passed
N10_4"	WRC-107/537	0.878	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	750	389195	593104	5.959E+09	10.87E+09
20	30	2500	377361	592284	5.384E+09	9.977E+09
30	40	3525	361583	589818	4.645E+09	8.794E+09
40	50	5050	361187	575223	4.627E+09	8.765E+09
50	60	8050	324707	552098	3.597E+09	7.073E+09
60	70	11050	281788	520848	2.687E+09	5.463E+09
70	80	14050	234712	477331	1912282368	3.965E+09
80	90	17050	205770	400050	1251291392	2.649E+09
90	100	20050	158308	330223	704954176	1552804480
100	110	22750	108198	252372	305034112	678557824
110	120	23975	83054.1	178106	75438944	161775376

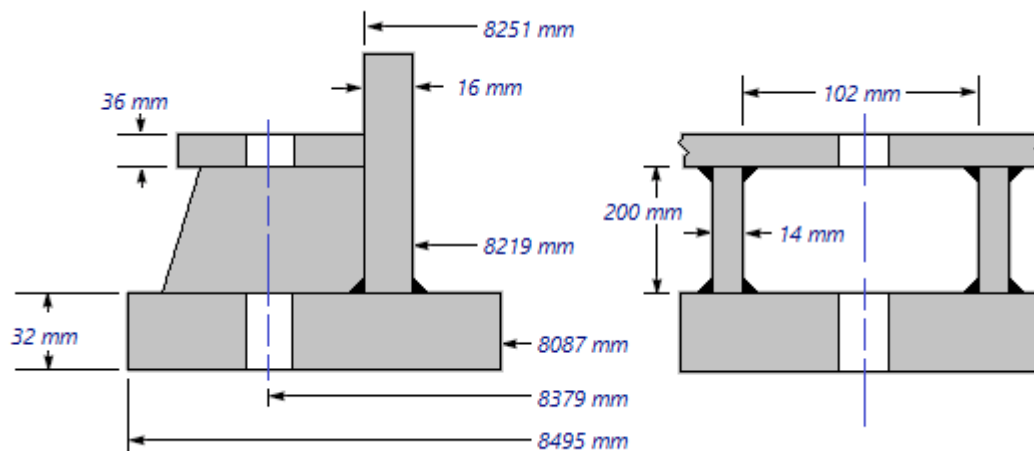
Abs Max of the all of the Stress Ratio's : 0.8054

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing 32.0000 mm
 Inside Diameter of Basing 8087.0000 mm
 Outside Diameter of Basing 8495.0000 mm
 Nominal Diameter of Bolts 36.0000 mm
 Diameter of Bolt Circle 8379.0000 mm
 Number of Bolts 16

Thickness of Gusset Plates 14.0000 mm
 Average Width of Gusset Plates 122.0000 mm
 Height of Gussets 200.0000 mm
 Distance between Gussets 102.0000 mm
 Thickness of Top Plate or Ring 36.0000 mm
 Radial Width of the Top Plate 122.0000 mm
 Number of Gussets 32

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	389195.	N
Total Earthquake Shear on Support	593104.	N
Wind Moment on Support	5995483648.	N-mm
Earthquake Moment on Support	10902293504.	N-mm
Maximum Initial Bolt Load	162005.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	229482.5	kgm
Shop Test - Fabricated + water (full)	1455909.6	kgm
Shipping - Fab. + rem. intls.+ shipping app.	272428.8	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	272428.8	kgm
Empty - Fab. + intls. + details + wghts.	272428.8	kgm
Operating - Empty + operating liquid (no ca)	278299.8	kgm
Field Test - Empty weight + test medium (full)	1498856.1	kgm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3_6"	1.13	...	...	OK	7.72	OK	Passed
M1B_24"	1.11	...	...	...		OK	Passed
N11A_2"	1.12	...	...	OK	6.30	OK	Passed
N11A_2"	1.12	...	...	OK	6.30	OK	No Calc[*]
M1A_24"	1.11	...	...	...		OK	Passed
N1_8"	1.11	...	...	OK	8.66	OK	Passed
N11B_2"	1.12	...	...	OK	6.30	OK	No Calc[*]
N7_2"	1.12	...	...	OK	6.30	OK	No Calc[*]
N5_2"	1.12	...	...	OK	6.30	OK	No Calc[*]
N9_8"	1.11	...	...	OK	8.66	OK	Passed
N2_8"	1.13	...	...	OK	8.66	OK	Passed
N4_6"	1.13	...	...	OK	7.72	OK	Passed
N6_2"	1.13	...	...	OK	6.30	OK	No Calc[*]
N8_6"	1.13	...	...	OK	7.72	OK	Passed
N10_4"	1.13	...	...	OK	6.76	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1.11 Nozzle : M1A24"
 Minimum MAWP Shells/Flanges : 1.11 Element : Shell-7
 Minimum MAPnc Shells/Flanges : 1.17 Element : Shell-7

Computed Vessel M.A.W.P. : 1.11 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3_6"	0.000	0.000	304.145
40	M1B_24"	1250.000	270.000	1145.200
50	N11A_2"	4000.000	0.000	144.724
60	M1A_24"	6600.000	270.000	1145.200
80	N1_8"	12149.999	180.000	399.700
80	N11B_2"	14999.999	0.000	144.724
90	N7_2"	17000.000	0.000	144.724
90	N5_2"	17999.998	0.000	144.724
100	N9_8"	19000.000	0.000	399.700
110	N2_8"	0.000	0.000	399.700
110	N4_6"	0.000	0.000	304.145
110	N6_2"	0.000	0.000	144.724
110	N8_6"	0.000	180.000	304.145
110	N10_4"	0.000	180.000	204.641

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!



Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N11A_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	362.04	150
N11B_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	335.18	150
N7_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	235.18	150
N5_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	235.18	150
N6_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	343.88	150
N10_4"	4.000 in	80	WNF	4.500	8.560	200.00	30.00	347.79	300
N3_6"	6.000 in	80	WNF	6.625	10.973	270.00	30.00	335.48	300
N4_6"	6.000 in	80	WNF	6.625	10.973	270.00	30.00	360.47	300
N8_6"	6.000 in	80	WNF	6.625	10.973	270.00	30.00	360.47	300
N1_8"	8.000 in	80	WNF	8.625	12.700	400.00	32.00	236.46	300
N9_8"	8.000 in	80	WNF	8.625	12.700	400.00	32.00	236.46	300
N2_8"	8.000 in	80	WNF	8.625	12.700	350.00	30.00	335.81	300
M1B_24"	24.000 in	Actual	WNF	24.000	20.000	1050.00	34.00	346.35	150
M1A_24"	24.000 in	Actual	WNF	24.000	20.000	1050.00	34.00	346.35	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grv Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grv Weld	Inside Weld
N11A_2"	SA-105	35.000	10.000	...	...	...
N11B_2"	SA-105	35.000	10.000	...	...	...
N7_2"	SA-105	35.000	10.000	...	...	...
N5_2"	SA-105	35.000	10.000	...	...	...
N6_2"	SA-105	35.000	10.000	...	...	...
N10_4"	SA-106 B	35.000	10.000	28.000	30.000	...
N3_6"	SA-106 B	35.000	10.000	28.000	30.000	...
N4_6"	SA-106 B	35.000	10.000	28.000	30.000	...
N8_6"	SA-106 B	35.000	10.000	28.000	30.000	...
N1_8"	SA-106 B	35.000	10.000	28.000	30.000	...
N9_8"	SA-106 B	35.000	10.000	28.000	30.000	...
N2_8"	SA-106 B	35.000	10.000	28.000	30.000	...
M1B_24"	SA-516 70	35.000	10.000	34.000	34.000	...
M1A_24"	SA-516 70	35.000	10.000	34.000	34.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N11A_2"	4000.000	0.0	300.00	0.00	Shell-2
N11B_2"	14999.999	0.0	300.00	0.00	Shell-5
N7_2"	16999.998	0.0	200.00	0.00	Shell-6
N5_2"	17999.998	0.0	200.00	0.00	Shell-6
N6_2"	...	0.0	300.00	0.00	Top_head
N10_4"	...	180.0	300.00	0.00	Top_head
N3_6"	...	0.0	300.00	0.00	Btm_head
N4_6"	...	0.0	300.00	0.00	Top_head
N8_6"	...	180.0	300.00	0.00	Top_head



N1_8"	12149.999	180.0	200.00	0.00	Shell-5
N9_8"	19000.000	0.0	200.00	0.00	Shell-7
N2_8"	...	0.0	300.00	0.00	Top_head
M1B_24"	1250.000	270.0	300.00	0.00	Shell-1
M1A_24"	6599.999	270.0	300.00	0.00	Shell-3

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 16.0mm THK X 8219.0mm ID X 1500.0mm	SA-285 C
1	SKIRT: 16.0mm THK X 8219.0mm ID X 2000.0mm	SA-516 70
2	ELLIPTICAL HEAD: 2.0 X 1, 42.0mm THK X 8200.0mm ID X 50.0mm	SA-516 70
6	CYLINDER: 35.0mm THK X 8200.0mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 35.0mm THK X 8200.0mm ID X 2400.0mm	SA-516 70
5	PLATFORM: 1100mm X 1500mm WIDE	
5	LADDER: 4000.0mm	...
1	PLATFORM: 1100mm X 10200mm WIDE	
1	LADDER: 7000.0mm	...
2	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 8495.0mm OD X 8087.0 ID X 32.0mm	SA-285 C
16	BASERING BOLTS: 36.0mm DIA	SA-36
16	BASERING BOLT NUTS: 36.0mm DIA	...
1	BASERING TOP RING: 8495.0mm OD X 8251.0 ID X 36.0mm	...
32	BASERING GUSSET PLATES: 122.0mm X 122.0mm X 200.0mm X 14.0mm	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Btmhead	[10]	D	-27	-28		0.986	35.000	1.00	No
Btmhead	[7]	D	-23	-33		0.819	42.000	1.00	No
Shell-1	[8]	D	-27	-28		0.990	35.000	1.00	No
Shell-2	[8]	D	-27	-28		0.990	35.000	1.00	No
Shell-3	[8]	D	-27	-28		0.990	35.000	1.00	No
Shell-4	[8]	D	-27	-28		0.990	35.000	1.00	No
Shell-5	[8]	D	-27	-28		0.990	35.000	1.00	No
Shell-6	[8]	D	-27	-28		0.990	35.000	1.00	No
Shell-7	[8]	D	-27	-28		0.990	35.000	1.00	No
Tophead	[10]	D	-27	-28		0.986	35.000	1.00	No
Tophead	[7]	D	-23	-33		0.819	42.000	1.00	No
N36"	[1]	B	-30	-37		0.879	30.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1B24"	[1]	D	-28	-29		0.981	34.000	1.00	No
Nozzle Flg	[5]	A	-18	-44					
N11A2"	[1]	A	1	-104		0.018	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-44					
M1A24"	[1]	D	-28	-29		0.981	34.000	1.00	No
Nozzle Flg	[5]	A	-18	-44					
N18"	[1]	B	-29	-30	-29	0.981	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N11B2"	[1]	A	1	-104		0.018	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-44					
N72"	[1]	A	1	-104		0.018	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-44					
N52"	[1]	A	1	-104		0.018	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-44					
N98"	[1]	B	-29	-30	-29	0.981	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N28"	[1]	B	-30	-37		0.879	30.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N46"	[1]	B	-30	-37		0.879	30.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N62"	[1]	A	1	-104		0.018	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-44					
N86"	[1]	B	-30	-37		0.879	30.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N104"	[1]	B	-30	-37		0.879	30.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Noz1. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: 1 -28

Required Minimum Design Metal Temperature -10.0 °C
 Warmest Computed Minimum Design Metal Temperature -28.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required



- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	1.1	MPa
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-10.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm
Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	1.500	
Site Class	D	
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	2.500
Force Factor		0.000
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg
Design Pressure + Static Head	Y	
Consider MAP New and Cold in Noz. Design	N	
Consider External Loads for Nozzle Des.	Y	
Use ASME VIII-1 Appendix 1-9	N	
Perform Blast Load Analysis	No	
Material Database Year	Current w/Addenda or Code Year	

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



Element#	1/11	
Element From Node	10	
Element To Node	20	
Element Type	Skirt Support	
Description	Skirt-2	
Distance "FROM" to "TO"	1500	mm
Skirt Outside Diameter	8251	mm
Diameter of Skirt at Base	8251	mm
Skirt Thickness	16	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	85	°C
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-285 C	
Allowable Stress, Ambient	108	MPa
Allowable Stress, Operating	108	MPa
Allowable Stress, Hydrotest	184.5	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	191.57	MPa
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-2	
UNS Number	K02801	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	

Element#	2/11	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt-1	
Distance "FROM" to "TO"	2000	mm
Skirt Outside Diameter	8251	mm
Diameter of Skirt at Base	8251	mm
Skirt Thickness	16	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	85	°C
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +N	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	242	MPa
UCS-66 Chart Curve Designation	D	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	Yes	

Element#	3/11	
Element From Node	30	
Element To Node	40	
Element Type	Elliptical	
Description	Btmhead	
Distance "FROM" to "TO"	50	mm



Inside Diameter	8200	mm
Element Thickness	35	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	42	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	Yes	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N36"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	413.79	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	4/11	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	8200	mm
Element Thickness	35	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	35	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	Yes	

Element From Node	40	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	450	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	8550720	mm ²
Platform Weight	80085	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1100	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.49	N/mm



Platform Length (top head platform)	0	mm
Element From Node	40	
Detail Type	Nozzle	
Detail ID	M1B24"	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	5503	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+N

Element#	5/11	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	8200	mm
Element Thickness	35	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	35	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	Yes	

Element From Node	50	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	1500	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	8550720	mm ²
Platform Weight	41023	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1100	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.49	N/mm
Platform Length (top head platform)	0	mm

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N11A2"	
Dist. from "FROM" Node / Offset dist	950	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	100.29	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+N



Element# 6/11
Element From Node 60
Element To Node 70
Element Type Cylinder
Description Shell-3
Distance "FROM" to "TO" 3000 mm
Inside Diameter 8200 mm
Element Thickness 35 mm
Internal Corrosion Allowance 1.5 mm
Nominal Thickness 35 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 1.1 MPa
Design Temperature Internal Pressure 85 °C
Design External Pressure 0 MPa
Design Temperature External Pressure 85 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 70 +N
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Weld is pre-Heated Yes

Element From Node 60
Detail Type Platform
Detail ID PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist 2500 mm
Platform Start Angle (degrees) 180.0
Platform End Angle (degrees) 0.0
Platform Wind Area 8550720 mm²
Platform Weight 41023 N
Platform Railing Weight 0 N/mm
Platform Grating Weight 0.017 kg/cm²
Platform Width 1500 mm
Platform Height 1100 mm
Platform Clearance or End Offset 85 mm
Platform Force Coefficient 2.0
Ladder Layout Angle 0.0
Ladder Start Elevation 0 mm
Ladder End Elevation 4000 mm
Unit Weight of Ladder 0.49 N/mm
Platform Length (top head platform) 0 mm

Element From Node 60
Detail Type Nozzle
Detail ID M1A24"
Dist. from "FROM" Node / Offset dist 550 mm
Nozzle Diameter 24 in.
Nozzle Schedule None
Nozzle Class 150
Layout Angle 270.0
Blind Flange (Y/N) Y
Weight of Nozzle (Used if > 0) 3690.2 N
Grade of Attached Flange GR 1.1
Nozzle Matl SA-516 70 +N

Element# 7/11
Element From Node 70
Element To Node 80
Element Type Cylinder
Description Shell-4
Distance "FROM" to "TO" 3000 mm
Inside Diameter 8200 mm
Element Thickness 35 mm
Internal Corrosion Allowance 1.5 mm
Nominal Thickness 35 mm
External Corrosion Allowance 0 mm



Design Internal Pressure	1.1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	Yes	

Element From Node	70	
Detail Type	Weight	
Detail ID	Clipwt	
Dist. from "FROM" Node / Offset dist	2500	mm
Miscellaneous Weight	46284	N
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	Misc	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	98060	N
Offset from Element Centerline	0	mm

Element#	8/11	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-5	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	8200	mm
Element Thickness	35	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	35	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	Yes	

Element From Node	80	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	500	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	8550720	mm ²
Platform Weight	41023	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1100	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.49	N/mm
Platform Length (top head platform)	0	mm

Element From Node	80	
Detail Type	Nozzle	
Detail ID	N18"	



Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	532.56	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	N11B2"	
Dist. from "FROM" Node / Offset dist	2950	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	100.29	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +N	

Element From Node	80	
Detail Type	Weight	
Detail ID	N1pipewt	
Dist. from "FROM" Node / Offset dist	100	mm
Miscellaneous Weight	13081	N
Offset from Element Centerline	5237	mm

Element#	9/11	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	Shell-6	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	8200	mm
Element Thickness	35	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	35	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70 +N	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	Yes	

Element From Node	90	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	1500	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	8550720	mm ²
Platform Weight	41023	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1100	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm



Unit Weight of Ladder	0.49	N/mm
Platform Length (top head platform)	0	mm
Element From Node	90	
Detail Type	Nozzle	
Detail ID	N72"	
Dist. from "FROM" Node / Offset dist	1950	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	79.631	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+N
Element From Node	90	
Detail Type	Nozzle	
Detail ID	N52"	
Dist. from "FROM" Node / Offset dist	2950	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	79.631	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+N

Element#	10/11
Element From Node	100
Element To Node	110
Element Type	Cylinder
Description	Shell-7
Distance "FROM" to "TO"	2400 mm
Inside Diameter	8200 mm
Element Thickness	35 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	35 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	Yes

Element From Node	100
Detail Type	Nozzle
Detail ID	N98"
Dist. from "FROM" Node / Offset dist	950 mm
Nozzle Diameter	8 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	532.56 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	100
Detail Type	Weight
Detail ID	N9pipe wt
Dist. from "FROM" Node / Offset dist	1200 mm
Miscellaneous Weight	13081 N



Offset from Element Centerline	5237	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	Trunnionwt	
Dist. from "FROM" Node / Offset dist	0.002326	mm
Miscellaneous Weight	29418	N
Offset from Element Centerline	0	mm

Element#	11/11
Element From Node	110
Element To Node	120
Element Type	Elliptical
Description	Tophead
Distance "FROM" to "TO"	50 mm
Inside Diameter	8200 mm
Element Thickness	35 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	42 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	Yes

Element From Node	110
Detail Type	Platform
Detail ID	TOPPF
Dist. from "FROM" Node / Offset dist	2500 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	25853800 mm ²
Platform Weight	176956 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	10200 mm
Platform Height	1100 mm
Platform Clearance or End Offset	5100 mm
Platform Force Coefficient	2.0
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm
Ladder End Elevation	7000 mm
Unit Weight of Ladder	0.49 N/mm
Platform Length (top head platform)	10200 mm

Element From Node	110
Detail Type	Nozzle
Detail ID	N28"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	8 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	643.33 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	110
Detail Type	Nozzle
Detail ID	N46"
Dist. from "FROM" Node / Offset dist	2000 mm



Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	413.79	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	110	
Detail Type	Nozzle	
Detail ID	N62"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	100.29	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +N	

Element From Node	110	
Detail Type	Nozzle	
Detail ID	N86"	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	413.79	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	110	
Detail Type	Nozzle	
Detail ID	N104"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	4	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	189.43	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	110	
Detail Type	Weight	
Detail ID	N2Pipe weight	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	11797	N
Offset from Element Centerline	5237	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	PSVwt	
Dist. from "FROM" Node / Offset dist	2000	mm
Miscellaneous Weight	19612	N
Offset from Element Centerline	5000	mm

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-2	...	...	1500	...	1500
Skirt-1	...	...	3500	...	2000
Btm_head	...	...	3550	...	50
Shell-1	...	...	6550	...	3000
Shell-2	...	...	9550	...	3000
Shell-3	...	...	12550	...	3000
Shell-4	...	...	15550	...	3000
Shell-5	...	...	18550	...	3000
Shell-6	...	...	21550	...	3000
Shell-7	...	...	23950	...	2400
Top_head	...	...	24000	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-2		...	16	1.5	8251	...	1
Skirt-1		...	16	1.5	8251	...	2
Btm_head		1.1	42	1.5	8200	138	3
Shell-1		1.1	35	1.5	8200	138	4
Shell-2		1.1	35	1.5	8200	138	5
Shell-3		1.1	35	1.5	8200	138	6
Shell-4		1.1	35	1.5	8200	138	7
Shell-5		1.1	35	1.5	8200	138	8
Shell-6		1.1	35	1.5	8200	138	9
Shell-7		1.1	35	1.5	8200	138	10
Top_head		1.1	42	1.5	8200	138	11

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-2		...	No Calc	No Calc	16	No Calc	1
Skirt-1		...	No Calc	No Calc	16	No Calc	2
Btm_head		1.1	1.13	1.18	35	34.2033	3
Shell-1		1.1	1.12	1.17	35	34.3508	4
Shell-2		1.1	1.12	1.17	35	34.3508	5
Shell-3		1.1	1.12	1.17	35	34.3508	6
Shell-4		1.1	1.12	1.17	35	34.3508	7
Shell-5		1.1	1.12	1.17	35	34.3508	8
Shell-6		1.1	1.12	1.17	35	34.3508	9
Shell-7		1.1	1.12	1.17	35	34.3508	10
Top_head		1.1	1.13	1.18	35	34.2038	11

Minimum 1.11 1.17

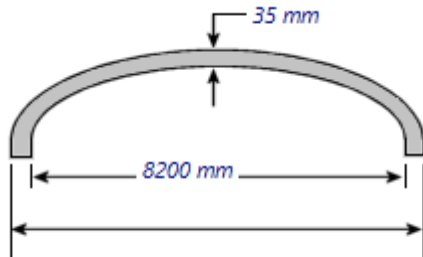
MAWP: 1.11 MPa, limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Btmhead

Elliptical Head Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	8200.000 mm	Actual Stress		134.7 MPa
Spec. Min. Thk t _e , or s _a		35.0000 mm	Spec. Nominal Thk.		42.000 mm
Surface Area		75689384.0 mm ²	SG of Contents		0.000
Element Volume		74827.8 ltr	Weight of Contents		0.0 N
Empty Weight		263642.6 N	Operating Weight		263642.6 N
			Comp. All. Ope. Stress		102.8 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 102.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 51.250 mm

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1.1 \cdot 8203 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$$

$$= 32.7033 + 1.5000 = 34.2033 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 33.5) / (1 \cdot 8203 + 0.2 \cdot 33.5)$$

$$= 1.13 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 35) / (1 \cdot 8200 + 0.2 \cdot 35)$$

$$= 1.18 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1.1 \cdot (1 \cdot 8203 + 0.2 \cdot 33.5)) / (2 \cdot 1 \cdot 33.5)$$

$$= 134.721 \text{ MPa}$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (1.1 \cdot 4102) / (138 \cdot 1 - 0.6 \cdot 1.1) + 1.5$
 $= 34.350 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 40.5) / (4102 + 0.6 \cdot 40.5)$
 $= 1.35 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (8203 / (2 \cdot 2052))^2) / 6$
 $= 0.999513$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.226 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 35$, $t_r = 33$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 42$, $t_r = 33.2$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.82$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -33 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.111 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	35.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.986	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

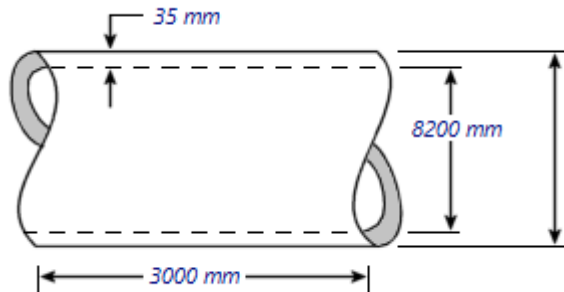
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 102.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 51.250 mm

Material treatment options: +N

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	8200.000 mm	Actual Stress		135.3 MPa
Spec. Min. Thk t _e , or s _a		35.0000 mm	Spec. Nominal Thk.		35.000 mm
Surface Area		77942920.0 mm ²	SG of Contents		0.000
Element Volume		158458.9 ltr	Weight of Contents		0.0 N
Empty Weight		206444.7 N	Operating Weight		206444.7 N
			Comp. All. Ope. Stress		102.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 4102) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 32.8508 + 1.5000 = 34.3508 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 33.5) / (4102 + 0.6 \cdot 33.5)$$

$$= 1.12 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 35) / (4100 + 0.6 \cdot 35)$$

$$= 1.17 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (4102 + 0.6 \cdot 33.5)) / (1 \cdot 33.5)$$

$$= 135.336 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.425 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 35, tr = 33.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

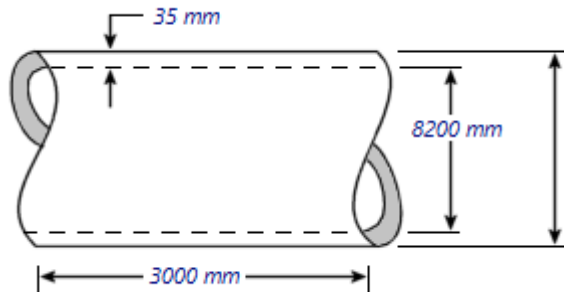
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.111 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	35.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.990	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	8200.000 mm	Actual Stress		135.3 MPa
Spec. Min. Thk t _e , or s _a		35.0000 mm	Spec. Nominal Thk.		35.000 mm
Surface Area		77942920.0 mm ²	SG of Contents		0.000
Element Volume		158458.9 ltr	Weight of Contents		0.0 N
Empty Weight		206444.7 N	Operating Weight		206444.7 N
			Comp. All. Ope. Stress		102.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 4102) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 32.8508 + 1.5000 = 34.3508 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 33.5) / (4102 + 0.6 \cdot 33.5)$$

$$= 1.12 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 35) / (4100 + 0.6 \cdot 35)$$

$$= 1.17 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (4102 + 0.6 \cdot 33.5)) / (1 \cdot 33.5)$$

$$= 135.336 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.425 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 35, tr = 33.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

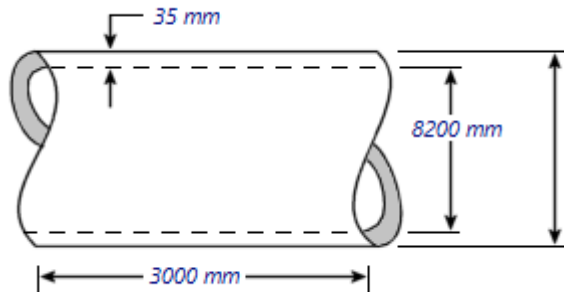
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.111 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	35.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.990	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	8200.000 mm	Actual Stress		135.3 MPa
Spec. Min. Thk t _e , or s _a		35.0000 mm	Spec. Nominal Thk.		35.000 mm
Surface Area		77942920.0 mm ²	SG of Contents		0.000
Element Volume		158458.9 ltr	Weight of Contents		0.0 N
Empty Weight		206444.7 N	Operating Weight		206444.7 N
			Comp. All. Ope. Stress		102.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 4102) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 32.8508 + 1.5000 = 34.3508 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 33.5) / (4102 + 0.6 \cdot 33.5)$$

$$= 1.12 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 35) / (4100 + 0.6 \cdot 35)$$

$$= 1.17 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (4102 + 0.6 \cdot 33.5)) / (1 \cdot 33.5)$$

$$= 135.336 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.425 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 35, tr = 33.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

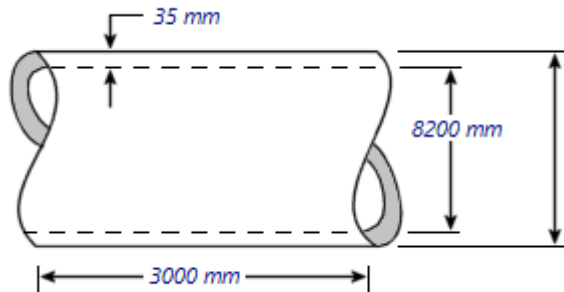
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.111 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	35.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.990	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	8200.000 mm	Actual Stress		135.3 MPa
Spec. Min. Thk t _e , or s _a		35.0000 mm	Spec. Nominal Thk.		35.000 mm
Surface Area		77942920.0 mm ²	SG of Contents		0.000
Element Volume		158458.9 ltr	Weight of Contents		0.0 N
Empty Weight		206444.7 N	Operating Weight		206444.7 N
			Comp. All. Ope. Stress		102.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 4102) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 32.8508 + 1.5000 = 34.3508 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 33.5) / (4102 + 0.6 \cdot 33.5)$$

$$= 1.12 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 35) / (4100 + 0.6 \cdot 35)$$

$$= 1.17 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (4102 + 0.6 \cdot 33.5)) / (1 \cdot 33.5)$$

$$= 135.336 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.425 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 35, tr = 33.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

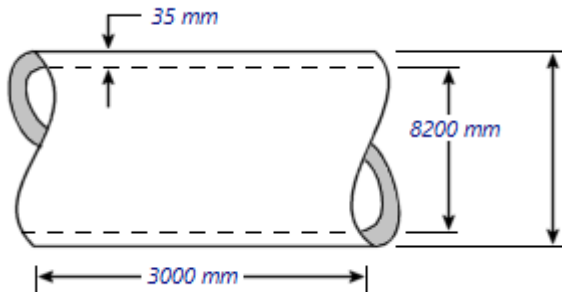
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.111 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	35.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.990	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	8200.000 mm	Actual Stress		135.3 MPa
Spec. Min. Thk t _e , or s _a		35.0000 mm	Spec. Nominal Thk.		35.000 mm
Surface Area		77942920.0 mm ²	SG of Contents		0.000
Element Volume		158458.9 ltr	Weight of Contents		0.0 N
Empty Weight		206444.7 N	Operating Weight		206444.7 N
			Comp. All. Ope. Stress		102.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 4102) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 32.8508 + 1.5000 = 34.3508 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 33.5) / (4102 + 0.6 \cdot 33.5)$$

$$= 1.12 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 35) / (4100 + 0.6 \cdot 35)$$

$$= 1.17 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (4102 + 0.6 \cdot 33.5)) / (1 \cdot 33.5)$$

$$= 135.336 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.425 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 35, tr = 33.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

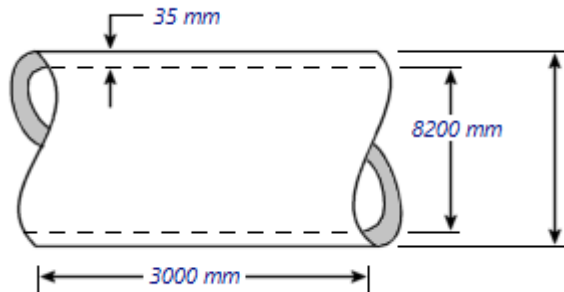
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.111 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	35.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.990	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	8200.000 mm	Actual Stress		135.3 MPa
Spec. Min. Thk t _e , or s _a		35.0000 mm	Spec. Nominal Thk.		35.000 mm
Surface Area		77942920.0 mm ²	SG of Contents		0.000
Element Volume		158458.9 ltr	Weight of Contents		0.0 N
Empty Weight		206444.7 N	Operating Weight		206444.7 N
			Comp. All. Ope. Stress		102.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 4102) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 32.8508 + 1.5000 = 34.3508 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 33.5) / (4102 + 0.6 \cdot 33.5)$$

$$= 1.12 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 35) / (4100 + 0.6 \cdot 35)$$

$$= 1.17 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (4102 + 0.6 \cdot 33.5)) / (1 \cdot 33.5)$$

$$= 135.336 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.425 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 35, tr = 33.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

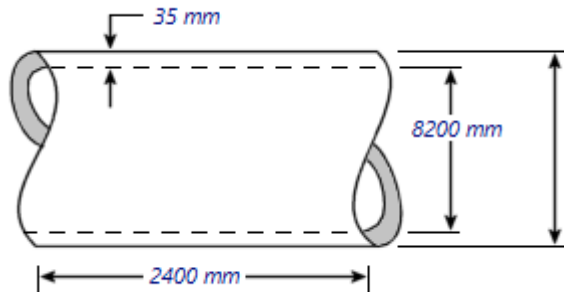
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.111 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	35.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.990	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		2400.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	8200.000 mm	Actual Stress		135.3 MPa
Spec. Min. Thk t _e , or s _a		35.0000 mm	Spec. Nominal Thk.		35.000 mm
Surface Area		62354340.0 mm ²	SG of Contents		0.000
Element Volume		126767.1 ltr	Weight of Contents		0.0 N
Empty Weight		165155.5 N	Operating Weight		165155.5 N
			Comp. All. Ope. Stress		102.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 4102) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 32.8508 + 1.5000 = 34.3508 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 33.5) / (4102 + 0.6 \cdot 33.5)$$

$$= 1.12 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 35) / (4100 + 0.6 \cdot 35)$$

$$= 1.17 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (4102 + 0.6 \cdot 33.5)) / (1 \cdot 33.5)$$

$$= 135.336 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.425 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 35, tr = 33.2, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

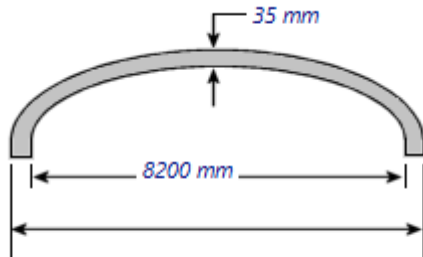
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.111 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	35.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.990	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Tophead

Elliptical Head Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	8200.000 mm	Actual Stress		134.7 MPa
Spec. Min. Thk t _e , or s _a		35.0000 mm	Spec. Nominal Thk.		42.000 mm
Surface Area		75689384.0 mm ²	SG of Contents		0.000
Element Volume		74827.8 ltr	Weight of Contents		0.0 N
Empty Weight		263642.6 N	Operating Weight		263642.6 N
			Comp. All. Ope. Stress		102.8 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 102.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 51.250 mm

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1.1 \cdot 8203 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$$

$$= 32.7038 + 1.5000 = 34.2038 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 33.5) / (1 \cdot 8203 + 0.2 \cdot 33.5)$$

$$= 1.13 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 35) / (1 \cdot 8200 + 0.2 \cdot 35)$$

$$= 1.18 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1.1 \cdot (1 \cdot 8203 + 0.2 \cdot 33.5)) / (2 \cdot 1 \cdot 33.5)$$

$$= 134.721 \text{ MPa}$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (1.1 \cdot 4102) / (138 \cdot 1 - 0.6 \cdot 1.1) + 1.5$
 $= 34.351 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 40.5) / (4102 + 0.6 \cdot 40.5)$
 $= 1.35 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (8203 / (2 \cdot 2052))^2) / 6$
 $= 0.999513$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.226 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 35$, $t_r = 33$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 42$, $t_r = 33.2$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.82$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -33 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.111 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	35.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.986	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 102.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 51.250 mm

Material treatment options: +N

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.444 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.430 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.443 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.222 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.573 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.573 MPa
App 27-4	= M.A.W.P.	1.111 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 1.11 • 1
 = 1.444 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm_head	178.7	234.0	0.764	1.52
Shell-1	179.5	234.0	0.767	1.52
Shell-2	179.5	234.0	0.767	1.52
Shell-3	179.5	234.0	0.767	1.52
Shell-4	179.5	234.0	0.767	1.52
Shell-5	179.5	234.0	0.767	1.52
Shell-6	179.5	234.0	0.767	1.52
Shell-7	179.5	234.0	0.767	1.52
Top_head	178.7	234.0	0.764	1.52

Theoretical Maximum Allowable test pressure is : 1.907 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_6"	Nozzle	118.00	118.00	1.000
N3_6"	Pad	138.00	138.00	1.000
M1B_24"	Nozzle	138.00	138.00	1.000
M1B_24"	Pad	138.00	138.00	1.000
N11A_2"	Nozzle	138.00	138.00	1.000
M1A_24"	Nozzle	138.00	138.00	1.000
M1A_24"	Pad	138.00	138.00	1.000
N1_8"	Nozzle	118.00	118.00	1.000
N1_8"	Pad	138.00	138.00	1.000
N11B_2"	Nozzle	138.00	138.00	1.000
N7_2"	Nozzle	138.00	138.00	1.000
N5_2"	Nozzle	138.00	138.00	1.000
N9_8"	Nozzle	118.00	118.00	1.000
N9_8"	Pad	138.00	138.00	1.000
N2_8"	Nozzle	118.00	118.00	1.000
N2_8"	Pad	138.00	138.00	1.000
N4_6"	Nozzle	118.00	118.00	1.000
N4_6"	Pad	138.00	138.00	1.000
N6_2"	Nozzle	138.00	138.00	1.000
N8_6"	Nozzle	118.00	118.00	1.000
N8_6"	Pad	138.00	138.00	1.000
N10_4"	Nozzle	118.00	118.00	1.000
N10_4"	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):



Description	Ambient	Operating	Ratio
Btm_head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Shell-6	138.00	138.00	1.000
Shell-7	138.00	138.00	1.000
Top_head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_6"	12.75	216.00	0.059
M1B_24"	22.62	234.00	0.097
N11A_2"	3.79	225.00	0.017
M1A_24"	22.62	234.00	0.097
N1_8"	14.42	216.00	0.067
N11B_2"	3.79	225.00	0.017
N7_2"	3.79	225.00	0.017
N5_2"	3.79	225.00	0.017
N9_8"	14.42	216.00	0.067
N2_8"	14.42	216.00	0.067
N4_6"	12.75	216.00	0.059
N6_2"	3.79	225.00	0.017
N8_6"	12.75	216.00	0.059
N10_4"	11.02	216.00	0.051

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.444 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.430 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.480 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.222 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.573 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.573 MPa
App 27-4	= M.A.W.P.	1.111 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 1.11 • 1
 = 1.444 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm_head	206.4	234.0	0.882	1.69
Shell-1	204.8	234.0	0.875	1.66
Shell-2	201.2	234.0	0.860	1.64
Shell-3	197.5	234.0	0.844	1.61
Shell-4	193.9	234.0	0.829	1.58
Shell-5	190.3	234.0	0.813	1.55
Shell-6	186.7	234.0	0.798	1.52
Shell-7	183.1	234.0	0.782	1.49
Top_head	179.4	234.0	0.767	1.46

Theoretical Maximum Allowable test pressure is : 1.669 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_6"	Nozzle	118.00	118.00	1.000
N3_6"	Pad	138.00	138.00	1.000
M1B_24"	Nozzle	138.00	138.00	1.000
M1B_24"	Pad	138.00	138.00	1.000
N11A_2"	Nozzle	138.00	138.00	1.000
M1A_24"	Nozzle	138.00	138.00	1.000
M1A_24"	Pad	138.00	138.00	1.000
N1_8"	Nozzle	118.00	118.00	1.000
N1_8"	Pad	138.00	138.00	1.000
N11B_2"	Nozzle	138.00	138.00	1.000
N7_2"	Nozzle	138.00	138.00	1.000
N5_2"	Nozzle	138.00	138.00	1.000
N9_8"	Nozzle	118.00	118.00	1.000
N9_8"	Pad	138.00	138.00	1.000
N2_8"	Nozzle	118.00	118.00	1.000
N2_8"	Pad	138.00	138.00	1.000
N4_6"	Nozzle	118.00	118.00	1.000
N4_6"	Pad	138.00	138.00	1.000
N6_2"	Nozzle	138.00	138.00	1.000
N8_6"	Nozzle	118.00	118.00	1.000
N8_6"	Pad	138.00	138.00	1.000
N10_4"	Nozzle	118.00	118.00	1.000
N10_4"	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):



Description	Ambient	Operating	Ratio
Btm_head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Shell-6	138.00	138.00	1.000
Shell-7	138.00	138.00	1.000
Top_head	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_6"	14.09	216.00	0.065
M1B_24"	24.70	234.00	0.106
N11A_2"	4.07	225.00	0.018
M1A_24"	23.83	234.00	0.102
N1_8"	14.63	216.00	0.068
N11B_2"	3.85	225.00	0.017
N7_2"	3.77	225.00	0.017
N5_2"	3.77	225.00	0.017
N9_8"	14.07	216.00	0.065
N2_8"	13.85	216.00	0.064
N4_6"	12.25	216.00	0.057
N6_2"	3.64	225.00	0.016
N8_6"	12.25	216.00	0.057
N10_4"	10.59	216.00	0.049

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	3500	...	...	No Calc	No Calc
20	30	3500	...	...	No Calc	No Calc
30	40	No Calc	8270	33.5	0.00056261	56.2483
40	50	21866.7	8270	33.5	0.00012435	12.4318
50	60	21866.7	8270	33.5	0.00012435	12.4318
60	70	21866.7	8270	33.5	0.00012435	12.4318
70	80	21866.7	8270	33.5	0.00012435	12.4318
80	90	21866.7	8270	33.5	0.00012435	12.4318
90	100	21866.7	8270	33.5	0.00012435	12.4318
100	110	21866.7	8270	33.5	0.00012435	12.4318
110	120	No Calc	8270	33.5	0.00056261	56.2483

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	35	No Calc	...	0.25
40	50	35	No Calc	...	0.067
50	60	35	No Calc	...	0.067
60	70	35	No Calc	...	0.067
70	80	35	No Calc	...	0.067
80	90	35	No Calc	...	0.067
90	100	35	No Calc	...	0.067
100	110	35	No Calc	...	0.067
110	120	35	No Calc	...	0.25

Minimum

0.067

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	3500	No Calc	No Calc	No Calc
20	30	3500	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	21866.7	No Calc	No Calc	No Calc
50	60	21866.7	No Calc	No Calc	No Calc
60	70	21866.7	No Calc	No Calc	No Calc
70	80	21866.7	No Calc	No Calc	No Calc
80	90	21866.7	No Calc	No Calc	No Calc
90	100	21866.7	No Calc	No Calc	No Calc
100	110	21866.7	No Calc	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	7110.38	...	6660.01	...	...
20	30	6416.11	...	5815.78	...	...
30	40	26885.9	74827.8	25925.6	74909	...
40	50	21052.9	158459	20154.4	158575	...
50	60	21052.9	158459	20154.4	158575	...
60	70	21052.9	158459	20154.4	158575	...
70	80	21052.9	158459	20154.4	158575	...
80	90	21052.9	158459	20154.4	158575	...
90	100	21052.9	158459	20154.4	158575	...
100	110	16842.3	126767	16123.5	126860	...
110	120	26885.9	74827.8	25925.6	74909	...
Total		210457	1227176.25	201376	1228127.38	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
30	Nozl	42.1976	...	-2232.5	...	N3_6"
40	Plat	8166.97	0.44035E-04	450	...	PLAT:[1 OF 1]
40	Nozl	561.19	...	1200	4250	M1B_24"
50	Plat	4183.42	...	1500	-3165.91	PLAT:[1 OF 1]
50	Nozl	10.227	4250	950	...	N11A_2"
60	Plat	4183.42	...	2500	3165.91	PLAT:[1 OF 1]
60	Nozl	376.316	...	550	4250	M1A_24"
70	Wght	4720	...	2500	...	Clip_wt
70	Wght	10000	...	...	...	Misc
80	Plat	4183.42	...	500	-3165.91	PLAT:[1 OF 1]
80	Nozl	54.3099	-4200	100	...	N1_8"
80	Nozl	10.227	4250	2950	...	N11B_2"
80	Wght	1334	-5237	100.001	...	N1_Pipe_wt
90	Plat	4183.42	...	1500	3165.91	PLAT:[1 OF 1]
90	Nozl	8.12068	4200	1950	...	N7_2"
90	Nozl	8.12068	4200	2950	...	N5_2"
100	Nozl	54.3099	4200	950	...	N9_8"
100	Wght	1334	5237	1200	...	N9_Pipe_wt
100	Wght	3000	...	0.0023256	...	Trunnion_wt
110	Plat	18045.7	...	2500	...	TOP_PF
110	Nozl	65.6061	...	2232.5	...	N2_8"
110	Nozl	42.1976	2000	1972.05	...	N4_6"
110	Nozl	10.227	1500	2090.38	...	N6_2"
110	Nozl	42.1976	-2000	1972.05	...	N8_6"
110	Nozl	19.3177	-1500	2090.38	...	N10_4"
110	Wght	1203	5237	...	...	N2_Pipe weight
110	Wght	2000	-5000	2000	...	PSV_wt

Total Weight of Each Detail Type:

Platforms	42946.3
Nozzles	1304.6
Weights	23591.0
Sum of the Detail Weights	67841.9 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	210457.9	210457.9	210457.9	210457.9	210457.9	210457.9
Nozzles	1304.6	1304.6	1304.6	1304.6	1304.6	1304.6



Wld Weights	17720.0	17720.0	17720.0	17720.0	17720.0	17720.0
Platforms	...	...	42946.3	42946.3	42946.3	42946.3
Ope Weights	...	...	...	...	...	5871.0
Test Liquid	...	1226427.2	...	...	...	...
Totals	229482.5	1455909.6	272428.8	272428.8	272428.8	278299.8

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	229482.5 kgm
Shop Test Wt.	- Fabricated weight + water (full)	1455909.6 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	272428.8 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	272428.8 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	272428.8 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	278299.8 kgm
Field Test Wt.	- Empty weight + test medium (full)	1498856.1 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	113544.6 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	588979.2 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	112348.9 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	38881924	38731124
20	30	51842564	51641504
30	40	75689384	74175240
40	50	77942920	77283184
50	60	77942920	77283184
60	70	77942920	77283184
70	80	77942920	77283184
80	90	77942920	77283184
90	100	77942920	77283184
100	110	62354340	61826552
110	120	75689384	74175240

Total	772115008.0	764248768.0 mm ²
Total	772.1	764.2 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	7110.38	7110.38	7110.38	...	7110.38
20	30	6416.11	6416.11	6416.11	...	6416.11
30	40	26928.1	26928.1	101710	...	26928.1
40	50	29781.1	29781.1	188143	23397346	29781.1
50	60	25246.5	25246.5	183609	129926360	25246.5
60	70	25612.6	25612.6	183975	145615872	25612.6
70	80	35772.9	35772.9	194135	...	35772.9
80	90	25300.9	26634.9	183663	129926360	26634.9
90	100	25252.6	25252.6	183615	129926360	25252.6
100	110	19896.6	21230.6	146586	...	21230.6
110	120	45111.1	48314.1	119893	...	48314.1

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
------	----	-----------------------------------	---------------------------	----------------------------

10	20	278300	278300	1498856
20	30	271189	271189	1491745
30	40	264773	264773	1485329
40	50	237845	237845	1383619
50	60	208064	208064	1195476
60	70	182818	182818	1011867
70	80	157205	157205	827892
80	90	121432	121432	633757
90	100	94797.3	94797.3	450094
100	110	69544.7	69544.7	266480
110	120	48314.1	48314.1	119893

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	558792320	558792320	558792320
20	30	558792320	558792320	558792320
30	40	558792320	558792320	558792320
40	50	558792320	558792320	558792320
50	60	535394944	535394944	535394944
60	70	405468608	405468608	405468608
70	80	259852720	259852720	259852720
80	90	259852720	259852720	259852720
90	100	129926360	129926360	129926360
100	110	...	...	...
110	120	...	...	...

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N3_6"	4.76	5.10	85	300	GR 1.1	3.508	3.637	4.763	3.637
M1B_24"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N11A_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
M1A_24"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N1_8"	4.76	5.10	85	300	GR 1.1	...	...	...	...
N11B_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N7_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N5_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N9_8"	4.76	5.10	85	300	GR 1.1	3.050	4.095	4.763	4.095
N2_8"	4.76	5.10	85	300	GR 1.1	3.049	4.096	4.763	4.096
N4_6"	4.76	5.10	85	300	GR 1.1	3.508	3.637	4.763	3.637
N6_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N8_6"	4.76	5.10	85	300	GR 1.1	3.508	3.637	4.763	3.637
N10_4"	4.76	5.10	85	300	GR 1.1	4.479	2.666	4.763	2.666
Min Rating	1.812	1.960, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N3_6"	196.088	0.50	4724851	9600	1.111	3.637	Pass
N9_8"	248.539	0.50	8400389	12799	1.111	4.095	Pass
N2_8"	248.539	0.50	8399774	12799	1.111	4.096	Pass
N4_6"	196.088	0.50	4724851	9600	1.111	3.637	Pass
N8_6"	196.088	0.50	4724851	9600	1.111	3.637	Pass
N10_4"	138.176	0.50	2099934	6400	1.111	2.666	Pass

Evaluating Flange on Nozzle [N36"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(4724851) + 4 \cdot \text{abs}(9600)196.088 \leq \pi \cdot 196.088^3 \cdot ((4.763 - 1.111) + .500 \cdot 4.763)$$

$$= 0.83130E+08 \leq 0.14298E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N36" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 83130120.0 / (\pi \cdot 196.088^3))$$

$$= \min(4.763, 3.637)$$

$$= 3.637 \text{ MPa}$$

Evaluating Flange on Nozzle [N98"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(8400389) + 4 \cdot \text{abs}(12799)248.539 \leq \pi \cdot 248.539^3 \cdot ((4.763 - 1.111) + .500 \cdot 4.763)$$

$$= 0.14714E+09 \leq 0.29114E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N98" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 147135952.0 / (\pi \cdot 248.539^3))$$

$$= \min(4.763, 4.095)$$

$$= 4.095 \text{ MPa}$$

Evaluating Flange on Nozzle [N28"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(8399774) + 4 \cdot \text{abs}(12799)248.539 \leq \pi \cdot 248.539^3 [(4.763 - 1.111) + .500 \cdot 4.763] \\ &= 0.14713E+09 \leq 0.29114E+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N28" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 147126128.0 / (\pi \cdot 248.539^3)) \\ &= \min(4.763, 4.096) \\ &= 4.096 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N46"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(4724851) + 4 \cdot \text{abs}(9600)196.088 \leq \pi \cdot 196.088^3 [(4.763 - 1.111) + .500 \cdot 4.763] \\ &= 0.83130E+08 \leq 0.14298E+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N46" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 83130120.0 / (\pi \cdot 196.088^3)) \\ &= \min(4.763, 3.637) \\ &= 3.637 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N86"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(4724851) + 4 \cdot \text{abs}(9600)196.088 \leq \pi \cdot 196.088^3 [(4.763 - 1.111) + .500 \cdot 4.763] \\ &= 0.83130E+08 \leq 0.14298E+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N86" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 83130120.0 / (\pi \cdot 196.088^3)) \\ &= \min(4.763, 3.637) \\ &= 3.637 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N104"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(2099934) + 4 \cdot \text{abs}(6400)138.176 \leq \pi \cdot 138.176^3 [(4.763 - 1.111) + .500 \cdot 4.763] \\ &= 0.37138E+08 \leq 0.50028E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N104" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 37137528.0 / (\pi \cdot 138.176^3)) \\ &= \min(4.763, 2.666) \\ &= 2.666 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 6.37269 Hz.

The Natural Frequency for the Vessel (Ope...) is 6.37269 Hz.

The Natural Frequency for the Vessel (Filled) is 2.88431 Hz.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.14 \cdot 0.91) / (1 + 1.7 \cdot 3.4 \cdot 0.14)))$
 $= \min(0.85, 0.89)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	6.373 Hz
Natural Frequency of Vessel (Empty)	6.373 Hz
Natural Frequency of Vessel (Test)	2.884 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.537
Structure Height to Diameter ratio	3.217
Height to top of Structure	26085.000 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5 : Z_g = 589788 \text{ mm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 750 + 300 = 1050 \text{ mm}$
 $= 3.44 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (3.44 ft.) $< 15 \text{ ft.}$
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15 / 1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[K_{zt}]$:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf } [1.59] \text{ kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.54 \cdot 160$$

$$= 2419 \text{ lbs. } [10758] \text{ N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	1050.0	0.000	0.000	0.000	1.035	1.000	1.587
Skirt-1	2800.0	0.000	0.000	0.000	1.035	1.000	1.587
Btmhead	3825.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-1	5350.0	0.000	0.000	0.000	1.064	1.000	1.631
Shell-2	8350.0	0.000	0.000	0.000	1.149	1.000	1.762
Shell-3	11350.0	0.000	0.000	0.000	1.212	1.000	1.859
Shell-4	14350.0	0.000	0.000	0.000	1.263	1.000	1.936
Shell-5	17350.0	0.000	0.000	0.000	1.305	1.000	2.001
Shell-6	20350.0	0.000	0.000	0.000	1.342	1.000	2.058
Shell-7	23050.0	0.000	0.000	0.000	1.371	1.000	2.103
Tophead	25157.9	0.000	0.000	0.000	1.392	1.000	2.135

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, $\leq 0.31416E-05$ (High); $0.31416E-05 < 0.39270E-05$ (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0/24000 = 0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[27.1/78.7^2] \cdot 10^4 = 43.7$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.39270E-05$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 2639957 / (24000 \cdot 8270^2)$$

$$= 0.16083E-05$$

Compute the damping factor Df which is a measure of instability [Df]:

$$= W \cdot DI / (L \cdot Dr^2)$$

$$= 2639957 \cdot 0.03 / (24000 \cdot 8270^2)$$

$$= 0.48250E-07$$

Compute the critical wind velocity [Vc]:

$$= 3.4 \cdot f \cdot Dr$$

$$= 3.4 \cdot 6.37 \cdot 27.1$$

$$= 587.885 \text{ mile/hr } [946.084 \text{ km/hr}]$$

Compute the velocity at the top of the tower [Vw]:

$$= V30 \cdot (L / (30 + \text{BaseHeight}))^{0.143}$$

$$= 112 \cdot (78.7 / (30 + 0.98))^{0.143}$$

$$= 127.808 \text{ mile/hr } [205.682 \text{ km/hr}]$$

Compute the maximum gust velocity using the gust response factor Gh [Vg]:

$$= Vw \cdot Gh$$

$$= 206 \cdot 0.85$$

$$= 174.829 \text{ km/hr}$$

Since Vc is greater than Vg the dynamic deflection Z, does not need to be computed.

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLAT:[1 OF 1]	8550720.00	4300.00	1.60	13687.95	2.00
PLAT:[1 OF 1]	8550720.00	8350.00	1.76	15069.49	2.00
PLAT:[1 OF 1]	8550720.00	12350.00	1.88	16116.41	2.00
PLAT:[1 OF 1]	8550720.00	16350.00	1.98	16928.11	2.00
PLAT:[1 OF 1]	8550720.00	20350.00	2.06	17594.70	2.00
TOPPF	25853800.00	26750.00	2.15	55662.48	2.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Clipwt	0.00	15350.00	1.96	0.00
Misc	0.00	12850.00	1.90	0.00
N1Pipewt	0.00	15950.00	1.97	0.00
N9Pipe wt	0.00	23050.00	2.10	0.00
Trunnionwt	0.00	21850.00	2.08	0.00
N2Pipe weigh	0.00	24250.00	2.12	0.00
PSVwt	0.00	26250.00	2.15	0.00

The Natural Frequency for the Vessel (Ope...) is 6.37269 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1050	9901.2	14851802	1.58714	11833.7
20	30	2800	9901.2	19802402	1.58714	15778.3
30	40	3825	9924	496200	1.58714	395.366
40	50	5350	9924	29772004	1.63111	36480.9
50	60	8350	9924	29772004	1.76241	42918.1
60	70	11350	9924	29772004	1.85905	47076.1
70	80	14350	9924	29772004	1.93644	28942.8
80	90	17350	9924	29772004	2.00144	47462
90	100	20350	9924	29772004	2.05773	50109.9
100	110	23050	9924	23817606	2.10281	25143.5
110	120	25157.9	9924	16747299	2.13505	83054.1



The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		1.500	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(85.50.75)$
 = 0.562 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/6.37)
 = 0.157

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.56$, 0.16) per 12.8.2
 = 0.157

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.22 / (1.5 / 1.5)$
 = 0.217

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.093 / (0.16 (1.5 / 1.5))$
 = 0.591

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.22 \cdot 1.5), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.22 \cdot 2729007$
 = 593104.250 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.22 \cdot 2729007$



= 118620.8 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 6.37269 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	750	69724.4	819.974	830.645	118621
20	30	2500	62916.4	2466.37	2498.47	115590
30	40	3525	264056	14595.2	14785.1	112855
40	50	5050	292033	23124.8	23425.7	101378
50	60	8050	247568	31249.5	31656.2	88684
60	70	11050	251157	43517.3	44083.7	77923.1
70	80	14050	350789	77281.6	78287.4	67006.1
80	90	17050	261181	69826.5	67192.5	51758.5
90	100	20050	247627	77851.2	78864.4	40405.8
100	110	22750	208187	74265.9	70505.3	29642.3
110	120	23975	473768	178106	168463	20593.1

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	750	389195	593104	5.959E+09	10.87E+09
20	30	2500	377361	592284	5.384E+09	9.977E+09
30	40	3525	361583	589818	4.645E+09	8.794E+09
40	50	5050	361187	575223	4.627E+09	8.765E+09
50	60	8050	324707	552098	3.597E+09	7.073E+09
60	70	11050	281788	520848	2.687E+09	5.463E+09
70	80	14050	234712	477331	1912282368	3.965E+09
80	90	17050	205770	400050	1251291392	2.649E+09
90	100	20050	158308	330223	704954176	1552804480
100	110	22750	108198	252372	305034112	678557824
110	120	23975	83054.1	178106	75438944	161775376

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	389195	0.0024751	0.009738	0.12768E-04
20	30	377361	0.026453	0.050692	0.27816E-04
30	40	361583	0.051388	0.052086	0.27974E-04
40	50	361187	0.097466	0.14915	0.36383E-04
50	60	324707	0.20637	0.26839	0.42802E-04
60	70	281788	0.33456	0.40423	0.47492E-04
70	80	234712	0.47685	0.55189	0.50719E-04
80	90	205770	0.62885	0.70729	0.52699E-04
90	100	158308	0.78683	0.86711	0.53709E-04
100	110	108198	0.93169	0.99645	0.54012E-04
110	120	83054.1	0.9978	0.99915	0.54012E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	1132.69	7079.33
20	30	1132.69	7079.33
30	40	1135.3	7095.63
40	50	1135.3	7095.63
50	60	1135.3	7095.63
60	70	1135.3	7095.63
70	80	1135.3	7095.63
80	90	1135.3	7095.63
90	100	1135.3	7095.63
100	110	1135.3	7095.63
110	120	1135.3	7095.63

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 120.000 Actual deflection : 0.999 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	413944	375205	850550336	771238592
20	30	413944	375205	850550336	771238592
30	40	905490	866846	1856321408	1777738624
40	50	905490	866846	1856321408	1777738624
50	60	905490	866846	1856321408	1777738624
60	70	905490	866846	1856321408	1777738624
70	80	905490	866846	1856321408	1777738624
80	90	905490	866846	1856321408	1777738624
90	100	905490	866846	1856321408	1777738624
100	110	905490	866846	1856321408	1777738624
110	120	905490	866846	1856321408	1777738624

Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	90.7184	154.977	-52.709	-52.709
20	30	115.92	196.56	-52.709	-52.709
30	40	165.6	280.8	-102.824	-102.824
40	50	165.597	280.795	-102.824	-102.824
50	60	165.597	280.795	-102.824	-102.824
60	70	165.597	280.795	-102.824	-102.824
70	80	165.597	280.795	-102.824	-102.824
80	90	165.597	280.795	-102.824	-102.824
90	100	165.597	280.795	-102.824	-102.824
100	110	165.597	280.795	-102.824	-102.824
110	120	165.597	280.795	-102.824	-102.824

Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	67.1181	...	88.1016
40	50	67.1181	...	88.1016
50	60	67.1181	...	88.1016
60	70	67.1181	...	88.1016
70	80	67.1181	...	88.1016
80	90	67.1181	...	88.1016
90	100	67.1181	...	88.1016
100	110	67.1181	...	88.1016
110	120	67.1181	...	88.1016

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-7.27399	-7.27399	-39.176	0.72431	0.72431
20	30	-7.08815	-7.08815	-38.9901	0.72431	0.72431
30	40	-2.99544	-2.99544	-2.99544	0.31423	0.31423
40	50	-2.6908	-2.6908	-2.6908	0.31423	0.31423
50	60	-2.35388	-2.35388	-2.35388	0.30107	0.30107
60	70	-2.06826	-2.06826	-2.06826	0.22801	0.22801
70	80	-1.7785	-1.7785	-1.7785	0.14612	0.14612
80	90	-1.37379	-1.37379	-1.37379	0.14612	0.14612
90	100	-1.07246	-1.07246	-1.07246	0.073062	0.073062
100	110	-0.78678	-0.78678	-0.78678	...	...
110	120	-0.54659	-0.54659	-0.54659	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	0.72431	7.7243	14.0845	2.54902	...
20	30	0.72431	6.97879	12.9317	2.303	...
30	40	0.31423	2.61192	4.94516	0.86193	...
40	50	0.31423	2.60175	4.92877	0.85858	...
50	60	0.30107	2.02297	3.97749	0.66758	...
60	70	0.22801	1.51118	3.0721	0.49869	...
70	80	0.14612	1.07534	2.22979	0.35486	...
80	90	0.14612	0.70364	1.48943	0.2322	...
90	100	0.073062	0.39642	0.87319	0.13082	...
100	110	...	0.17153	0.38157	0.056605	...
110	120	...	0.042422	0.090971	0.013999	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case	Vortex Shedding Empty Case	Vortex Shedding Test Case	Earthquake Empty Case
------	----	-----------------------------------	-------------------------------	------------------------------	--------------------------



		MPa	MPa	MPa	MPa
10	20	...	...	...	13.6645
20	30	...	...	...	12.536
30	40	...	...	...	4.7876
40	50	...	...	...	4.77158
50	60	...	...	...	3.84206
60	70	...	...	...	2.95902
70	80	...	...	...	2.1399
80	90	...	...	...	1.42404
90	100	...	...	...	0.83094
100	110	...	...	...	0.36109
110	120	...	...	...	0.086046

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	0.31618
20	30	...	0.3081
30	40	...	0.1302
40	50	...	0.11696
50	60	...	0.10232
60	70	...	0.0899
70	80	...	0.077305
80	90	...	0.059714
90	100	...	0.046616
100	110	...	0.034199
110	120	...	0.023758

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...
100	110	...	...	...	...
110	120	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.17	90.72	-15.72	52.71	0.0129	0.2983
20	0.61	115.92	-14.79	52.71	0.0053	0.2806
30		165.60	-5.92	102.82		0.0576
40	0.23	165.60	-5.61	102.82	0.0014	0.0545
50		165.60	-4.68	102.82		0.0455
60		165.60	-3.81	102.82		0.0370
70		165.60	-3.00	102.82		0.0292
80		165.60	-2.22	102.82		0.0216
90		165.60	-1.54	102.82		0.0150
100		165.60	-0.96	102.82		0.0093
110		165.60	-0.59	102.82		0.0057

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	7.43	90.72	-21.35	52.71	0.0819	0.4050
20	6.48	115.92	-20.04	52.71	0.0559	0.3802
30	2.24	165.60	-7.97	102.82	0.0135	0.0775
40	2.51	165.60	-7.66	102.82	0.0152	0.0745
50	1.89	165.60	-6.39	102.82	0.0114	0.0622
60	1.21	165.60	-5.17	102.82	0.0073	0.0502
70	0.58	165.60	-3.99	102.82	0.0035	0.0388
80	0.26	165.60	-2.88	102.82	0.0015	0.0281



90	165.60	-1.93	102.82	0.0188
100	165.60	-1.11	102.82	0.0108
110	165.60	-0.61	102.82	0.0059

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.17	90.72	-15.72	52.71	0.0129	0.2983
20	0.61	115.92	-14.79	52.71	0.0053	0.2806
30		165.60	-5.92	102.82		0.0576
40	0.23	165.60	-5.61	102.82	0.0014	0.0545
50		165.60	-4.68	102.82		0.0455
60		165.60	-3.81	102.82		0.0370
70		165.60	-3.00	102.82		0.0292
80		165.60	-2.22	102.82		0.0216
90		165.60	-1.54	102.82		0.0150
100		165.60	-0.96	102.82		0.0093
110		165.60	-0.59	102.82		0.0057

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	7.85	90.72	-21.77	52.71	0.0865	0.4130
20	6.88	115.92	-20.44	52.71	0.0593	0.3877
30	2.39	165.60	-8.12	102.82	0.0145	0.0790
40	2.67	165.60	-7.82	102.82	0.0161	0.0760
50	2.03	165.60	-6.53	102.82	0.0122	0.0635
60	1.32	165.60	-5.28	102.82	0.0080	0.0513
70	0.67	165.60	-4.08	102.82	0.0041	0.0397
80	0.32	165.60	-2.95	102.82	0.0019	0.0287
90		165.60	-1.97	102.82		0.0192
100		165.60	-1.13	102.82		0.0110
110		165.60	-0.61	102.82		0.0060

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-42.45	52.71		0.8054
20		196.56	-42.02	52.71		0.7972
30		280.80	-4.17	102.82		0.0406
40		280.79	-3.86	102.82		0.0376
50		280.79	-3.32	102.82		0.0323
60		280.79	-2.79	102.82		0.0272
70		280.79	-2.28	102.82		0.0222
80		280.79	-1.75	102.82		0.0170
90		280.79	-1.28	102.82		0.0124
100		280.79	-0.84	102.82		0.0082
110		280.79	-0.56	102.82		0.0055

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-39.90	52.71		0.7570
20		196.56	-39.71	52.71		0.7535
30		280.80	-3.31	102.82		0.0322
40		280.79	-3.01	102.82		0.0292
50		280.79	-2.65	102.82		0.0258
60		280.79	-2.30	102.82		0.0223
70		280.79	-1.92	102.82		0.0187
80		280.79	-1.52	102.82		0.0148
90		280.79	-1.15	102.82		0.0111
100		280.79	-0.79	102.82		0.0077
110		280.79	-0.55	102.82		0.0053

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
-----------	----------------	-------------------	--------------	-------------------	-------------	-------------

Node	Stress	Stress	Stress	Stress	Ratio	Ratio
10	1.17	90.72	-15.72	52.71	0.0129	0.2983
20	0.61	115.92	-14.79	52.71	0.0053	0.2806
30	67.05	165.60		102.82	0.4049	
40	67.34	165.60		102.82	0.4067	
50	67.09	165.60		102.82	0.4051	
60	66.79	165.60		102.82	0.4033	
70	66.56	165.60		102.82	0.4019	
80	66.59	165.60		102.82	0.4021	
90	66.52	165.60		102.82	0.4017	
100	66.50	165.60		102.82	0.4016	
110	66.61	165.60		102.82	0.4023	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	7.85	90.72	-21.77	52.71	0.0865	0.4130
20	6.88	115.92	-20.44	52.71	0.0593	0.3877
30	69.51	165.60		102.82	0.4198	
40	69.79	165.60		102.82	0.4214	
50	69.15	165.60		102.82	0.4176	
60	68.44	165.60		102.82	0.4133	
70	67.79	165.60		102.82	0.4094	
80	67.44	165.60		102.82	0.4073	
90	67.04	165.60		102.82	0.4048	
100	66.75	165.60		102.82	0.4031	
110	66.69	165.60		102.82	0.4027	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.17	90.72	-15.72	52.71	0.0129	0.2983
20	0.61	115.92	-14.79	52.71	0.0053	0.2806
30		165.60	-5.92	102.82		0.0576
40	0.23	165.60	-5.61	102.82	0.0014	0.0545
50		165.60	-4.68	102.82		0.0455
60		165.60	-3.81	102.82		0.0370
70		165.60	-3.00	102.82		0.0292
80		165.60	-2.22	102.82		0.0216
90		165.60	-1.54	102.82		0.0150
100		165.60	-0.96	102.82		0.0093
110		165.60	-0.59	102.82		0.0057

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	7.85	90.72	-21.77	52.71	0.0865	0.4130
20	6.88	115.92	-20.44	52.71	0.0593	0.3877
30	2.39	165.60	-8.12	102.82	0.0145	0.0790
40	2.67	165.60	-7.82	102.82	0.0161	0.0760
50	2.03	165.60	-6.53	102.82	0.0122	0.0635
60	1.32	165.60	-5.28	102.82	0.0080	0.0513
70	0.67	165.60	-4.08	102.82	0.0041	0.0397
80	0.32	165.60	-2.95	102.82	0.0019	0.0287
90		165.60	-1.97	102.82		0.0192
100		165.60	-1.13	102.82		0.0110
110		165.60	-0.61	102.82		0.0060

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-42.45	52.71		0.8054
20		196.56	-42.02	52.71		0.7972
30	86.28	280.80		102.82	0.3073	
40	86.58	280.79		102.82	0.3084	



50	86.72	280.79	102.82	0.3088
60	86.76	280.79	102.82	0.3090
70	86.82	280.79	102.82	0.3092
80	87.11	280.79	102.82	0.3102
90	87.23	280.79	102.82	0.3107
100	87.37	280.79	102.82	0.3112
110	87.57	280.79	102.82	0.3119

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-39.90	52.71		0.7570
20		196.56	-39.71	52.71		0.7535
30	85.42	280.80		102.82	0.3042	
40	85.73	280.79		102.82	0.3053	
50	86.05	280.79		102.82	0.3064	
60	86.26	280.79		102.82	0.3072	
70	86.47	280.79		102.82	0.3079	
80	86.87	280.79		102.82	0.3094	
90	87.10	280.79		102.82	0.3102	
100	87.31	280.79		102.82	0.3110	
110	87.56	280.79		102.82	0.3118	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-8.00	52.71		0.1517
20		115.92	-7.81	52.71		0.1482
30	64.44	165.60		102.82	0.3891	
40	64.74	165.60		102.82	0.3910	
50	65.07	165.60		102.82	0.3929	
60	65.28	165.60		102.82	0.3942	
70	65.49	165.60		102.82	0.3955	
80	65.89	165.60		102.82	0.3979	
90	66.12	165.60		102.82	0.3993	
100	66.33	165.60		102.82	0.4006	
110	66.57	165.60		102.82	0.4020	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-6.59	52.71		0.1251
20		115.92	-6.42	52.71		0.1219
30	64.25	165.60		102.82	0.3880	
40	64.54	165.60		102.82	0.3898	
50	64.86	165.60		102.82	0.3917	
60	65.14	165.60		102.82	0.3934	
70	65.42	165.60		102.82	0.3950	
80	65.80	165.60		102.82	0.3974	
90	66.09	165.60		102.82	0.3991	
100	66.36	165.60		102.82	0.4008	
110	66.59	165.60		102.82	0.4021	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-8.00	52.71		0.1517
20		115.92	-7.81	52.71		0.1482
30	64.44	165.60		102.82	0.3891	
40	64.74	165.60		102.82	0.3910	
50	65.07	165.60		102.82	0.3929	
60	65.28	165.60		102.82	0.3942	
70	65.49	165.60		102.82	0.3955	
80	65.89	165.60		102.82	0.3979	
90	66.12	165.60		102.82	0.3993	
100	66.33	165.60		102.82	0.4006	

110 66.57 165.60 102.82 0.4020

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-8.00	52.71		0.1517
20		115.92	-7.81	52.71		0.1482
30	64.44	165.60		102.82	0.3891	
40	64.74	165.60		102.82	0.3910	
50	65.07	165.60		102.82	0.3929	
60	65.28	165.60		102.82	0.3942	
70	65.49	165.60		102.82	0.3955	
80	65.89	165.60		102.82	0.3979	
90	66.12	165.60		102.82	0.3993	
100	66.33	165.60		102.82	0.4006	
110	66.57	165.60		102.82	0.4020	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-8.00	52.71		0.1517
20		115.92	-7.81	52.71		0.1482
30		165.60	-3.31	102.82		0.0322
40		165.60	-3.01	102.82		0.0292
50		165.60	-2.65	102.82		0.0258
60		165.60	-2.30	102.82		0.0223
70		165.60	-1.92	102.82		0.0187
80		165.60	-1.52	102.82		0.0148
90		165.60	-1.15	102.82		0.0111
100		165.60	-0.79	102.82		0.0077
110		165.60	-0.55	102.82		0.0053

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.60	-7.68	43.92		0.1749
20		96.60	-7.50	43.92		0.1708
30	64.57	138.00		85.69	0.4679	
40	64.86	138.00		85.69	0.4700	
50	65.17	138.00		85.69	0.4722	
60	65.37	138.00		85.69	0.4737	
70	65.56	138.00		85.69	0.4751	
80	65.95	138.00		85.69	0.4779	
90	66.17	138.00		85.69	0.4795	
100	66.37	138.00		85.69	0.4809	
110	66.60	138.00		85.69	0.4826	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.60	-7.68	43.92		0.1749
20		96.60	-7.50	43.92		0.1708
30		138.00	-3.18	85.69		0.0371
40		138.00	-2.89	85.69		0.0337
50		138.00	-2.55	85.69		0.0298
60		138.00	-2.21	85.69		0.0257
70		138.00	-1.85	85.69		0.0216
80		138.00	-1.46	85.69		0.0170
90		138.00	-1.10	85.69		0.0128
100		138.00	-0.75	85.69		0.0088
110		138.00	-0.52	85.69		0.0061

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
-----------	----------------	-------------------	--------------	-------------------	-------------	-------------



10		90.72	-8.00	52.71	0.1517
20		115.92	-7.81	52.71	0.1482
30	64.44	165.60		102.82	0.3891
40	64.74	165.60		102.82	0.3910
50	65.07	165.60		102.82	0.3929
60	65.28	165.60		102.82	0.3942
70	65.49	165.60		102.82	0.3955
80	65.89	165.60		102.82	0.3979
90	66.12	165.60		102.82	0.3993
100	66.33	165.60		102.82	0.4006
110	66.57	165.60		102.82	0.4020

Absolute Maximum of the all of the Stress Ratio's 0.8054

Governing Element: Skirt-2

Governing Load Case 5 : NP+HW+HI

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-2	...	-39.2	0.7	2.5	...	+ Tens
Skirt-2	...	-39.2	-0.7	-2.5	-42.4	- Comp
Skirt-1	...	-39.0	0.7	2.3	...	+ Tens
Skirt-1	...	-39.0	-0.7	-2.3	-42.0	- Comp
Btm_head	...	-3.0	0.3	0.9	...	+ Tens
Btm_head	...	-3.0	-0.3	-0.9	-4.2	- Comp
Shell-1	...	-2.7	0.3	0.9	...	+ Tens
Shell-1	...	-2.7	-0.3	-0.9	-3.9	- Comp
Shell-2	...	-2.4	0.3	0.7	...	+ Tens
Shell-2	...	-2.4	-0.3	-0.7	-3.3	- Comp
Shell-3	...	-2.1	0.2	0.5	...	+ Tens
Shell-3	...	-2.1	-0.2	-0.5	-2.8	- Comp
Shell-4	...	-1.8	0.1	0.4	...	+ Tens
Shell-4	...	-1.8	-0.1	-0.4	-2.3	- Comp
Shell-5	...	-1.4	0.1	0.2	...	+ Tens
Shell-5	...	-1.4	-0.1	-0.2	-1.8	- Comp
Shell-6	...	-1.1	0.1	0.1	...	+ Tens
Shell-6	...	-1.1	-0.1	-0.1	-1.3	- Comp
Shell-7	...	-0.8	...	0.1	...	+ Tens
Shell-7	...	-0.8	...	-0.1	-0.8	- Comp
Top_head	...	-0.5	...	...	...	+ Tens
Top_head	...	-0.5	...	...	-0.6	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

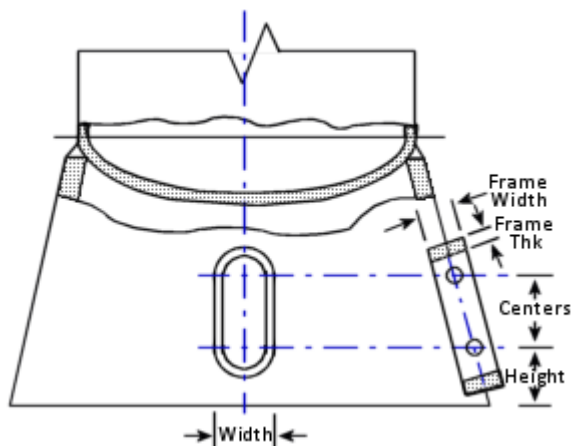
Total vessel (empty) weight	Fg	2596366.500 N
Head Weight below section 2-2	DeltaFg	264056.406 N
Weight of vessel contents (liquid)	Ff	0.000 N
Vessel Gross Weight	Fg + Ff	2596366.500 N
Skirt Mean Diameter	Dz	8236.499 mm
Skirt Analysis Thickness (Corroded)	ez	14.500 mm
Skirt Height	H	2000.000 mm
Additional Axial Force at section 1-1	F1	0.000 N
Moment at section 1-1	M1	8794014720.000 N-mm
Applied Moment - Bottom of Skirt	Mb	9976596480.000 N-mm
Weight at Bottom of Skirt	W	2659282.750 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	8236.500 mm
Head Corroded Thickness	eb	33.500 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 2052 \text{ mm} \\
 K &= D_i / (2 * h_i) = 8203 / (2 * 2052) = 2 \\
 R &= D_i (0.44 * K + 0.02) = 8203 * (0.44 * 2 + 0.02) = 7380 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 8203 * ((0.5 / 2) - 0.08) = 1395 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	3181672136704.000 mm ⁴
Moment of Inertia	Iyy =	3181672136704.000 mm ⁴
Section Modulus	W =	772579520.000 mm ³
Section Metal Area	A =	375198.000 mm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(3.18E+12, 3.18E+12) / 375198)} \\
 &= 2912 \text{ mm}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:

$$= rg / L$$

$$= 2912/2000$$

$$= 1.456$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 8.79E+09 N-mm

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -8.79E+09/3.09E+09 - 2659282/375198$$

$$= -9.93 \text{ MPa}$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +8.79E+09/3.09E+09 - 2659282/375198$$

$$= -4.24 \text{ MPa}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F1 - Fg - Ff + 4 \cdot M1 / Dz$$

$$= 0 - 2596366 - 0 + 4 \cdot 8.79E+09/8236$$

$$= 1672655 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F1 - Fg - Ff - 4 \cdot M1 / Dz$$

$$= 0 - 2596366 - 0 - 4 \cdot 8.79E+09/8236$$

$$= -6865389 \text{ N}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (1672655 + 264056 + 0) / (\pi \cdot 8236 \cdot 33.5) + 1.1 \cdot 8236 / (4 \cdot 33.5)$$

$$= 69.85 \text{ MPa}$$

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-6865389 + 264056 + 0) / (\pi \cdot 8236 \cdot 33.5) + 1.1 \cdot 8236 / (4 \cdot 33.5)$$

$$= 60.00 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (0 + 264056) / (\pi \cdot 8236 \cdot 33.5) + 1.1 \cdot 8236 / (4 \cdot 33.5)$$

$$= 67.92 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= 1672655 / (\pi \cdot 8236 \cdot 14.5)$$

$$= 4.46 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -6865389 / (\pi \cdot 8236 \cdot 14.5)$$

$$= -18.30 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(8236 + 33.5 - 8236 + 14.5) / (1395 + 33.5)$$

$$= 0.98$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.98) = 10.5 \text{ deg}$$

$$\begin{aligned} \text{Dimension [a]:} \\ &= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))} \\ &= 0.5 \cdot \sqrt{(33.5^2 + 14.5^2 + 2 \cdot 33.5 \cdot 14.5 \cdot 0.98)} \\ &= 23.9 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Bending moment at verification points p and q [Mp \& Mq]:} \\ M_p &= a \cdot F_{zp} = 23.91 \cdot 1672655.9 = 40.0175 \times 10^6 \text{ N-mm} \\ M_q &= a \cdot F_{zq} = 23.91 \cdot -6865389.0 = -164251328 \text{ N-mm} \end{aligned}$$

$$\begin{aligned} \text{Correction factor [C]:} \\ &= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{33.5}{14.5} \right)^2 \\ &= 0.326 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

$$\begin{aligned} \text{Stress } [\sigma_{1pb(o)} \& \sigma_{2pb(o)}]: \\ &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ &= 0.33 \cdot 6 \cdot 40017536 / (\pi \cdot 8236 \cdot 33.5^2) \\ &= 2.69 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qb(o)} \& \sigma_{2qb(o)}]: \\ &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ &= 0.33 \cdot 6 \cdot -164251328 / (\pi \cdot 8236 \cdot 33.5^2) \\ &= -11.05 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3pb(o)}]: \\ &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ &= 0.33 \cdot 6 \cdot 40017536 / (\pi \cdot 8236 \cdot 14.5^2) \\ &= 14.37 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qb(o)}]: \\ &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ &= 0.33 \cdot 6 \cdot -164251328 / (\pi \cdot 8236 \cdot 14.5^2) \\ &= -58.99 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Geometry variables, 16.12-32 to 36:} \\ eb/D_b &= 1.32/324 = 0.0041 \\ y &= 125 \cdot eb/D_b = 125 \cdot 33.5/8236 = 0.51 \end{aligned}$$

As this an elliptical head, it will be treated as a Korbogen type head

$$\begin{aligned} \text{Geometric value } [\alpha]: \\ &= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994 \\ &= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.51 + 3.994 = 4.17 \end{aligned}$$

$$\begin{aligned} \text{Bending stress in sections 1-1 and 2-2 Equation 16.12-31 } [\sigma_{1b(p)} \& \sigma_{2b(p)}]: \\ &= (P + Ph) \cdot D_b \cdot ((\gamma / \gamma_A) \cdot \alpha - 1) / (4 \cdot eb) \\ &= (1.1 + 0) \cdot 8236 \cdot ((10.5/40) \cdot 4.17 - 1) / (4 \cdot 33.5) \\ &= 6.52 \text{ MPa} \end{aligned}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

$$\begin{aligned} \text{Stress } [\sigma_{1pitot}]: \\ &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 69.8 - 2.69 + 6.52 \\ &= 73.68 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1potot}]: \\ &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 69.8 + 2.69 - 6.52 \\ &= 66.02 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qitot}]: \\ &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \end{aligned}$$



$$= 60 - -11.1 + 6.52$$
$$= 77.57 \text{ MPa}$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$
$$= 60 + -11.1 - 6.52$$
$$= 42.42 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$
$$= 67.9 + 2.69 + 6.52$$
$$= 77.13 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$
$$= 67.9 - 2.69 - 6.52$$
$$= 58.70 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$
$$= 67.9 + -11.1 + 6.52$$
$$= 63.39 \text{ MPa}$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$
$$= 67.9 - -11.1 - 6.52$$
$$= 72.45 \text{ MPa}$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$
$$= 4.46 - 14.4$$
$$= -9.91 \text{ MPa}$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$
$$= 4.46 + 14.4$$
$$= 18.83 \text{ MPa}$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$
$$= -18.3 - -59$$
$$= 40.69 \text{ MPa}$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$
$$= -18.3 + -59$$
$$= -77.29 \text{ MPa}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$
$$= 138 (3 - (1/1.5)) \cdot (69.8/138)^2$$
$$= 390.43 \text{ MPa}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$
$$= 138 (3 - (1/1.5)) \cdot (60/138)^2$$
$$= 396.61 \text{ MPa}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$
$$= 138 (3 - (1/1.5)) \cdot (67.9/138)^2$$
$$= 391.72 \text{ MPa}$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$
$$= 138 (3 - (1/1.5)) \cdot (67.9/138)^2$$
$$= 391.72 \text{ MPa}$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138(3 - (1/1.5) \cdot (4.46/138)^2)$$

$$= 413.90 \text{ MPa}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138(3 - (1/1.5) \cdot (-18.3/138)^2)$$

$$= 412.38 \text{ MPa}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	73.68	390.43	Pass	46
Stot1po	66.02	390.43	Pass	47
Stot1qi	77.57	396.61	Pass	48
Stot1qo	42.42	396.61	Pass	49
Stot2pi	77.13	391.72	Pass	50
Stot2po	58.70	391.72	Pass	51
Stot2qi	63.39	391.72	Pass	52
Stot2qo	72.45	391.72	Pass	53
Stot3pi	-9.91	413.90	Pass	54
Stot3po	18.83	413.90	Pass	55
Stot3qi	40.69	412.38	Pass	56
Stot3qo	-77.29	412.38	Pass	57

The stresses are satisfactory in the region without openings.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	17349.199 mm
Center of Gravity of the Nozzles	10634.662 mm
Center of Gravity of the Added Weights (Operating)	16664.127 mm
Center of Gravity of the Added Weights (Empty)	14739.616 mm
Center of Gravity of Bare Shell New and Cold	12967.817 mm
Center of Gravity of Bare Shell Corroded	12995.155 mm
Vessel CG in the Operating Condition	13999.787 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	13762.583 mm
Vessel CG in the Test Condition	13752.285 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.

Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	8251.0000	mm
Skirt Thickness	STHK	16.0000	mm
Skirt Internal Corrosion Allowance	SCA	1.5000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-285 C	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-285 C	
Basing Ope. All. Stress	BASOPE, f4	108.00	MPa
Basing Yield Stress		207.00	MPa
Inside Diameter of Basing	DI, D4	8087.0000	mm
Outside Diameter of Basing	DOU, D5	8495.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	36.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	150.00	MPa
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	8379.0000	mm
Bolt Allowable Shear Stress		138.000	MPa
Ultimate Comp. Strength of Concrete	FPC	35.0	MPa
Allowable Comp. Strength of Concrete	FC	15.8	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	14.0000	mm
Width of Gussets at Top Plate	TWDT	122.0000	mm
Width of Gussets at Base Plate	BWDT	122.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	207.0	MPa
Height of Chair	h1	268.0000	mm
Distance between Gussets	RG, b6	102.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	51.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	36.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	122.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	42.0000	mm
External Corrosion Allowance	CA	1.5000	mm
Dead Weight of Vessel	DW	2671436.2	N
Operating Weight of Vessel	ROW	2729007.0	N
Earthquake Moment on Basing	EQMOM	10902293504.0	N-mm
Wind Moment on Basing	WIMOM	5995483648.0	N-mm
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 10.90E+09/8379) - 2610386) / 16$$

$$= 162005.3 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 6.00E+09/8379) - 2671436) / 16$$

$$= 11847.0 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/8379) - 2729007) / 16$$

$$= -170562.9 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	600.6298	mm ²
Area Available in all the Bolts	Abss * RN :	9610.0771	mm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.243	0.364	1.348	2.567	0.450	0.778	161879.8	2772266.0	26598.8
4	0.303	0.273	1.518	2.435	0.437	0.781	204796.6	2815182.8	35475.9
6	0.288	0.296	1.477	2.468	0.440	0.781	193999.4	2804385.5	33156.6
8	0.292	0.290	1.487	2.460	0.439	0.781	196694.2	2807080.2	33729.8
10	0.291	0.291	1.485	2.462	0.440	0.781	196019.8	2806405.8	33586.0
12	0.291	0.291	1.485	2.461	0.440	0.781	196188.7	2806574.8	33622.0

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 196189 / (0.37 \cdot 8379 \cdot 2.46)$$

$$= 52.1 \text{ MPa, Should be } \leq 150 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 16 \cdot 601 / (\pi \cdot 8379)$$

$$= 0.37 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 2.18[(2 \cdot 0.29 \cdot 8379 + 204) / (2 \cdot 0.29 \cdot 8379)]$$

$$= 2.27 \text{ MPa, Should be } \leq 15.8 \text{ MPa}$$

Values for table 10.3, l = 122, b = 823, l/b = 0.14830896

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(5325, 15794) = 15794 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 15794 / 137)} + 1.5$$

$$= 27.8 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 122, b = 102, b/l = 0.83606553

As b/l (0.84) is less than 1, inverting b/l = 1.2 0.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3 (\ln((2 \cdot l \sin(\pi \cdot a / l) / (\pi \cdot g))) + 1)] - [(0.7 - g2)P / (4 \cdot \pi)]$$

$$= 90087 / (4 \cdot \pi) [1.3 (\ln((2 \cdot 122 \cdot \sin(\pi \cdot 64 / 122))) / (\pi \cdot 30)) + 1] - [(0.7 - 0.11) \cdot 90087 / (4 \cdot \pi)]$$

$$= 11771.0078 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot Abs(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot Abs(11771) / 138)} + 1.5$$

$$= 24.1238 \text{ mm}$$

Required Gusset Plate Thickness [tg]:
 $= P / (\text{Stress Term} \cdot l_{avg}) + Ca$
 $= 90087 / (124 \cdot 122) + 1.5$
 $= 11 \text{ (not less than } 9.52 + 1.5 \text{) mm}$

Bolt spacing [m]:
 $= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$
 $= \pi \cdot 8379 / 16 = 1645 \text{ mm}$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:
 $= 1.76(P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot Sktope))^{2/3} \cdot r^{1/3} + Ca$
 $= 1.76(90087 \cdot 64 / (1645 \cdot 232 \cdot 162))^{2/3} \cdot 41261^{1/3} + Ca$
 $= 5.8 + 1.5 = 7.3 \text{ mm}$

Shear Stress in a Single Bolt [tb]:
 $= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 593104 / (601 \cdot 16)$
 $= 61.7 \text{ MPa. Must be less than } 138.0 \text{ MPa.}$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	27.8379	mm
Actual Basing Thickness as entered by user	32.0000	mm
Required Thickness of Chair Cap	24.1238	mm
Actual Top Ring Thickness as entered by user	36.0000	mm
Required Gusset thickness, + CA	11.0250	mm
Actual Gusset Thickness as entered by user	14.0000	mm
Required Thickness of Skirt for Local Stress	7.3047	mm
Given Thickness of Skirt	16.0000	mm
Required Gusset Height to meet local stress	26.7630	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:
 $= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$
 $= 42.4 \cdot (16 - 1.5)$
 $= 615 \text{ N/mm}$

Results for Computed Minimum Basing Weld Size [BWeld]:
 $= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 615 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$
 $= 5.68 \text{ mm}$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:
 $= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (Hg + Tta))$
 $= 90087 / (174 + 2 \cdot (200 + 36))$
 $= 139 \text{ N/mm}$

Horizontal Plate Load [Wh]:
 $= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$
 $= 90087 \cdot 64 / (174 \cdot (236) + 0.6667 \cdot (236)^2)$
 $= 73.7 \text{ N/mm}$

Resultant Weld Load [Wr]:
 $= \sqrt{ Wv^2 + Wh^2 }$
 $= \sqrt{ 139^2 + 73.7^2 }$
 $= 158 \text{ N/mm}$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
 $= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 158 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$
 $= 1.46 \text{ mm}$

Results for Computed Minimum Gusset to Top Plate Weld Size



Weld Load [Wv]:
 = Bolt Load / (2 • TopWth)
 = 90087/(2 • 122)
 = 369 N/mm

Weld Load [Wh]:
 = Bolt Load • e / (2 • Hgt • TopWth)
 = 90087 • 64/(2 • 236 • 122)
 = 100 N/mm

Resultant Weld Load [Wr]:
 = $\sqrt{Wv^2 + Wh^2}$
 = $\sqrt{369^2 + 100^2}$
 = 383 N/mm

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
 = Wr / [(0.4 • Yield) • 2 • 0.707]
 = 383/[(0.4 • 192) • 2 • 0.707]
 = 3.53 mm

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	3.5309	mm

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N36"

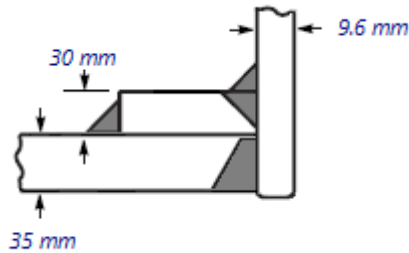
From: 30

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	8200.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	35.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	270.0000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	28.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		50.8625	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N36"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.1 \cdot 8203 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$
 $= 29.4370 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 76 / (118 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.7128 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	304.1452	mm
Parallel to Vessel Wall, opening length	d	152.0726	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		50.2530	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

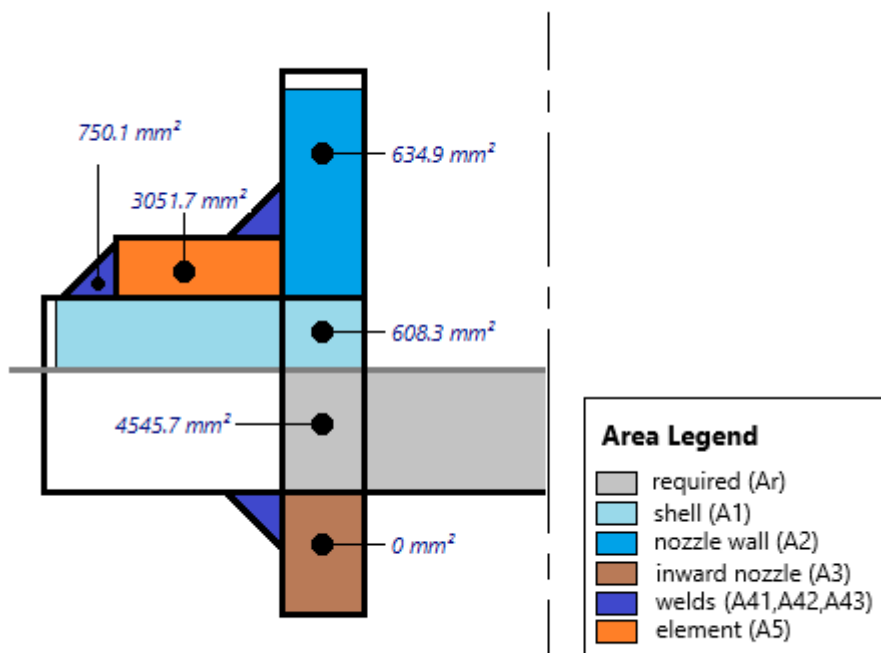


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4545.690	NA	NA
Area in Shell	A1	608.331	NA	NA
Area in Nozzle Wall	A2	634.946	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	750.098	NA	NA
Area in Element	A5	3051.750	NA	NA
TOTAL AREA AVAILABLE	Atot	5045.124	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	253.3522	30.0000 mm
Based on given Pad Diameter:	270.0000	25.0904 mm
Based on the Estimated Diameter Limit:	302.5577	19.0070 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152 \cdot 29.4 \cdot 1 + 2 \cdot 8.1 \cdot 29.4 \cdot 1 \cdot (1 - 0.86))$
 $= 4545.690 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152(1 \cdot 33.5 - 1 \cdot 29.4) - 2 \cdot 8.1$
 $(1 \cdot 33.5 - 1 \cdot 29.4) \cdot (1 - 0.86)$
 $= 608.331 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 50.3) \cdot (8.1 - 0.71) \cdot 0.86$

= 634.946 mm²

Area Available in Welds [A41 + A42 + A43]:

= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (100) • 0.86 + (0) • 0.86 + 665² • 1
 = 750.098 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (270 - 168) • 30 • 1
 = 3051.750 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 2.2128 mm
 Wall Thickness per UG16(b), tr16b = 3.0000 mm
 Wall Thickness, shell/head, internal pressure trb1 = 34.2033 mm
 Wall Thickness tb1 = max(trb1, tr16b) = 34.2033 mm
 Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm
 Wall Thickness per table UG-45 tb3 = 7.7200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[7.72, max(34.2, 3)]
 = 7.7200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(2.21, 7.72)
 = 7.7200 mm

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.1 MPa	Passed
Occasional	:	4.9,	Allowable	:	156.9 MPa	Passed
Shear	:	20.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.088, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.088, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.088, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 30, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.88$, Temp. Reduction = 7 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -30 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.088$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -37 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.10/5.10 = 0.216$

Weld Size Calculations, Description: N36"

Intermediate Calc. for nozzle/shell Welds Tmin 8.1012 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6708 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$19.7960 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (4546 - 608 + 2 \cdot 8.1 \cdot 0.86 \cdot (1 \cdot 33.5 - 29.4)) \cdot 138)$
 $= 551076.62 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (635 + 3052 + 750 - 0 \cdot 0.86) \cdot 138$
 $= 612225.44 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (635 + 0 + 85.5 + (464)) \cdot 138$
 $= 163455.34 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (635 + 0 + 750 + 3052 + (464)) \cdot 138$
 $= 676266.88 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118$
 $= 152817. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 270 \cdot 28 \cdot 0.49 \cdot 138$
 $= 802934. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 80.1) \cdot (9.6 - 1.5) \cdot 0.7 \cdot 118$
 $= 168343. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 168 \cdot 30 \cdot 0.74 \cdot 138$
 $= 809720. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 168 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$
 $= 904187. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (802934 + 168343) = 971278 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (152817 + 809720 + 904187 + 0) = 1866723 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (802934 + 904187 + 0) = 1707121 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 971277 N	, must exceed W = 551076 N	or W1 = 612225 N
Path 2-2 = 1866723 N	, must exceed W = 551076 N	or W2 = 163455 N
Path 3-3 = 1707121 N	, must exceed W = 551076 N	or W3 = 676266 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.13 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.4775 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.4777 mm

Input Echo, WRC107/537 Item 1, Description: N36" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	14760.000	mm
Vessel Thickness	Tv	35.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID

Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	149.073	mm
Nozzle Thickness	Tn	9.601	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	30.000	mm
Diameter of Reinforcing Pad	Dpad	270.000	mm
Design Internal Pressure	Dp	1.100	MPa
Include Pressure Thrust		No	

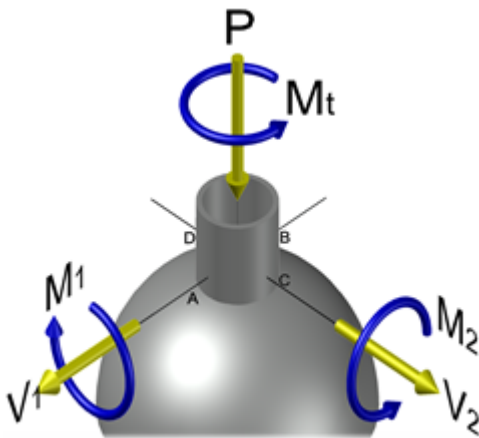
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-9599.6	N
Longitudinal Shear (SUS)	(V1) V1	8484.9	N
Circumferential Shear (SUS)	(Vc) V2	8484.9	N
Circumferential Moment (SUS)	(Mc) M1	3340974.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	3340974.0	N-mm
Torsional Moment (SUS)	Mt	4321525.0	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(7398.25 (35.0 - 1.5))}$$

$$= 1811.136 \text{ mm}$$

WRC 107 Stress Calculation for SUS tained loads:

Radial Load	P	-9599.6	N
Circumferential Shear	(VC) V2	8484.9	N
Longitudinal Shear	(VL) V1	8484.9	N
Circumferential Moment	(MC) M1	3340974.0	N-mm
Longitudinal Moment	(ML) M2	3340974.0	N-mm
Torsional Moment	MT	4321525.0	N-mm

Dimensionless Param: U = 0.12 TAU = 9.89 RHO = 4.00 (7.84)



Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.09313	(A,B,C,D)
$M(x) / P$	SP 4	0.04668	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.41530	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.22398	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.41530	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.22398	(A,B,C,D)
$N(y) * T / P$	SP 4	0.34154	(A,B,C,D)
$M(y) / P$	SP 4	0.28206	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10147	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.51251	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10147	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.51251	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad.	Bend. P	0.7	-0.7	0.7	-0.7	0.7	-0.7	0.7	-0.7
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-1.6	1.6	1.6	-1.6
Rad.	Memb. ML	-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-1.6	1.6	1.6	-1.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.2	0.7	3.0	-1.6	-1.2	0.7	3.0	-1.6
Tang.	Memb. P	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Tang.	Bend. P	4.0	-4.0	4.0	-4.0	4.0	-4.0	4.0	-4.0
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-11.0	11.0	11.0	-11.0
Tang.	Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-11.0	11.0	11.0	-11.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-6.2	7.6	15.9	-14.0	-6.2	7.6	15.9	-14.0
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tot. Shear		2.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0
Str. Int.		7.0	8.2	16.0	14.1	6.4	7.8	16.2	14.4

Unitless Prm: $U = 0.27$ TAU = 0.00 (16.16) RHO = 0.00 (4.14)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SR 2	0.20113	(A,B,C,D)
$M(x) / P$	SR 2	0.15428	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SR 3	0.16308	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SR 3	0.54429	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SR 3	0.16308	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SR 3	0.54429	(A,B,C,D)
$N(y) * T / P$	SR 2	0.06035	(A,B,C,D)
$M(y) / P$	SR 2	0.04632	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SR 3	0.04978	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SR 3	0.16279	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SR 3	0.04978	(A,B,C,D)



$M(y) * \sqrt{R_m * T} / ML$ SR 3 0.16279 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Rad. Bend. P		7.9	-7.9	7.9	-7.9	7.9	-7.9	7.9	-7.9
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Rad. Bend. MC		0.0	0.0	0.0	0.0	-19.5	19.5	19.5	-19.5
Rad. Memb. ML		-1.0	-1.0	1.0	1.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-19.5	19.5	19.5	-19.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-10.9	12.3	30.1	-24.7	-10.9	12.3	30.1	-24.7
Tang. Memb. P		0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Tang. Bend. P		2.4	-2.4	2.4	-2.4	2.4	-2.4	2.4	-2.4
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-5.8	5.8	5.8	-5.8
Tang. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-5.8	5.8	5.8	-5.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-3.2	3.7	9.0	-7.4	-3.2	3.7	9.0	-7.4
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Tot. Shear		1.7	1.7	0.5	0.5	0.5	0.5	1.7	1.7
Str. Int.		11.2	12.7	30.2	24.8	10.9	12.4	30.3	24.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Rad. Pl (SUS)		-0.3	-0.3	0.7	0.7	-0.3	-0.3	0.7	0.7
Rad. Q (SUS)		-1.0	1.0	2.3	-2.3	-1.0	1.0	2.3	-2.3
Long. Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Long. Pl (SUS)		0.7	0.7	0.9	0.9	0.7	0.7	0.9	0.9
Long. Q (SUS)		-6.9	6.9	15.0	-15.0	-6.9	6.9	15.0	-15.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Pm+Pl (SUS)		64.8	64.8	65.3	65.3	64.8	64.8	65.3	65.3
Pm+Pl+Q (Total)		63.4	72.1	79.9	62.5	62.9	71.7	80.2	62.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.94	138.00	Passed



Pm+Pl (SUS)	65.28	207.00	Passed
Pm+Pl+Q (TOTAL)	80.17	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Rad.	Pl (SUS)	0.7	0.7	2.7	2.7	0.7	0.7	2.7	2.7
Rad.	Q (SUS)	-11.6	11.6	27.4	-27.4	-11.6	11.6	27.4	-27.4
Long.	Pm (SUS)	121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Long.	Pl (SUS)	0.2	0.2	0.8	0.8	0.2	0.2	0.8	0.8
Long.	Q (SUS)	-3.5	3.5	8.2	-8.2	-3.5	3.5	8.2	-8.2
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear	Q (SUS)	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Pm (SUS)		121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Pm+Pl (SUS)		122.3	122.3	124.1	124.1	122.3	122.3	124.1	124.1
Pm+Pl+Q (Total)		118.3	133.9	151.3	113.8	118.0	133.6	151.5	114.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.19	138.00	Passed
Pm+Pl (SUS)	124.06	207.00	Passed
Pm+Pl+Q (TOTAL)	151.47	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: M1B24"

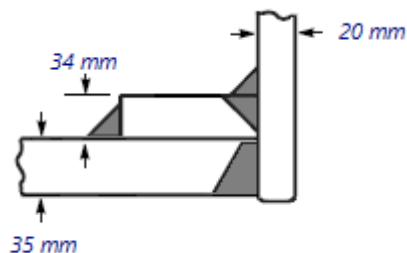
From: 40

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	8200.00	mm
Shell Finished (Minimum) Thickness	t	35.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1250.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+N
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-105	+N
Flange Type		Weld Neck	Flange
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1050.0000	mm
Thickness of Pad	te	34.0000	mm
Weld leg size between Pad and Shell	Wp	34.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	34.0000	mm
Reinforcing Pad Width		220.2000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M1B24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 4102 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 32.8508 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.1 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 1.1)$
 $= 2.4218 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1145.2001	mm
Parallel to Vessel Wall, opening length	d	572.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		80.2500	mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

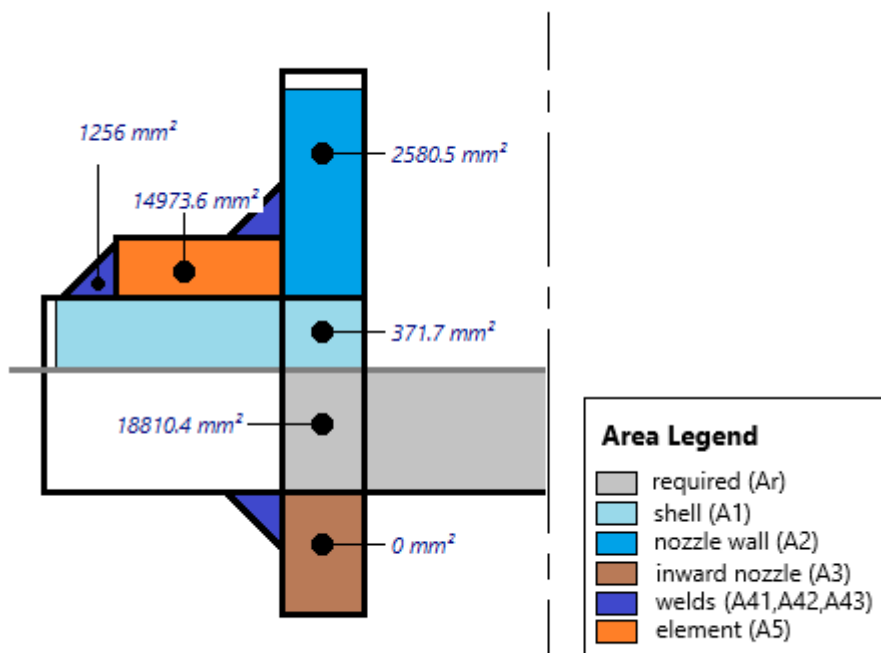


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	18810.381	NA	NA
Area in Shell	A1	371.720	NA	NA
Area in Nozzle Wall	A2	2580.544	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1256.000	NA	NA
Area in Element	A5	14973.603	NA	NA
TOTAL AREA AVAILABLE	Atot	19181.865	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	1039.0741	34.0000 mm
Based on given Pad Diameter:	1050.0000	33.1565 mm
Based on the Estimated Diameter Limit:	1143.6125	27.3441 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (573 \cdot 32.9 \cdot 1 + 2 \cdot 18.5 \cdot 32.9 \cdot 1 \cdot (1 - 1))$
 $= 18810.381 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 573(1 \cdot 33.5 - 1 \cdot 32.9) - 2 \cdot 18.5$
 $(1 \cdot 33.5 - 1 \cdot 32.9) \cdot (1 - 1)$
 $= 371.720 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 80.2) \cdot (18.5 - 2.42) \cdot 1$

= 2580.544 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 = $10^2 \cdot 1 + (0)^2 \cdot 1 + 34^2 \cdot 1$
 = 1256.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 = $(1050 - 610) \cdot 34 \cdot 1$
 = 14973.603 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.9218 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 3.9218 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 20$, $t_r = 2.42$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 20$, $t_r = 2.42$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 20$, $t_r = 2.42$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 34$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-28 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, $t_g = 20$, $t_r = 2.42$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1.10/1.96 = 0.561

Weld Size Calculations, Description: M1B24"

Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	24.0380 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (18810 - 372 + 2 \cdot 18.5 \cdot 1 \cdot (1 \cdot 33.5 - 32.9)) \cdot 138)$
 $= 2547587.25 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (2581 + 14974 + 1256 - 0 \cdot 1) \cdot 138$
 $= 2595532.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (2581 + 0 + 100 + (1240)) \cdot 138$
 $= 540910.31 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (2581 + 0 + 1256 + 14974 + (1240)) \cdot 138$
 $= 2766566.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$
 $= 647445. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 1050 \cdot 34 \cdot 0.49 \cdot 138$
 $= 3791565. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 296) \cdot (20 - 1.5) \cdot 0.7 \cdot 138$
 $= 1659177. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 610 \cdot 34 \cdot 0.74 \cdot 138$
 $= 3324434. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 610 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$
 $= 3275545. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.



Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (3791565 + 1659177) = 5450742 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (647445 + 3324433 + 3275544 + 0) = 7247423 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (3791565 + 3275544 + 0) = 7067110 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 5450742 N , must exceed W = 2547587 N or W1 = 2595532 N
Path 2-2 = 7247423 N , must exceed W = 2547587 N or W2 = 540910 N
Path 3-3 = 7067110 N , must exceed W = 2547587 N or W3 = 2766566 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.11 MPa

The Drop for this Nozzle is : 11.3454 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 346.3456 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 * t_{nom} / R_f * (1 - R_f / R_o)$) 3.392 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N11A2"

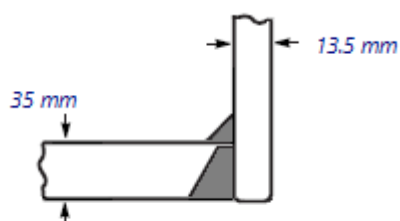
From: 50

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	8200.00	mm
Shell Finished (Minimum) Thickness	t	35.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	2000.0000	mm
Distance from Bottom/Left Tangent		4000.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+N
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	+N
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: N11A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 4102 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 32.8508 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 26.9 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.2155 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 152.4715 mm
 Parallel to Vessel Wall Rn+tn+t 76.2357 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 29.9050 mm

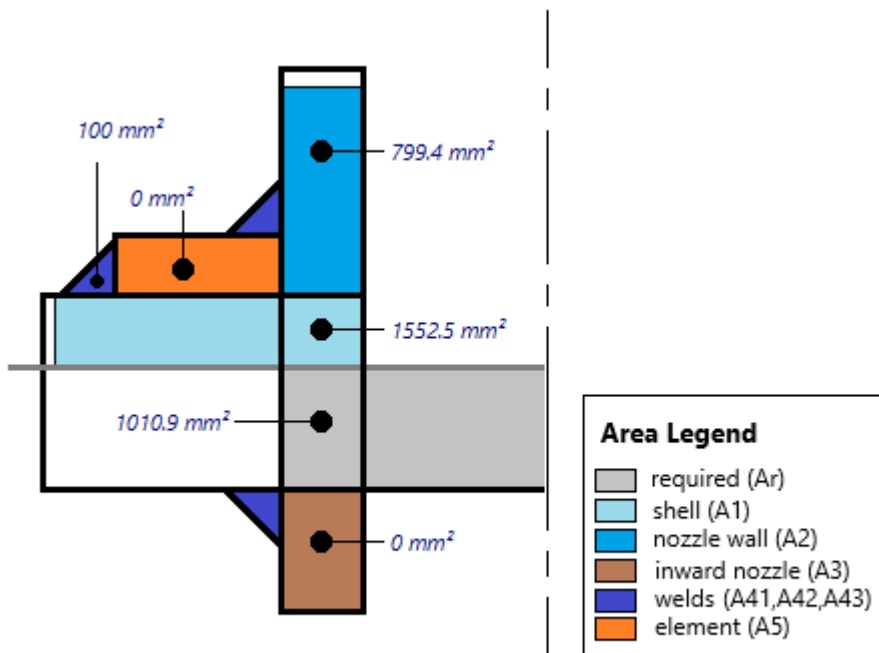


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1010.942	NA	NA
Area in Shell	A1	1552.490	NA	NA
Area in Nozzle Wall	A2	799.445	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	2451.935	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 60.94 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$
 $= (61.5 \cdot 32.9 \cdot 0.5 + 2 \cdot 12 \cdot 32.9 \cdot 0.5 \cdot (1 - 1))$
 $= 1010.942 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 90.9(1 \cdot 33.5 - 0.5 \cdot 32.9) - 2 \cdot 12$
 $(1 \cdot 33.5 - 0.5 \cdot 32.9) \cdot (1 - 1)$
 $= 1552.490 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 29.9)(12 - 0.22)/\sin(61.5)$
 $= 799.445 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 100.000 \text{ mm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.22, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.018$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.10/1.96 = 0.561$

Weld Size Calculations, Description: N11A2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.9620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot Wo \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (1011 - 1552 + 2 \cdot 12 \cdot 1 \cdot$
 $(1 \cdot 33.5 - 16.4)) \cdot 138)$
 $= \max(0, -18359.76) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can/0.707)^2 \cdot fr2) \cdot Sv$
 $= (799 + 0 + 100 - 0 \cdot 1) \cdot 138$
 $= 124110.66 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$

$$= (799 + 0 + 100 + (801)) \cdot 138$$

$$= 234699.92 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (799 + 0 + 100 + 0 + (801)) \cdot 138$$

$$= 234699.92 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 88.9 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 94435. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 37.6) \cdot (13.5 - 1.5) \cdot 0.7 \cdot 138$$

$$= 136542. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 88.9 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$$

$$= 477773. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (94435 + 136542) = 230977 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (94435 + 0 + 477773 + 0) = 572208 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (94435 + 477773 + 0) = 572208 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 230977 N , must exceed W = 0 N	or W1 = 124110 N
Path 2-2 = 572207 N , must exceed W = 0 N	or W2 = 234699 N
Path 3-3 = 572207 N , must exceed W = 0 N	or W3 = 234699 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.12 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: N11A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1.1 \cdot 4102 / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 32.8508 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1.1 \cdot 26.9 / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 0.2155 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	144.7240	mm
Parallel to Vessel Wall	Rn+tn+t	72.3620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	29.9050	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N11A2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7155	mm
Wall Thickness per UG16(b),	tr16b = 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 34.3508	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 34.3508	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000	mm
Wall Thickness per table UG-45	tb3 = 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[6.3, max(34.4, 3)]
 = 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.72, 6.3)
 = 6.3000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Weld Size Calculations, Description: N11A2"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.9620	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 21.9944 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 362.0354 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: M1A24"

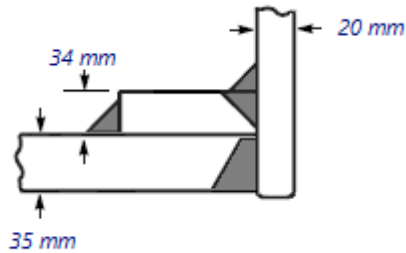
From: 60

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	8200.00	mm
Shell Finished (Minimum) Thickness	t	35.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6600.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+N
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-105	+N
Flange Type		Weld Neck	Flange
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1050.0000	mm
Thickness of Pad	te	34.0000	mm
Weld leg size between Pad and Shell	Wp	34.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	34.0000	mm
Reinforcing Pad Width		220.2000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M1A24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 4102 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 32.8508 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.1 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 1.1)$
 $= 2.4218 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1145.2001 mm
Parallel to Vessel Wall, opening length	d	572.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		80.2500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

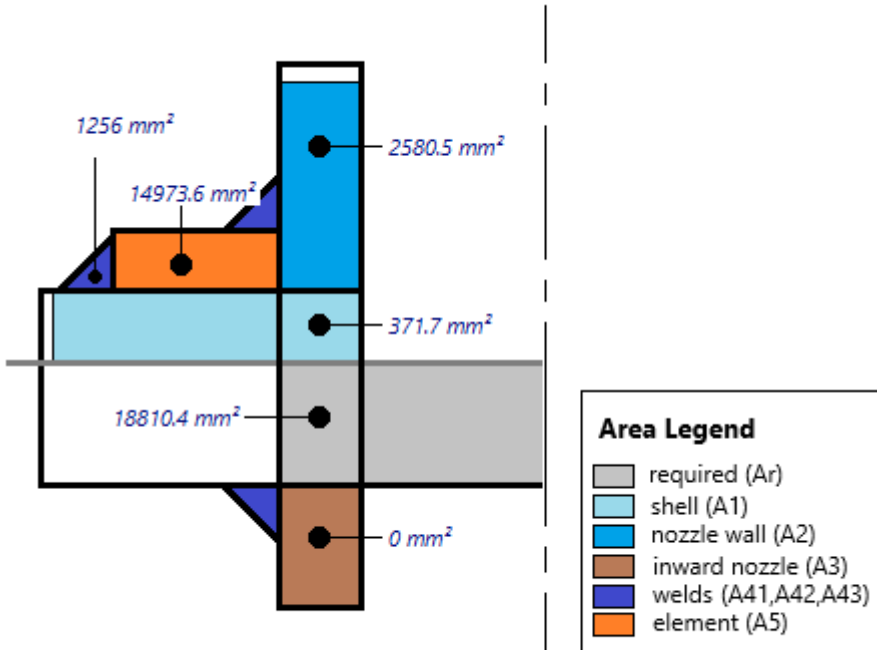


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	18810.381	NA	NA
Area in Shell	A1	371.720	NA	NA
Area in Nozzle Wall	A2	2580.544	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1256.000	NA	NA
Area in Element	A5	14973.603	NA	NA
TOTAL AREA AVAILABLE	Atot	19181.865	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1039.0741 34.0000 mm
 Based on given Pad Diameter: 1050.0000 33.1565 mm
 Based on the Estimated Diameter Limit: 1143.6125 27.3441 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (573 \cdot 32.9 \cdot 1 + 2 \cdot 18.5 \cdot 32.9 \cdot 1 \cdot (1 - 1))$
 $= 18810.381 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 573(1 \cdot 33.5 - 1 \cdot 32.9) - 2 \cdot 18.5$
 $(1 \cdot 33.5 - 1 \cdot 32.9) \cdot (1 - 1)$
 $= 371.720 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 80.2) \cdot (18.5 - 2.42) \cdot 1$



= 2580.544 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 = $10^2 \cdot 1 + (0)^2 \cdot 1 + 34^2 \cdot 1$
 = 1256.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 = $(1050 - 610) \cdot 34 \cdot 1$
 = 14973.603 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.9218 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 3.9218 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 20$, $t_r = 2.42$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 12 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 20$, $t_r = 2.42$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 20$, $t_r = 2.42$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 34$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D -28 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, $t_g = 20$, $t_r = 2.42$, $c = 1.5 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -40 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1.10/1.96 = 0.561

Weld Size Calculations, Description: M1A24"

Intermediate Calc. for nozzle/shell Welds Tmin 18.5000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	24.0380 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (18810 - 372 + 2 \cdot 18.5 \cdot 1 \cdot (1 \cdot 33.5 - 32.9)) \cdot 138)$
 $= 2547587.25 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (2581 + 14974 + 1256 - 0 \cdot 1) \cdot 138$
 $= 2595532.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (2581 + 0 + 100 + (1240)) \cdot 138$
 $= 540910.31 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (2581 + 0 + 1256 + 14974 + (1240)) \cdot 138$
 $= 2766566.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$
 $= 647445. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 1050 \cdot 34 \cdot 0.49 \cdot 138$
 $= 3791565. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 296) \cdot (20 - 1.5) \cdot 0.7 \cdot 138$
 $= 1659177. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 610 \cdot 34 \cdot 0.74 \cdot 138$
 $= 3324434. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 610 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$
 $= 3275545. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (3791565 + 1659177) = 5450742 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (647445 + 3324433 + 3275544 + 0) = 7247423 \text{ N}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (3791565 + 3275544 + 0) = 7067110 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 5450742 N , must exceed W = 2547587 N or W1 = 2595532 N
Path 2-2 = 7247423 N , must exceed W = 2547587 N or W2 = 540910 N
Path 3-3 = 7067110 N , must exceed W = 2547587 N or W3 = 2766566 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.11 MPa

The Drop for this Nozzle is : 11.3454 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 346.3456 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.392 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N18"

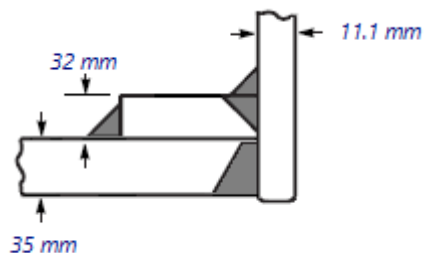
From: 80

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	8200.00	mm
Shell Finished (Minimum) Thickness	t	35.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12150.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	+N
Flange Type		Weld Neck	Flange
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	400.0000	mm
Thickness of Pad	te	32.0000	mm
Weld leg size between Pad and Shell	Wp	28.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		90.4625	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.750 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 4102 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 32.8508 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 99.9 / (118 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.9368 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.7000 mm
Parallel to Vessel Wall, opening length	d	199.8500 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		56.0312 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

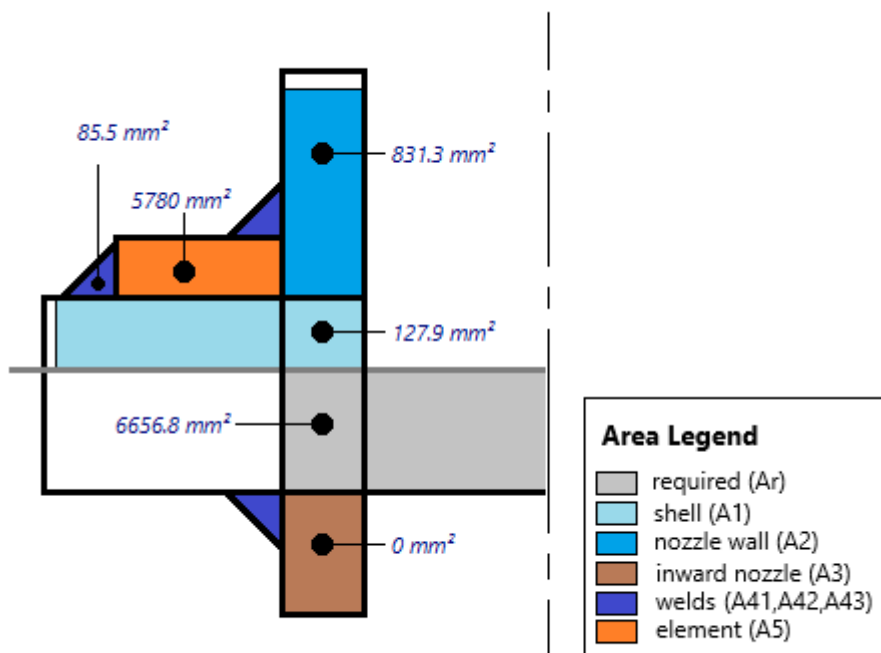


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	6656.767	NA	NA
Area in Shell	A1	127.930	NA	NA
Area in Nozzle Wall	A2	831.323	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	5780.000	NA	NA
TOTAL AREA AVAILABLE	Atot	6824.759	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 394.4502 32.0000 mm
 Based on given Pad Diameter: 400.0000 31.0699 mm
 Based on the Estimated Diameter Limit: 398.1125 31.3454 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (200 \cdot 32.9 \cdot 1 + 2 \cdot 9.61 \cdot 32.9 \cdot 1 \cdot (1 - 0.86))$
 $= 6656.767 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 200(1 \cdot 33.5 - 1 \cdot 32.9) - 2 \cdot 9.61$
 $(1 \cdot 33.5 - 1 \cdot 32.9) \cdot (1 - 0.86)$
 $= 127.930 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 56) \cdot (9.61 - 0.94) \cdot 0.86$



= 831.323 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$
= 85.507 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(400 - 219) \cdot 32 \cdot 1$
= 5780.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.4368 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 34.3508 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 34.3508 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 8.6600 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.66, max(34.4, 3)]
= 8.6600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.44, 8.66)
= 8.6600 mm

Available Nozzle Neck Thickness = 11.1125 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29.9,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 265.1 MPa	Passed
Occasional	: 5.5,	Allowable	: 156.9 MPa	Passed
Shear	: 18.0,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 32$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1°C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, $t_g = 11.1$, $t_r = 0.94$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.097$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -30 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.10/5.10 = 0.216$

Weld Size Calculations, Description: N18"

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.6125 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.7287 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$19.7960 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (6657 - 128 + 2 \cdot 9.61 \cdot 0.86 \cdot (1 \cdot 33.5 - 32.9)) \cdot 138)$
 $= 902359.12 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v$
 $= (831 + 5780 + 85.5 - 0 \cdot 0.86) \cdot 138$
 $= 924067.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (831 + 0 + 85.5 + (551)) \cdot 138$
 $= 202498.09 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (831 + 0 + 85.5 + 5780 + (551)) \cdot 138$
 $= 1000055.94 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$

$$= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 198951. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 400 \cdot 28 \cdot 0.49 \cdot 138$$

$$= 1189511. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 105) \cdot (11.1 - 1.5) \cdot 0.7 \cdot 118$$

$$= 261215. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 219 \cdot 30 \cdot 0.74 \cdot 138$$

$$= 1054163. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 219 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$$

$$= 1177128. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1189510 + 261215) = 1450725 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (198951 + 1054163 + 1177127 + 0) = 2430241 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (1189510 + 1177127 + 0) = 2366638 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 1450725 N	, must exceed W = 902359 N	or W1 = 924067 N
Path 2-2 = 2430241 N	, must exceed W = 902359 N	or W2 = 202498 N
Path 3-3 = 2366638 N	, must exceed W = 902359 N	or W3 = 1000055 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.11 MPa

The Drop for this Nozzle is : 1.4635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.4637 mm

Input Echo, WRC297 Item 1, Description: N18" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	8200.000	mm
Shell Thickness	35.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	196.850	mm
Nozzle Thickness	11.1125	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	



Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	400.000	mm
Thickness of Reinforcing Pad	32.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	1.10	MPa
Radial Load	P	-12799.39	N
Circumferential Shear	Vc	9599.58	N
Longitudinal Shear	Vl	12799.39	N
Circumferential Moment	Mc	0.512E+07	N-mm
Longitudinal Moment	Ml	0.666E+07	N-mm
Torsional Moment	Mt	0.768E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(4134.25(35.0 - 1.5))}$$

$$= 1447.178 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	8268.500	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	65.500	mm
Nozzle Thickness used	(t) =	9.613	mm
T / t	=	6.814	
d / t	=	22.791	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.298$$

Nr/P	=	0.112
Mr/P	=	0.078
M0/P	=	0.289
N0/P	=	0.368
MrD/Mc	=	0.089
NrDL/Mc	=	0.100
M0d/Mc	=	0.323
N0DL/Mc	=	-0.134
MrD/Ml	=	0.090
NrDL/Ml	=	0.102
M0D/Ml	=	0.315
N0DL/Ml	=	0.060

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-13	13	13	-13
Normal Pressure Stress	68	69	68	69



Outplane Stress Summary 60 78 88 52

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	34	34	34	34

Inplane Stress Summary 31 36 40 29

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary 1 1 1 1

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 60 78 88 52

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-10	10	10	-10
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34

Outplane Stress Summary 30 41 50 18

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-2	2	2	-2
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	69	68	69

Inplane Stress Summary 66 70 73 66

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	66	70	73	66

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	14	13	14
Hoop Stress Summary	13	15	14	15

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-77	77	77	-77
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-72	47	125	-66

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	13	13	7	7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	89	52	125	83

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	13	14	13	14
----------------------	----	----	----	----

Hoop Stress Summary	14	16	13	14
---------------------	----	----	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-59	59	59	-59
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-48	33	101	-52
----------------------	-----	----	-----	-----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	10	10	10	10

Shear Stress Summary	6	6	14	14
----------------------	---	---	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	64	35	104	73
------------------------	----	----	-----	----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{4118.25(35.0 - 1.5)}$$

$$= 1444.799 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	8236.500 mm
Nozzle Outside Diameter	(d) =	400.000 mm
Vessel Thickness used	(T) =	33.500 mm
Nozzle Thickness used	(t) =	100.075 mm
T / t	=	0.335
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.761$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.120
Mr/P	=	0.265
M0/P	=	0.077
N0/P	=	0.140
MrD/Mc	=	0.313
NrDL/Mc	=	0.071
M0d/Mc	=	0.097
N0DL/Mc	=	0.028
MrD/Ml	=	0.230
NrDL/Ml	=	0.088
M0D/Ml	=	0.073
N0DL/Ml	=	0.094

Vessel Stresses (MPa):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	134	135	134	135
Outplane Stress Summary	133	136	148	126

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-20	20	20	-20
Inplane Pressure Stress	67	67	67	67
Inplane Stress Summary	64	69	108	31

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	133	136	148	126

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-6	6	6	-6
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	67	67	67	67
Outplane Stress Summary	66	69	80	57

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0

Inplane Bending (Mc)	-21	21	21	-21
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	134	135	134	135

Inplane Stress Summary	131	139	175	97
------------------------	-----	-----	-----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	131	139	175	97

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	69	68	69	68	69	68	69
Circ. Pl (SUS)	1	1	1	1	0	0	0	0
Circ. Q (SUS)	-8	8	18	-18	-1	1	3	-3
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	-2	2	4	-4	-5	5	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	68.0	69.0	68.0	69.0	68.0	69.0	68.0	69.0
Pm+Pl (SUS)	69.0	70.0	69.0	70.0	68.0	69.0	68.0	69.0
Pm+Pl+Q (Total)	61.0	78.0	87.0	52.0	67.0	70.0	71.0	66.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	69.00	138.00	Passed
Pm+Pl (SUS)	70.00	207.00	Passed
Pm+Pl+Q (TOTAL)	87.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	134	135	134	135	134	135	134	135
Circ. Pl (SUS)	0	0	2	2	1	1	1	1
Circ. Q (SUS)	-1	1	11	-11	-3	3	39	-39
Long. Pm (SUS)	67	67	67	67	67	67	67	67
Long. Pl (SUS)	0	0	2	2	1	1	1	1
Long. Q (SUS)	-2	2	38	-38	-1	1	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	134.0	135.0	134.0	135.0	134.0	135.0	134.0	135.0
Pm+Pl (SUS)	134.0	135.0	136.0	137.0	135.0	136.0	135.0	136.0
Pm+Pl+Q (Total)	133.0	136.0	147.0	126.0	132.0	139.0	174.0	97.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	135.00	138.00	Passed
Pm+Pl (SUS)	137.00	207.00	Passed
Pm+Pl+Q (TOTAL)	174.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	14	13	14	13	14	13	14
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-18	-18	22	22	-14	-14	18	18
Long. Q (SUS)	-59	59	95	-95	-41	41	77	-77
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	13.0	14.0	13.0	14.0	13.0	14.0	13.0	14.0
Pm+Pl (SUS)	26.3	27.3	28.3	28.3	22.8	23.8	24.8	24.9
Pm+Pl+Q (Total)	88.3	51.0	123.6	83.5	64.5	35.4	102.9	72.8

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+Pl (SUS)	28.30	177.00	Passed
Pm+Pl+Q (TOTAL)	123.58	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N11B2"

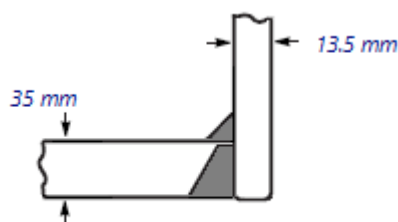
From: 80

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	8200.00	mm
Shell Finished (Minimum) Thickness	t	35.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		15000.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+N
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	+N
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N11B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.1 \cdot 4102 / (138 \cdot 1 - 0.6 \cdot 1.1)$
= 32.8508 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.1 \cdot 26.9 / (138 \cdot 1 - 0.6 \cdot 1.1)$
= 0.2155 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	144.7240	mm
Parallel to Vessel Wall	Rn+tn+t	72.3620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	29.9050	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N11B2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7155	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 34.3508	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 34.3508	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(34.4, 3)]
= 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.72, 6.3)
= 6.3000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.22, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.018$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.10 / 1.96 = 0.561$

Weld Size Calculations, Description: N11B2"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.9620	mm
---	------	---------	----

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1841 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.1843 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N72"

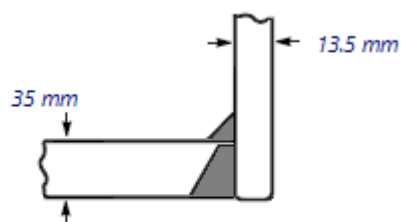
From: 90

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	8200.00	mm
Shell Finished (Minimum) Thickness	t	35.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		17000.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+N
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	+N
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N72"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 4102 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 32.8508 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 26.9 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.2155 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	144.7240	mm
Parallel to Vessel Wall	Rn+tn+t	72.3620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	29.9050	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N7_2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7155	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 34.3508	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 34.3508	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(34.4, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.72, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.22, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.018$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.10 / 1.96 = 0.561$

Weld Size Calculations, Description: N7_2"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.9620	mm
---	------	---------	----

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1841 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 235.1843 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N52"

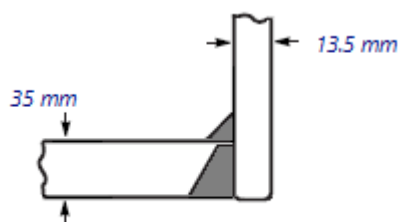
From: 90

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	8200.00	mm
Shell Finished (Minimum) Thickness	t	35.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		18000.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+N
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	+N
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.1 \cdot 4102 / (138 \cdot 1 - 0.6 \cdot 1.1)$
= 32.8508 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.1 \cdot 26.9 / (138 \cdot 1 - 0.6 \cdot 1.1)$
= 0.2155 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	144.7240	mm
Parallel to Vessel Wall	Rn+tn+t	72.3620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	29.9050	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N52".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7155	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 34.3508	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 34.3508	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.3, max(34.4, 3)]
= 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.72, 6.3)
= 6.3000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.22, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.018$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1.10 / 1.96 = 0.561$

Weld Size Calculations, Description: N52"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.9620	mm
---	------	---------	----

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1841 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 235.1843 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N98"

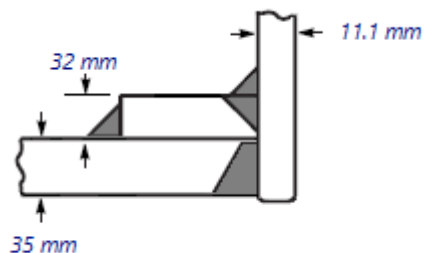
From: 100

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	8200.00	mm
Shell Finished (Minimum) Thickness	t	35.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		19000.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	+N
Flange Type		Weld Neck	Flange
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	400.0000	mm
Thickness of Pad	te	32.0000	mm
Weld leg size between Pad and Shell	Wp	28.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		90.4625	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N98"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.750 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 4102 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 32.8508 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 99.9 / (118 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.9368 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.7000 mm
Parallel to Vessel Wall, opening length	d	199.8500 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		56.0312 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

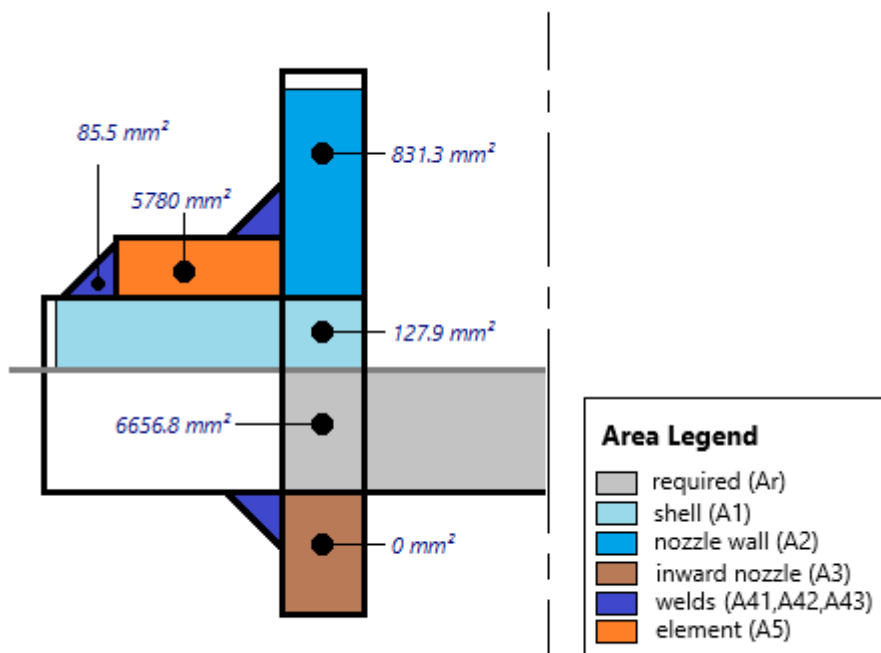


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	6656.767	NA	NA
Area in Shell	A1	127.930	NA	NA
Area in Nozzle Wall	A2	831.323	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	5780.000	NA	NA
TOTAL AREA AVAILABLE	Atot	6824.759	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	394.4502	32.0000 mm
Based on given Pad Diameter:	400.0000	31.0699 mm
Based on the Estimated Diameter Limit:	398.1125	31.3454 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (200 \cdot 32.9 \cdot 1 + 2 \cdot 9.61 \cdot 32.9 \cdot 1 \cdot (1 - 0.86))$
 $= 6656.767 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 200(1 \cdot 33.5 - 1 \cdot 32.9) - 2 \cdot 9.61$
 $(1 \cdot 33.5 - 1 \cdot 32.9) \cdot (1 - 0.86)$
 $= 127.930 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 56) \cdot (9.61 - 0.94) \cdot 0.86$



= 831.323 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$
= 85.507 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(400 - 219) \cdot 32 \cdot 1$
= 5780.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.4368 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 34.3508 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 34.3508 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 8.6600 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.66, max(34.4, 3)]
= 8.6600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.44, 8.66)
= 8.6600 mm

Available Nozzle Neck Thickness = 11.1125 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29.9,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 265.1 MPa	Passed
Occasional	: 5.5,	Allowable	: 156.9 MPa	Passed
Shear	: 18.0,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 32, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.98$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -30 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.10/5.10 = 0.216$

Weld Size Calculations, Description: N98"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.6125 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.7287 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$19.7960 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (6657 - 128 + 2 \cdot 9.61 \cdot 0.86 \cdot (1 \cdot 33.5 - 32.9)) \cdot 138)$
 $= 902359.12 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (831 + 5780 + 85.5 - 0 \cdot 0.86) \cdot 138$
 $= 924067.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (831 + 0 + 85.5 + (551)) \cdot 138$
 $= 202498.09 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (831 + 0 + 85.5 + 5780 + (551)) \cdot 138$
 $= 1000055.94 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$



$$= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 198951. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 400 \cdot 28 \cdot 0.49 \cdot 138$$

$$= 1189511. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 105) \cdot (11.1 - 1.5) \cdot 0.7 \cdot 118$$

$$= 261215. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 219 \cdot 30 \cdot 0.74 \cdot 138$$

$$= 1054163. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 219 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$$

$$= 1177128. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1189510 + 261215) = 1450725 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (198951 + 1054163 + 1177127 + 0) = 2430241 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (1189510 + 1177127 + 0) = 2366638 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 1450725 N	, must exceed W = 902359 N	or W1 = 924067 N
Path 2-2 = 2430241 N	, must exceed W = 902359 N	or W2 = 202498 N
Path 3-3 = 2366638 N	, must exceed W = 902359 N	or W3 = 1000055 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.11 MPa

The Drop for this Nozzle is : 1.4635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.4637 mm

Input Echo, WRC297 Item 1, Description: N98" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	8200.000	mm
Shell Thickness	35.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	196.850	mm
Nozzle Thickness	11.1125	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	



Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	400.000	mm
Thickness of Reinforcing Pad	32.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	1.10	MPa
Radial Load	P	-12799.39	N
Circumferential Shear	Vc	9599.58	N
Longitudinal Shear	Vl	12799.39	N
Circumferential Moment	Mc	0.512E+07	N-mm
Longitudinal Moment	Ml	0.666E+07	N-mm
Torsional Moment	Mt	0.768E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{4134.25(35.0 - 1.5)}$$

$$= 1447.178 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	8268.500	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	65.500	mm
Nozzle Thickness used	(t) =	9.613	mm
T / t	=	6.814	
d / t	=	22.791	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.298$$

Nr/P	=	0.112
Mr/P	=	0.078
M0/P	=	0.289
N0/P	=	0.368
MrD/Mc	=	0.089
NrDL/Mc	=	0.100
M0d/Mc	=	0.323
N0DL/Mc	=	-0.134
MrD/Ml	=	0.090
NrDL/Ml	=	0.102
M0D/Ml	=	0.315
N0DL/Ml	=	0.060

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-13	13	13	-13
Normal Pressure Stress	68	69	68	69



Outplane Stress Summary 60 78 88 52

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-3	3	3	-3
Inplane Pressure Stress	34	34	34	34

Inplane Stress Summary 31 36 40 29

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary 1 1 1 1

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 60 78 88 52

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-10	10	10	-10
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34

Outplane Stress Summary 30 41 50 18

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-2	2	2	-2
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	69	68	69

Inplane Stress Summary 66 70 73 66

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	66	70	73	66

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	14	13	14
Hoop Stress Summary	13	15	14	15

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-77	77	77	-77
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-72	47	125	-66

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	13	13	7	7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	89	52	125	83

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	13	14	13	14
----------------------	----	----	----	----

Hoop Stress Summary	14	16	13	14
---------------------	----	----	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-59	59	59	-59
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-48	33	101	-52
----------------------	-----	----	-----	-----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	10	10	10	10

Shear Stress Summary	6	6	14	14
----------------------	---	---	----	----

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	64	35	104	73
------------------------	----	----	-----	----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{4118.25(35.0 - 1.5)}$$

$$= 1444.799 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	8236.500 mm
Nozzle Outside Diameter	(d) =	400.000 mm
Vessel Thickness used	(T) =	33.500 mm
Nozzle Thickness used	(t) =	100.075 mm
T / t	=	0.335
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.761$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.120
Mr/P	=	0.265
M0/P	=	0.077
N0/P	=	0.140
MrD/Mc	=	0.313
NrDL/Mc	=	0.071
M0d/Mc	=	0.097
N0DL/Mc	=	0.028
MrD/Ml	=	0.230
NrDL/Ml	=	0.088
M0D/Ml	=	0.073
N0DL/Ml	=	0.094

Vessel Stresses (MPa):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	134	135	134	135
Outplane Stress Summary	133	136	148	126

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-20	20	20	-20
Inplane Pressure Stress	67	67	67	67
Inplane Stress Summary	64	69	108	31

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	133	136	148	126

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-6	6	6	-6
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	67	67	67	67
Outplane Stress Summary	66	69	80	57

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0



Inplane Bending (Mc)	-21	21	21	-21
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	134	135	134	135

Inplane Stress Summary	131	139	175	97
------------------------	-----	-----	-----	----

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	1	1
----------------------	---	---	---	---

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	131	139	175	97

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	69	68	69	68	69	68	69
Circ. Pl (SUS)	1	1	1	1	0	0	0	0
Circ. Q (SUS)	-8	8	18	-18	-1	1	3	-3
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	-2	2	4	-4	-5	5	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	68.0	69.0	68.0	69.0	68.0	69.0	68.0	69.0
Pm+Pl (SUS)	69.0	70.0	69.0	70.0	68.0	69.0	68.0	69.0
Pm+Pl+Q (Total)	61.0	78.0	87.0	52.0	67.0	70.0	71.0	66.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	69.00	138.00	Passed
Pm+Pl (SUS)	70.00	207.00	Passed
Pm+Pl+Q (TOTAL)	87.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	134	135	134	135	134	135	134	135
Circ. Pl (SUS)	0	0	2	2	1	1	1	1
Circ. Q (SUS)	-1	1	11	-11	-3	3	39	-39
Long. Pm (SUS)	67	67	67	67	67	67	67	67
Long. Pl (SUS)	0	0	2	2	1	1	1	1
Long. Q (SUS)	-2	2	38	-38	-1	1	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	134.0	135.0	134.0	135.0	134.0	135.0	134.0	135.0
Pm+Pl (SUS)	134.0	135.0	136.0	137.0	135.0	136.0	135.0	136.0
Pm+Pl+Q (Total)	133.0	136.0	147.0	126.0	132.0	139.0	174.0	97.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	135.00	138.00	Passed
Pm+Pl (SUS)	137.00	207.00	Passed
Pm+Pl+Q (TOTAL)	174.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	14	13	14	13	14	13	14
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-18	-18	22	22	-14	-14	18	18
Long. Q (SUS)	-59	59	95	-95	-41	41	77	-77
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	13.0	14.0	13.0	14.0	13.0	14.0	13.0	14.0
Pm+Pl (SUS)	26.3	27.3	28.3	28.3	22.8	23.8	24.8	24.9
Pm+Pl+Q (Total)	88.3	51.0	123.6	83.5	64.5	35.4	102.9	72.8

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+Pl (SUS)	28.30	177.00	Passed
Pm+Pl+Q (TOTAL)	123.58	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N28"

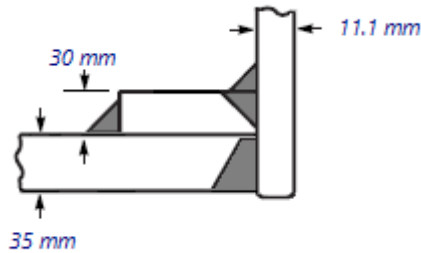
From: 110

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	8200.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	35.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	350.0000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	28.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		65.4625	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N28"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.750 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.1 \cdot 8203 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$
 $= 29.4375 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 99.9 / (118 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.9368 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.7000	mm
Parallel to Vessel Wall, opening length	d	199.8500	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		54.0312	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

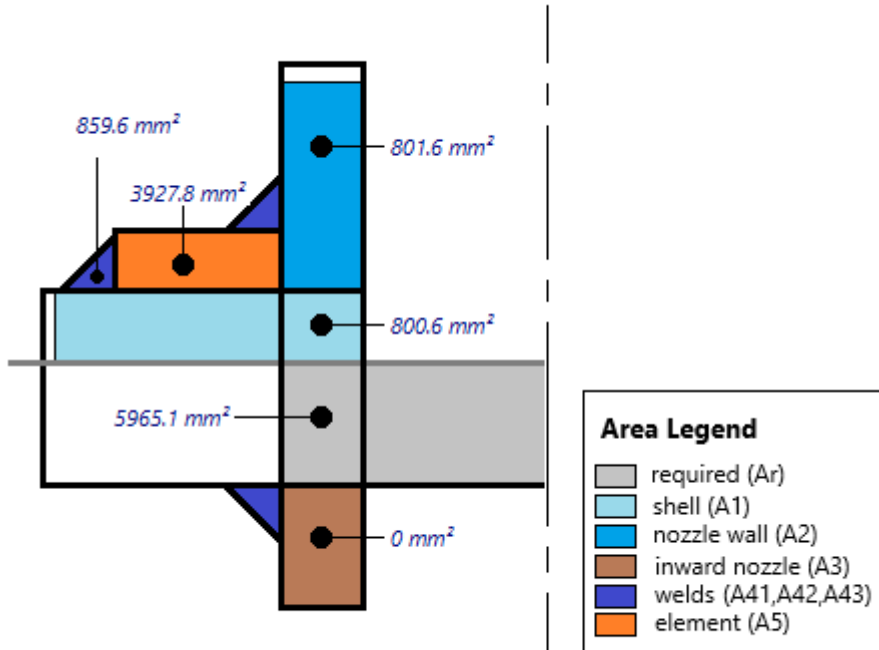


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5965.109	NA	NA
Area in Shell	A1	800.567	NA	NA
Area in Nozzle Wall	A2	801.649	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	859.585	NA	NA
Area in Element	A5	3927.750	NA	NA
TOTAL AREA AVAILABLE	Atot	6389.551	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	335.8519	30.0000 mm
Based on given Pad Diameter:	350.0000	26.7581 mm
Based on the Estimated Diameter Limit:	398.1125	19.5675 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (200 \cdot 29.4 \cdot 1 + 2 \cdot 9.61 \cdot 29.4 \cdot 1 \cdot (1 - 0.86))$
 $= 5965.109 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 200(1 \cdot 33.5 - 1 \cdot 29.4) - 2 \cdot 9.61$
 $(1 \cdot 33.5 - 1 \cdot 29.4) \cdot (1 - 0.86)$
 $= 800.567 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 54) \cdot (9.61 - 0.94) \cdot 0.86$

= 801.649 mm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (100) • 0.86 + (0) • 0.86 + 774² • 1
 = 859.585 mm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (350 - 219) • 30 • 1
 = 3927.750 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.4368 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 34.2038 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 34.2038 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 8.6600 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[8.66, max(34.2, 3)]
 = 8.6600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(2.44, 8.66)
 = 8.6600 mm

Available Nozzle Neck Thickness = 11.1125 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	29.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	265.1 MPa	Passed
Occasional	:	5.5,	Allowable	:	156.9 MPa	Passed
Shear	:	18.0,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.097, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.097, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.097, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 30, c = 1.5 mm, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.88$, Temp. Reduction = 7 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-30 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 11.1, tr = 0.94, c = 1.5 mm, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.10/5.10 = 0.216$

Weld Size Calculations, Description: N28"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.6125 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{\text{minPad}}$	$19.7960 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (5965 - 801 + 2 \cdot 9.61 \cdot 0.86 \cdot (1 \cdot 33.5 - 29.4)) \cdot 138)$
 $= 721848.31 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (802 + 3928 + 860 - 0 \cdot 0.86) \cdot 138$
 $= 771200.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (802 + 0 + 85.5 + (551)) \cdot 138$
 $= 198403.58 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (802 + 0 + 860 + 3928 + (551)) \cdot 138$
 $= 847188.88 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$



$$= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 198951. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 350 \cdot 28 \cdot 0.49 \cdot 138$$

$$= 1040822. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 105) \cdot (11.1 - 1.5) \cdot 0.7 \cdot 118$$

$$= 261215. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 219 \cdot 30 \cdot 0.74 \cdot 138$$

$$= 1054163. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 219 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$$

$$= 1177128. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1040821 + 261215) = 1302036 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (198951 + 1054163 + 1177127 + 0) = 2430241 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (1040821 + 1177127 + 0) = 2217949 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 1302036 N	, must exceed W = 721848 N	or W1 = 771200 N
Path 2-2 = 2430241 N	, must exceed W = 721848 N	or W2 = 198403 N
Path 3-3 = 2217949 N	, must exceed W = 721848 N	or W3 = 847188 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.13 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.8100 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.8102 mm

Input Echo, WRC107/537 Item 1, Description: N28" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.5000 mm
Vessel Diameter	Dv	14760.000 mm
Vessel Thickness	Tv	35.000 mm
Design Temperature	T1	85.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow



Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	196.850	mm
Nozzle Thickness	Tn	11.113	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

Thickness of Reinforcing Pad	Tpad	30.000	mm
Diameter of Reinforcing Pad	Dpad	350.000	mm

Design Internal Pressure	Dp	1.100	MPa
Include Pressure Thrust		No	

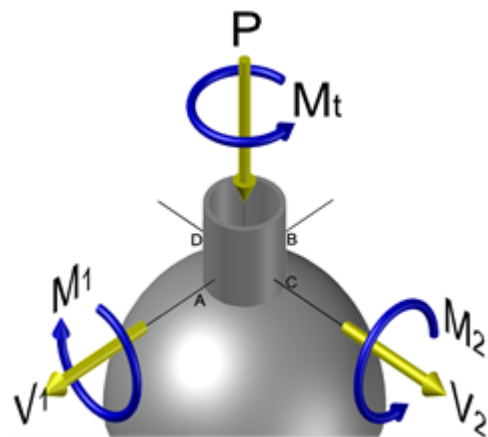
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-12799.4	N
Longitudinal Shear (SUS)	(V1) V1	11313.2	N
Circumferential Shear (SUS)	(Vc) V2	11313.2	N
Circumferential Moment (SUS)	(Mc) M1	5939537.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	5939537.0	N-mm
Torsional Moment (SUS)	Mt	7682707.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))} \\
 &= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(7398.25 (35.0 - 1.5))} \\
 &= 1861.936 \text{ mm}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-12799.4	N
Circumferential Shear	(VC) V2	11313.2	N
Longitudinal Shear	(VL) V1	11313.2	N
Circumferential Moment	(MC) M1	5939537.0	N-mm
Longitudinal Moment	(ML) M2	5939537.0	N-mm
Torsional Moment	MT	7682707.0	N-mm



Dimensionless Param: U = 0.16 TAU = 10.90 RHO = 4.00 (6.61)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 4	0.08180	(A,B,C,D)
M(x) / P	SP 4	0.04181	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 4	0.31541	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 4	0.17769	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 4	0.31541	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 4	0.17769	(A,B,C,D)
N(y) * T / P	SP 4	0.35274	(A,B,C,D)
M(y) / P	SP 4	0.23379	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 4	0.10105	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 4	1.13764	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 4	0.10105	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 4	1.13764	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Rad.	Bend. P	0.8	-0.8	0.8	-0.8	0.8	-0.8	0.8	-0.8
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-2.3	2.3	2.3	-2.3
Rad.	Memb. ML	-0.7	-0.7	0.7	0.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-2.3	2.3	2.3	-2.3	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-1.9	1.1	4.0	-2.1	-1.9	1.1	4.0	-2.1
Tang.	Memb. P	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Tang.	Bend. P	4.5	-4.5	4.5	-4.5	4.5	-4.5	4.5	-4.5
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-14.6	14.6	14.6	-14.6
Tang.	Memb. ML	-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-14.6	14.6	14.6	-14.6	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-9.3	11.1	20.4	-17.8	-9.3	11.1	20.4	-17.8
	Shear VC	0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
	Shear MT	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
	Tot. Shear	2.1	2.1	1.1	1.1	1.1	1.1	2.1	2.1
	Str. Int.	9.9	11.5	20.5	17.8	9.5	11.2	20.7	18.0

Unitless Prm: U = 0.35 TAU = 0.00 (17.71) RHO = 0.00 (3.49)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.17415	(A,B,C,D)
M(x) / P	SR 2	0.12075	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.16099	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.39533	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.16099	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.39533	(A,B,C,D)
N(y) * T / P	SR 2	0.05221	(A,B,C,D)
M(y) / P	SR 2	0.03632	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04941	(A,B,C,D)



$M(y) * \sqrt{R_m * T} / MC$ SR 3 0.11805 (A,B,C,D)
 $N(y) * T * \sqrt{R_m * T} / ML$ SR 3 0.04941 (A,B,C,D)
 $M(y) * \sqrt{R_m * T} / ML$ SR 3 0.11805 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Rad. Bend. P		8.3	-8.3	8.3	-8.3	8.3	-8.3	8.3	-8.3
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-25.2	25.2	25.2	-25.2
Rad. Memb. ML		-1.7	-1.7	1.7	1.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-25.2	25.2	25.2	-25.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-16.7	17.2	37.2	-29.8	-16.7	17.2	37.2	-29.8
Tang. Memb. P		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Tang. Bend. P		2.5	-2.5	2.5	-2.5	2.5	-2.5	2.5	-2.5
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-7.5	7.5	7.5	-7.5
Tang. Memb. ML		-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Tang. Bend. ML		-7.5	7.5	7.5	-7.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-5.0	5.1	11.1	-8.9	-5.0	5.1	11.1	-8.9
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Tot. Shear		1.8	1.8	0.6	0.6	0.6	0.6	1.8	1.8
Str. Int.		16.9	17.5	37.2	29.8	16.7	17.2	37.3	29.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Rad. Pl (SUS)		-0.4	-0.4	0.9	0.9	-0.4	-0.4	0.9	0.9
Rad. Q (SUS)		-1.5	1.5	3.1	-3.1	-1.5	1.5	3.1	-3.1
Long. Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Long. Pl (SUS)		0.9	0.9	1.3	1.3	0.9	0.9	1.3	1.3
Long. Q (SUS)		-10.2	10.2	19.1	-19.1	-10.2	10.2	19.1	-19.1
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Pm+Pl (SUS)		65.0	65.0	65.6	65.6	65.0	65.0	65.6	65.6
Pm+Pl+Q (Total)		62.6	75.5	84.4	61.9	62.2	75.2	84.6	62.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
---------------------	-----------	----------------	--------



Pm (SUS)	63.94	138.00	Passed
Pm+Pl (SUS)	65.63	207.00	Passed
Pm+Pl+Q (TOTAL)	84.65	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Rad.	Pl (SUS)	0.3	0.3	3.7	3.7	0.3	0.3	3.7	3.7
Rad.	Q (SUS)	-16.9	16.9	33.5	-33.5	-16.9	16.9	33.5	-33.5
Long.	Pm (SUS)	121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Long.	Pl (SUS)	0.1	0.1	1.1	1.1	0.1	0.1	1.1	1.1
Long.	Q (SUS)	-5.0	5.0	10.0	-10.0	-5.0	5.0	10.0	-10.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear	Q (SUS)	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Pm (SUS)		121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Pm+Pl (SUS)		122.0	122.0	125.0	125.0	122.0	122.0	125.0	125.0
Pm+Pl+Q (Total)		116.5	138.7	158.4	112.3	116.2	138.4	158.5	112.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.19	138.00	Passed
Pm+Pl (SUS)	125.03	207.00	Passed
Pm+Pl+Q (TOTAL)	158.48	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N46"

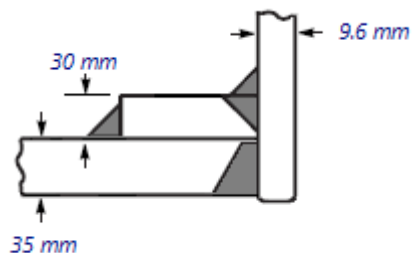
From: 110

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	8200.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	35.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2000.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	270.0000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	28.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		50.8625	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N46"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.1 \cdot 8203 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$
 $= 29.4375 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 76 / (118 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.7128 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	315.8214 mm
Parallel to Vessel Wall, opening length	d	157.9107 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		50.2530 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:

$= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

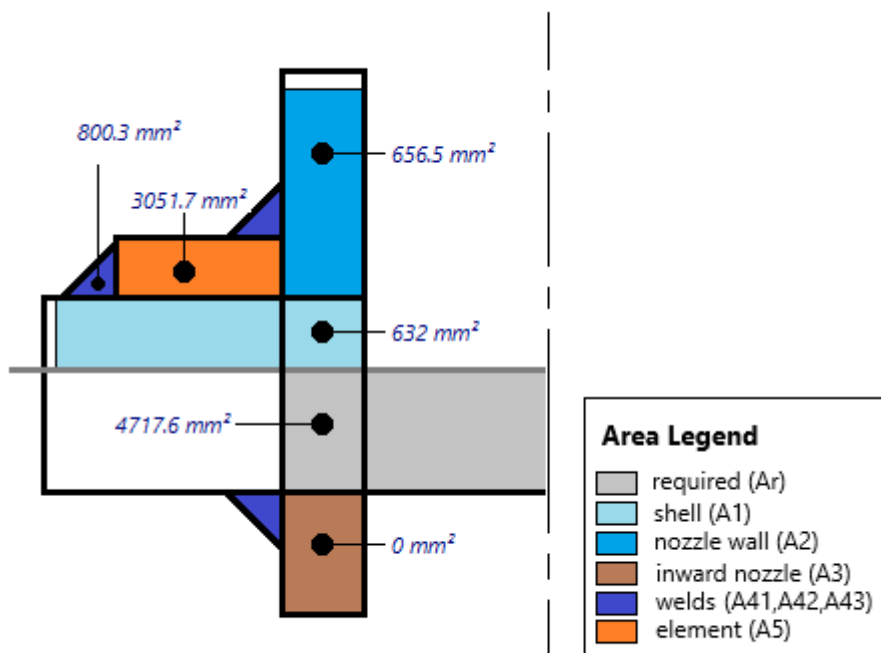


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4717.624	NA	NA
Area in Shell	A1	631.969	NA	NA
Area in Nozzle Wall	A2	656.522	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	800.296	NA	NA
Area in Element	A5	3051.750	NA	NA
TOTAL AREA AVAILABLE	Atot	5140.536	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 74.37 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	255.9029	30.0000 mm
Based on given Pad Diameter:	270.0000	25.8426 mm
Based on the Estimated Diameter Limit:	314.2339	18.8449 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (158 \cdot 29.4 \cdot 1 + 2 \cdot 8.1 \cdot 29.4 \cdot 1 \cdot (1 - 0.86))$
 $= 4717.624 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 158(1 \cdot 33.5 - 1 \cdot 29.4) - 2 \cdot 8.1$
 $(1 \cdot 33.5 - 1 \cdot 29.4) \cdot (1 - 0.86)$
 $= 631.969 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 50.3) \cdot (8.1 - 0.71) \cdot 0.86 / \sin(75.3)$

= 656.522 mm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$$

$$= (100) \cdot 0.86 + (0) \cdot 0.86 + 715^2 \cdot 1$$

$$= 800.296 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (276 - 175) \cdot 30 \cdot 1$$

$$= 3051.750 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.088$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.088$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.088$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 30, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.88$, Temp. Reduction = 7 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-30 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.088$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.10 / 5.10 = 0.216$

Weld Size Calculations, Description: N46"

Intermediate Calc. for nozzle/shell Welds t_{min} 8.1012 mm
 Intermediate Calc. for pad/shell Welds t_{minPad} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6708 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$9.5000 = 0.5 * t_{minPad}$	$19.7960 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (4718 - 632 + 2 \cdot 8.1 \cdot 0.86 \cdot (1 \cdot 33.5 - 29.4)) \cdot 138)$$

$$= 571528.44 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (657 + 3052 + 800 - 0 \cdot 0.86) \cdot 138$$

$$= 622118.12 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (657 + 0 + 85.5 + (464)) \cdot 138$$

$$= 166430.94 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (657 + 0 + 800 + 3052 + (464)) \cdot 138$$

$$= 686159.56 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 175 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 158684. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 270 \cdot 28 \cdot 0.49 \cdot 138$$

$$= 802920. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 83.2) \cdot (9.6 - 1.5) \cdot 0.7 \cdot 118$$

$$= 174806. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 175 \cdot 30 \cdot 0.74 \cdot 138$$

$$= 840805. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 175 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$$

$$= 938882. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (802920 + 174806) = 977726 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (158684 + 840805 + 938882 + 0) = 1938370 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$
$$= (802920 + 938882 + 0) = 1741801 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 977725 N , must exceed W = 571528 N or W1 = 622118 N
Path 2-2 = 1938370 N , must exceed W = 571528 N or W2 = 166430 N
Path 3-3 = 1741801 N , must exceed W = 571528 N or W3 = 686159 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.13 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N46"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$
$$= 1.1 \cdot 8203 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$$
$$= 29.4375 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$
$$= 1.1 \cdot 76 / (118 \cdot 1 - 0.6 \cdot 1.1)$$
$$= 0.7128 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	304.1452	mm
Parallel to Vessel Wall, opening length	d	152.0726	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		50.2530	mm

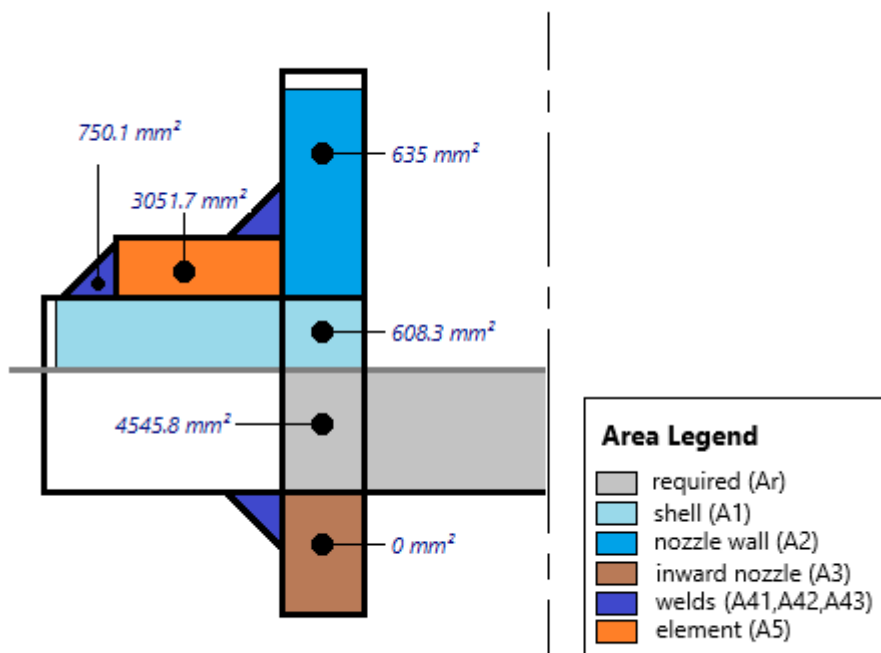


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4545.765	NA	NA
Area in Shell	A1	608.252	NA	NA
Area in Nozzle Wall	A2	634.957	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	750.099	NA	NA
Area in Element	A5	3051.750	NA	NA
TOTAL AREA AVAILABLE	Atot	5045.058	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 253.3569 30.0000 mm
 Based on given Pad Diameter: 270.0000 25.0917 mm
 Based on the Estimated Diameter Limit: 302.5577 19.0081 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152 \cdot 29.4 \cdot 1 + 2 \cdot 8.1 \cdot 29.4 \cdot 1 \cdot (1 - 0.86))$
 $= 4545.765 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152(1 \cdot 33.5 - 1 \cdot 29.4) - 2 \cdot 8.1$
 $(1 \cdot 33.5 - 1 \cdot 29.4) \cdot (1 - 0.86)$
 $= 608.252 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 50.3) \cdot (8.1 - 0.71) \cdot 0.86$

= 634.957 mm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (100) • 0.86 + (0) • 0.86 + 665² • 1
 = 750.099 mm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (270 - 168) • 30 • 1
 = 3051.750 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.2128 mm
Wall Thickness per UG16(b),	tr16b = 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 34.2038 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 34.2038 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm
Wall Thickness per table UG-45	tb3 = 7.7200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[7.72, max(34.2, 3)]
 = 7.7200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(2.21, 7.72)
 = 7.7200 mm

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.1 MPa	Passed
Occasional	:	4.9,	Allowable	:	156.9 MPa	Passed
Shear	:	20.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N46"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.1012 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6708 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	19.7960 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
 = max(0, (4546 - 608 + 2 • 8.1 • 0.86 •
 (1 • 33.5 - 29.4))138)
 = 551087.00 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
 = (635 + 3052 + 750 - 0 • 0.86) • 138
 = 612216.19 N

Weld Load [W2]:
 = (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv

$$= (635 + 0 + 85.5 + (464)) \cdot 138$$

$$= 163455.34 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (635 + 0 + 750 + 3052 + (464)) \cdot 138$$

$$= 676257.62 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 152817. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 270 \cdot 28 \cdot 0.49 \cdot 138$$

$$= 802920. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 80.1) \cdot (9.6 - 1.5) \cdot 0.7 \cdot 118$$

$$= 168343. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 168 \cdot 30 \cdot 0.74 \cdot 138$$

$$= 809720. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 168 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$$

$$= 904170. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (802920 + 168343) = 971263 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (152817 + 809720 + 904170 + 0) = 1866707 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (802920 + 904170 + 0) = 1707090 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 971263 N	, must exceed W = 551087 N	or W1 = 612216 N
Path 2-2 = 1866707 N	, must exceed W = 551087 N	or W2 = 163455 N
Path 3-3 = 1707090 N	, must exceed W = 551087 N	or W3 = 676257 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.13 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 24.1288 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 360.4715 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N46" :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	14760.000	mm
Vessel Thickness	Tv	35.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	149.073	mm
Nozzle Thickness	Tn	9.601	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	30.000	mm
Diameter of Reinforcing Pad	Dpad	270.000	mm
Design Internal Pressure	Dp	1.100	MPa
Include Pressure Thrust		No	

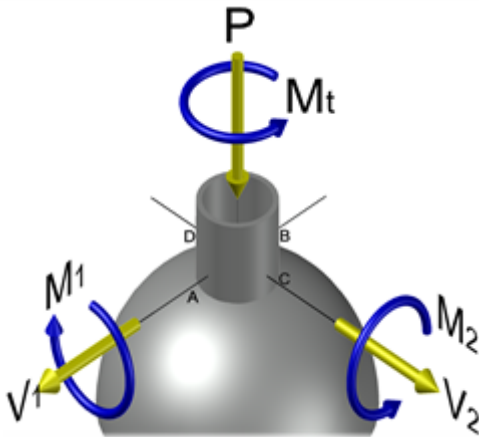
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-9599.6	N
Longitudinal Shear (SUS)	(V1) V1	8484.9	N
Circumferential Shear (SUS)	(Vc) V2	8484.9	N
Circumferential Moment (SUS)	(Mc) M1	3340974.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	3340974.0	N-mm
Torsional Moment (SUS)	Mt	4321525.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(7398.25 (35.0 - 1.5))}$$

$$= 1811.136 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-9599.6	N
Circumferential Shear	(VC) V2	8484.9	N
Longitudinal Shear	(VL) V1	8484.9	N
Circumferential Moment	(MC) M1	3340974.0	N-mm
Longitudinal Moment	(ML) M2	3340974.0	N-mm
Torsional Moment	MT	4321525.0	N-mm

Dimensionless Param: U = 0.12 TAU = 9.89 RHO = 4.00 (7.84)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.09313	(A,B,C,D)
$M(x) / P$	SP 4	0.04668	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.41530	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.22398	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.41530	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.22398	(A,B,C,D)
$N(y) * T / P$	SP 4	0.34154	(A,B,C,D)
$M(y) / P$	SP 4	0.28206	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10147	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.51251	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10147	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.51251	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad. Bend.	P	0.7	-0.7	0.7	-0.7	0.7	-0.7	0.7	-0.7
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-1.6	1.6	1.6	-1.6
Rad. Memb.	ML	-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-1.6	1.6	1.6	-1.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.2	0.7	3.0	-1.6	-1.2	0.7	3.0	-1.6



Tang. Memb. P	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Tang. Bend. P	4.0	-4.0	4.0	-4.0	4.0	-4.0	4.0	-4.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-11.0	11.0	11.0	-11.0
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-11.0	11.0	11.0	-11.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-6.2	7.6	15.9	-14.0	-6.2	7.6	15.9	-14.0
Shear VC	0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tot. Shear	2.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0
Str. Int.	7.0	8.2	16.0	14.1	6.4	7.8	16.2	14.4

Unitless Prm: U = 0.27 TAU = 0.00 (16.16) RHO = 0.00 (4.14)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.20113	(A,B,C,D)
M(x) / P	SR 2	0.15428	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.16308	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.54429	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.16308	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.54429	(A,B,C,D)
N(y) * T / P	SR 2	0.06035	(A,B,C,D)
M(y) / P	SR 2	0.04632	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04978	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.16279	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04978	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.16279	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Rad. Bend. P		7.9	-7.9	7.9	-7.9	7.9	-7.9	7.9	-7.9
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Rad. Bend. MC		0.0	0.0	0.0	0.0	-19.5	19.5	19.5	-19.5
Rad. Memb. ML		-1.0	-1.0	1.0	1.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-19.5	19.5	19.5	-19.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-10.9	12.3	30.1	-24.7	-10.9	12.3	30.1	-24.7
Tang. Memb. P		0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Tang. Bend. P		2.4	-2.4	2.4	-2.4	2.4	-2.4	2.4	-2.4
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-5.8	5.8	5.8	-5.8
Tang. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-5.8	5.8	5.8	-5.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-3.2	3.7	9.0	-7.4	-3.2	3.7	9.0	-7.4
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Tot. Shear		1.7	1.7	0.5	0.5	0.5	0.5	1.7	1.7



Str. Int.	11.2	12.7	30.2	24.8	10.9	12.4	30.3	24.9
-----------	------	------	------	------	------	------	------	------

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Rad.	Pl (SUS)	-0.3	-0.3	0.7	0.7	-0.3	-0.3	0.7	0.7
Rad.	Q (SUS)	-1.0	1.0	2.3	-2.3	-1.0	1.0	2.3	-2.3
Long.	Pm (SUS)	63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Long.	Pl (SUS)	0.7	0.7	0.9	0.9	0.7	0.7	0.9	0.9
Long.	Q (SUS)	-6.9	6.9	15.0	-15.0	-6.9	6.9	15.0	-15.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear	Q (SUS)	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Pm+Pl (SUS)		64.8	64.8	65.3	65.3	64.8	64.8	65.3	65.3
Pm+Pl+Q (Total)		63.4	72.1	79.9	62.5	62.9	71.7	80.2	62.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.94	138.00	Passed
Pm+Pl (SUS)	65.28	207.00	Passed
Pm+Pl+Q (TOTAL)	80.17	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Rad.	Pl (SUS)	0.7	0.7	2.7	2.7	0.7	0.7	2.7	2.7
Rad.	Q (SUS)	-11.6	11.6	27.4	-27.4	-11.6	11.6	27.4	-27.4
Long.	Pm (SUS)	121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Long.	Pl (SUS)	0.2	0.2	0.8	0.8	0.2	0.2	0.8	0.8
Long.	Q (SUS)	-3.5	3.5	8.2	-8.2	-3.5	3.5	8.2	-8.2
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear	Q (SUS)	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Pm (SUS)		121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Pm+Pl (SUS)		122.3	122.3	124.1	124.1	122.3	122.3	124.1	124.1
Pm+Pl+Q (Total)		118.3	133.9	151.3	113.8	118.0	133.6	151.5	114.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.19	138.00	Passed
Pm+Pl (SUS)	124.06	207.00	Passed
Pm+Pl+Q (TOTAL)	151.47	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N62"

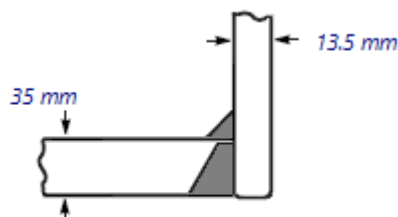
From: 110

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	8200.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	35.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+N
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	+N
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.1 \cdot 8203 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$
 $= 29.4375$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 26.9 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.2155$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.7479	mm
Parallel to Vessel Wall	Rn+tn+t	72.8739	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	29.9050	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N62".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.22, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.018$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.10 / 1.96 = 0.561$

Weld Size Calculations, Description: N62"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.9620	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.1 \cdot 8203 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$
 $= 29.4375 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 26.9 / (138 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.2155 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	144.7240	mm
Parallel to Vessel Wall	Rn+tn+t	72.3620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	29.9050	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N62".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.7155	mm
Wall Thickness per UG16(b),	tr16b = 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 34.2038	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 34.2038	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000	mm
Wall Thickness per table UG-45	tb3 = 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(34.2, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.72, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Weld Size Calculations, Description: N62"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.9620	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 8.1445 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 343.8810 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N86"

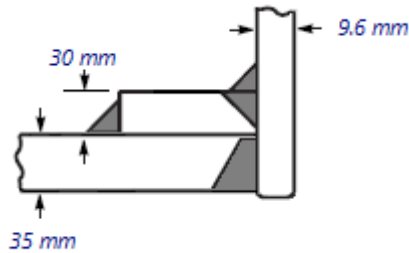
From: 110

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	8200.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	35.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	2000.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	270.0000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	28.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		50.8625	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N86"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.1 \cdot 8203 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$
 $= 29.4375 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 76 / (118 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.7128 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	315.8214 mm
Parallel to Vessel Wall, opening length	d	157.9107 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		50.2530 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

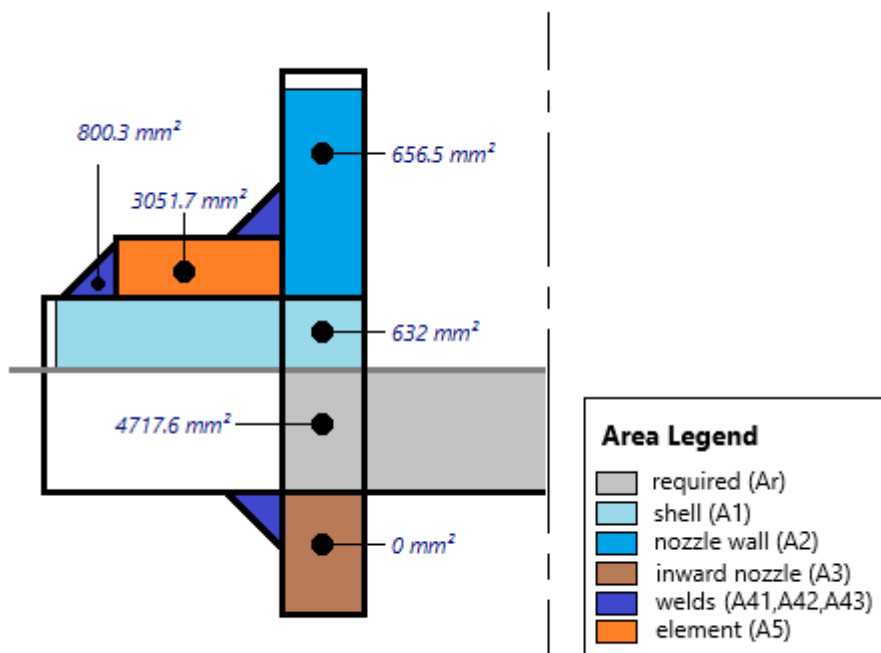


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4717.624	NA	NA
Area in Shell	A1	631.969	NA	NA
Area in Nozzle Wall	A2	656.522	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	800.296	NA	NA
Area in Element	A5	3051.750	NA	NA
TOTAL AREA AVAILABLE	Atot	5140.536	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 74.37 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	255.9029	30.0000 mm
Based on given Pad Diameter:	270.0000	25.8426 mm
Based on the Estimated Diameter Limit:	314.2339	18.8449 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (158 \cdot 29.4 \cdot 1 + 2 \cdot 8.1 \cdot 29.4 \cdot 1 \cdot (1 - 0.86))$
 $= 4717.624 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 158(1 \cdot 33.5 - 1 \cdot 29.4) - 2 \cdot 8.1$
 $(1 \cdot 33.5 - 1 \cdot 29.4) \cdot (1 - 0.86)$
 $= 631.969 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 50.3) \cdot (8.1 - 0.71) \cdot 0.86 / \sin(75.3)$

= 656.522 mm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr}3 + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr}2 + \text{Trapfr}4$$

$$= (100) \cdot 0.86 + (0) \cdot 0.86 + 715^2 \cdot 1$$

$$= 800.296 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot \text{fr}4$$

$$= (276 - 175) \cdot 30 \cdot 1$$

$$= 3051.750 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = $\text{tr} \cdot E^* / (t_g - c) = 0.088$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = $\text{tr} \cdot E^* / (t_g - c) = 0.088$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = $\text{tr} \cdot E^* / (t_g - c) = 0.088$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 30, c = 1.5 mm, E* = 1
 Thickness Ratio = $\text{tr} \cdot E^* / (t_g - c) = 0.88$, Temp. Reduction = 7 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-30 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 9.6, tr = 0.71, c = 1.5 mm, E* = 1
 Thickness Ratio = $\text{tr} \cdot E^* / (t_g - c) = 0.088$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.10/5.10 = 0.216$

Weld Size Calculations, Description: N86"

Intermediate Calc. for nozzle/shell Welds t_{min} 8.1012 mm
 Intermediate Calc. for pad/shell Welds t_{minPad} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6708 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$9.5000 = 0.5 * t_{minPad}$	$19.7960 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (4718 - 632 + 2 \cdot 8.1 \cdot 0.86 \cdot (1 \cdot 33.5 - 29.4)) \cdot 138)$$

$$= 571528.44 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (657 + 3052 + 800 - 0 \cdot 0.86) \cdot 138$$

$$= 622118.12 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (657 + 0 + 85.5 + (464)) \cdot 138$$

$$= 166430.94 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (657 + 0 + 800 + 3052 + (464)) \cdot 138$$

$$= 686159.56 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 175 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 158684. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 270 \cdot 28 \cdot 0.49 \cdot 138$$

$$= 802920. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 83.2) \cdot (9.6 - 1.5) \cdot 0.7 \cdot 118$$

$$= 174806. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 175 \cdot 30 \cdot 0.74 \cdot 138$$

$$= 840805. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 175 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$$

$$= 938882. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (802920 + 174806) = 977726 \text{ N}$$

Failure Path [Path 2-2]:



$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (158684 + 840805 + 938882 + 0) = 1938370 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (802920 + 938882 + 0) = 1741801 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 977725 N , must exceed W = 571528 N or W1 = 622118 N
 Path 2-2 = 1938370 N , must exceed W = 571528 N or W2 = 166430 N
 Path 3-3 = 1741801 N , must exceed W = 571528 N or W3 = 686159 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.13 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N86"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 1.1 \cdot 8203 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$$

$$= 29.4375 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1.1 \cdot 76 / (118 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 0.7128 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	304.1452 mm
Parallel to Vessel Wall, opening length	d	152.0726 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		50.2530 mm

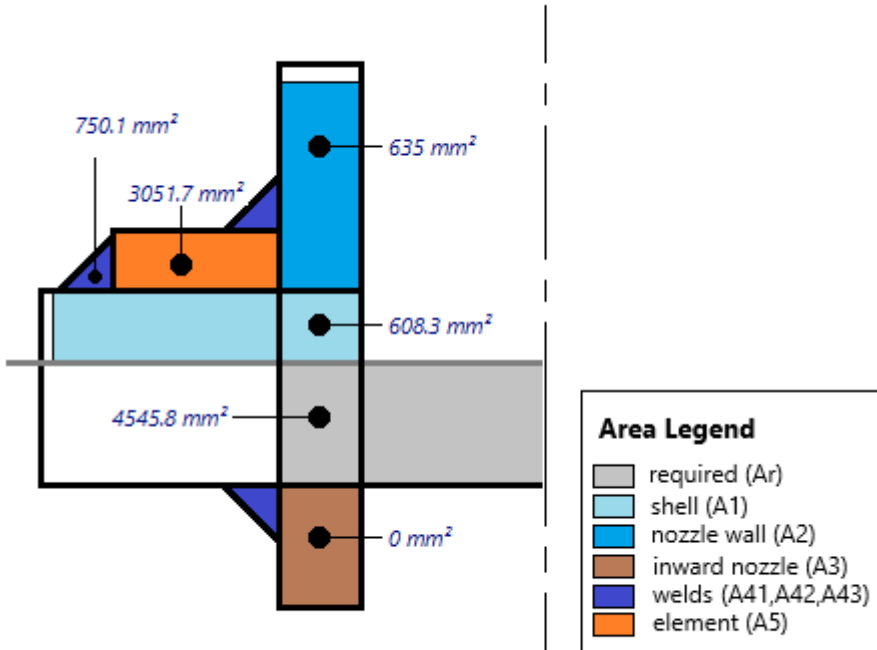


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4545.765	NA	NA
Area in Shell	A1	608.252	NA	NA
Area in Nozzle Wall	A2	634.957	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	750.099	NA	NA
Area in Element	A5	3051.750	NA	NA
TOTAL AREA AVAILABLE	Atot	5045.058	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 253.3569 30.0000 mm
 Based on given Pad Diameter: 270.0000 25.0917 mm
 Based on the Estimated Diameter Limit: 302.5577 19.0081 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (152 \cdot 29.4 \cdot 1 + 2 \cdot 8.1 \cdot 29.4 \cdot 1 \cdot (1 - 0.86))$
 $= 4545.765 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152(1 \cdot 33.5 - 1 \cdot 29.4) - 2 \cdot 8.1$
 $(1 \cdot 33.5 - 1 \cdot 29.4) \cdot (1 - 0.86)$
 $= 608.252 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 50.3) \cdot (8.1 - 0.71) \cdot 0.86$

= 634.957 mm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (100) • 0.86 + (0) • 0.86 + 665² • 1
 = 750.099 mm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (270 - 168) • 30 • 1
 = 3051.750 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 2.2128 mm
 Wall Thickness per UG16(b), tr16b = 3.0000 mm
 Wall Thickness, shell/head, internal pressure trb1 = 34.2038 mm
 Wall Thickness tb1 = max(trb1, tr16b) = 34.2038 mm
 Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm
 Wall Thickness per table UG-45 tb3 = 7.7200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[7.72, max(34.2, 3)]
 = 7.7200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(2.21, 7.72)
 = 7.7200 mm

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	32.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	262.1 MPa	Passed
Occasional	:	4.9,	Allowable	:	156.9 MPa	Passed
Shear	:	20.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N86"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.1012 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.6708 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	19.7960 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
 = max(0, (4546 - 608 + 2 • 8.1 • 0.86 •
 (1 • 33.5 - 29.4))138)
 = 551087.00 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
 = (635 + 3052 + 750 - 0 • 0.86) • 138
 = 612216.19 N

Weld Load [W2]:
 = (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv



$$= (635 + 0 + 85.5 + (464)) \cdot 138$$
$$= 163455.34 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$
$$= (635 + 0 + 750 + 3052 + (464)) \cdot 138$$
$$= 676257.62 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$
$$= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118$$
$$= 152817. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$
$$= (3.14/2.0) \cdot 270 \cdot 28 \cdot 0.49 \cdot 138$$
$$= 802920. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$
$$= (3.14 \cdot 80.1) \cdot (9.6 - 1.5) \cdot 0.7 \cdot 118$$
$$= 168343. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$
$$= (3.14/2) \cdot 168 \cdot 30 \cdot 0.74 \cdot 138$$
$$= 809720. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$
$$= (3.14/2.0) \cdot 168 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$$
$$= 904170. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (802920 + 168343) = 971263 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$
$$= (152817 + 809720 + 904170 + 0) = 1866707 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$
$$= (802920 + 904170 + 0) = 1707090 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 971263 N	, must exceed W = 551087 N	or W1 = 612216 N
Path 2-2 = 1866707 N	, must exceed W = 551087 N	or W2 = 163455 N
Path 3-3 = 1707090 N	, must exceed W = 551087 N	or W3 = 676257 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.13 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 24.1288 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 360.4715 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N86" :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	14760.000	mm
Vessel Thickness	Tv	35.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	149.073	mm
Nozzle Thickness	Tn	9.601	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	30.000	mm
Diameter of Reinforcing Pad	Dpad	270.000	mm
Design Internal Pressure	Dp	1.100	MPa
Include Pressure Thrust		No	

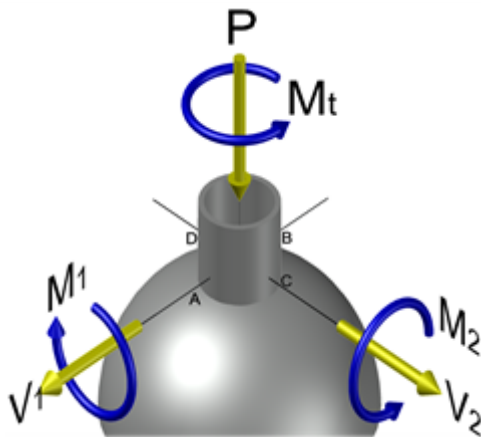
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-9599.6	N
Longitudinal Shear (SUS)	(V1) V1	8484.9	N
Circumferential Shear (SUS)	(Vc) V2	8484.9	N
Circumferential Moment (SUS)	(Mc) M1	3340974.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	3340974.0	N-mm
Torsional Moment (SUS)	Mt	4321525.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(7398.25 (35.0 - 1.5))}$$

$$= 1811.136 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-9599.6	N
Circumferential Shear	(VC) V2	8484.9	N
Longitudinal Shear	(VL) V1	8484.9	N
Circumferential Moment	(MC) M1	3340974.0	N-mm
Longitudinal Moment	(ML) M2	3340974.0	N-mm
Torsional Moment	MT	4321525.0	N-mm

Dimensionless Param: U = 0.12 TAU = 9.89 RHO = 4.00 (7.84)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.09313	(A,B,C,D)
$M(x) / P$	SP 4	0.04668	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.41530	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.22398	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.41530	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.22398	(A,B,C,D)
$N(y) * T / P$	SP 4	0.34154	(A,B,C,D)
$M(y) / P$	SP 4	0.28206	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10147	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.51251	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10147	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.51251	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad. Bend.	P	0.7	-0.7	0.7	-0.7	0.7	-0.7	0.7	-0.7
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-1.6	1.6	1.6	-1.6
Rad. Memb.	ML	-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-1.6	1.6	1.6	-1.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.2	0.7	3.0	-1.6	-1.2	0.7	3.0	-1.6



Tang. Memb. P	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Tang. Bend. P	4.0	-4.0	4.0	-4.0	4.0	-4.0	4.0	-4.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-11.0	11.0	11.0	-11.0
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-11.0	11.0	11.0	-11.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-6.2	7.6	15.9	-14.0	-6.2	7.6	15.9	-14.0
Shear VC	0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tot. Shear	2.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0
Str. Int.	7.0	8.2	16.0	14.1	6.4	7.8	16.2	14.4

Unitless Prm: U = 0.27 TAU = 0.00 (16.16) RHO = 0.00 (4.14)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.20113	(A,B,C,D)
M(x) / P	SR 2	0.15428	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.16308	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.54429	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.16308	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.54429	(A,B,C,D)
N(y) * T / P	SR 2	0.06035	(A,B,C,D)
M(y) / P	SR 2	0.04632	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04978	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.16279	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04978	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.16279	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Rad. Bend. P		7.9	-7.9	7.9	-7.9	7.9	-7.9	7.9	-7.9
Rad. Memb. MC		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Rad. Bend. MC		0.0	0.0	0.0	0.0	-19.5	19.5	19.5	-19.5
Rad. Memb. ML		-1.0	-1.0	1.0	1.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-19.5	19.5	19.5	-19.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-10.9	12.3	30.1	-24.7	-10.9	12.3	30.1	-24.7
Tang. Memb. P		0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Tang. Bend. P		2.4	-2.4	2.4	-2.4	2.4	-2.4	2.4	-2.4
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang. Bend. MC		0.0	0.0	0.0	0.0	-5.8	5.8	5.8	-5.8
Tang. Memb. ML		-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML		-5.8	5.8	5.8	-5.8	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-3.2	3.7	9.0	-7.4	-3.2	3.7	9.0	-7.4
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Tot. Shear		1.7	1.7	0.5	0.5	0.5	0.5	1.7	1.7



Str. Int.	11.2	12.7	30.2	24.8	10.9	12.4	30.3	24.9
-----------	------	------	------	------	------	------	------	------

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Rad.	Pl (SUS)	-0.3	-0.3	0.7	0.7	-0.3	-0.3	0.7	0.7
Rad.	Q (SUS)	-1.0	1.0	2.3	-2.3	-1.0	1.0	2.3	-2.3
Long.	Pm (SUS)	63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Long.	Pl (SUS)	0.7	0.7	0.9	0.9	0.7	0.7	0.9	0.9
Long.	Q (SUS)	-6.9	6.9	15.0	-15.0	-6.9	6.9	15.0	-15.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear	Q (SUS)	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Pm+Pl (SUS)		64.8	64.8	65.3	65.3	64.8	64.8	65.3	65.3
Pm+Pl+Q (Total)		63.4	72.1	79.9	62.5	62.9	71.7	80.2	62.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.94	138.00	Passed
Pm+Pl (SUS)	65.28	207.00	Passed
Pm+Pl+Q (TOTAL)	80.17	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Rad.	Pl (SUS)	0.7	0.7	2.7	2.7	0.7	0.7	2.7	2.7
Rad.	Q (SUS)	-11.6	11.6	27.4	-27.4	-11.6	11.6	27.4	-27.4
Long.	Pm (SUS)	121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Long.	Pl (SUS)	0.2	0.2	0.8	0.8	0.2	0.2	0.8	0.8
Long.	Q (SUS)	-3.5	3.5	8.2	-8.2	-3.5	3.5	8.2	-8.2
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear	Q (SUS)	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Pm (SUS)		121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Pm+Pl (SUS)		122.3	122.3	124.1	124.1	122.3	122.3	124.1	124.1
Pm+Pl+Q (Total)		118.3	133.9	151.3	113.8	118.0	133.6	151.5	114.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.19	138.00	Passed
Pm+Pl (SUS)	124.06	207.00	Passed
Pm+Pl+Q (TOTAL)	151.47	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input, Nozzle Desc: N104"

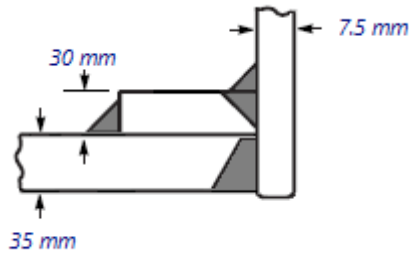
From: 110

Pressure for Reinforcement Calculations	P	1.1000	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	8200.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	35.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	35.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	28.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N104"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.910 in.
Actual Thickness Used in Calculation	0.295 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.1 \cdot 8203 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$
 $= 29.4375 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 51.2 / (118 \cdot 1 - 0.6 \cdot 1.1)$
 $= 0.4796 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	208.5536 mm
Parallel to Vessel Wall, opening length	d	104.2768 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		44.9746 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

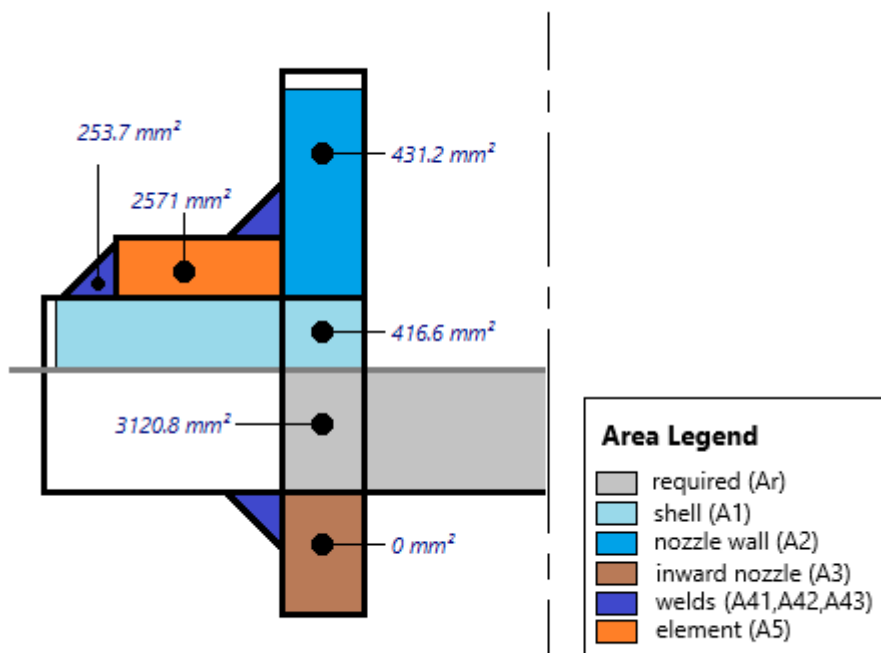


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3120.760	NA	NA
Area in Shell	A1	416.569	NA	NA
Area in Nozzle Wall	A2	431.158	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	253.676	NA	NA
Area in Element	A5	2571.000	NA	NA
TOTAL AREA AVAILABLE	Atot	3672.403	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 78.88 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	181.6119	30.0000 mm
Based on given Pad Diameter:	200.0000	23.5631 mm
Based on the Estimated Diameter Limit:	206.9661	22.3181 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (104 \cdot 29.4 \cdot 1 + 2 \cdot 5.99 \cdot 29.4 \cdot 1 \cdot (1 - 0.86))$
 $= 3120.760 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 104(1 \cdot 33.5 - 1 \cdot 29.4) - 2 \cdot 5.99$
 $(1 \cdot 33.5 - 1 \cdot 29.4) \cdot (1 - 0.86)$
 $= 416.569 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 45) \cdot (5.99 - 0.48) \cdot 0.86 / \sin(79.4)$

= 431.158 mm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (100) • 0.86 + (0) • 0.86 + 168² • 1
 = 253.676 mm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (202 - 116) • 30 • 1
 = 2571.000 mm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.49, tr = 0.48, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.08, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 7.49, tr = 0.48, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.08, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 7.49, tr = 0.48, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.08, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 30, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.88, Temp. Reduction = 7 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve D	-30 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 7.49, tr = 0.48, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.08, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1.10/5.10 = 0.216



Weld Size Calculations, Description: N104"

Intermediate Calc. for nozzle/shell Welds T_{min} 5.9898 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.1929 = $0.7 \cdot t_{min}$	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	9.5000 = $0.5 \cdot T_{minPad}$	19.7960 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (3121 - 417 + 2 \cdot 5.99 \cdot 0.86 \cdot (1 \cdot 33.5 - 29.4)) \cdot 138)$$

$$= 378882.09 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (431 + 2571 + 254 - 0 \cdot 0.86) \cdot 138$$

$$= 449258.78 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (431 + 0 + 85.5 + (343)) \cdot 138$$

$$= 118643.09 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (431 + 0 + 254 + 2571 + (343)) \cdot 138$$

$$= 496609.47 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 116 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 105785. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 200 \cdot 28 \cdot 0.49 \cdot 138$$

$$= 594755. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 55.2) \cdot (7.49 - 1.5) \cdot 0.7 \cdot 118$$

$$= 85776. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 116 \cdot 30 \cdot 0.74 \cdot 138$$

$$= 560515. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 116 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$$

$$= 625897. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (594755 + 85776) = 680531 \text{ N}$$

Failure Path [Path 2-2]:



$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (105785 + 560515 + 625897 + 0) = 1292196 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$
$$= (594755 + 625897 + 0) = 1220652 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 680531 N , must exceed W = 378882 N or W1 = 449258 N
Path 2-2 = 1292196 N , must exceed W = 378882 N or W2 = 118643 N
Path 3-3 = 1220652 N , must exceed W = 378882 N or W3 = 496609 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.13 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N104"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.910 in.
Actual Thickness Used in Calculation 0.295 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$
$$= 1.1 \cdot 8203 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$$
$$= 29.4375 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$
$$= 1.1 \cdot 51.2 / (118 \cdot 1 - 0.6 \cdot 1.1)$$
$$= 0.4796 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	204.6407	mm
Parallel to Vessel Wall, opening length	d	102.3204	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		44.9746	mm

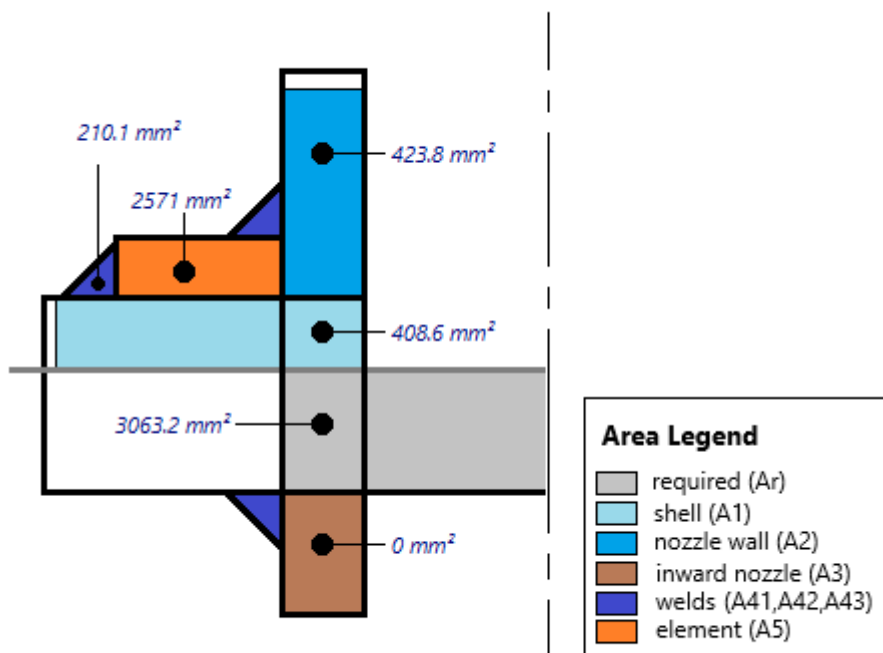


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3063.167	NA	NA
Area in Shell	A1	408.621	NA	NA
Area in Nozzle Wall	A2	423.807	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	210.063	NA	NA
Area in Element	A5	2571.000	NA	NA
TOTAL AREA AVAILABLE	Atot	3613.491	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	181.6558	30.0000 mm
Based on given Pad Diameter:	200.0000	23.5785 mm
Based on the Estimated Diameter Limit:	203.0532	22.7674 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (102 \cdot 29.4 \cdot 1 + 2 \cdot 5.99 \cdot 29.4 \cdot 1 \cdot (1 - 0.86))$
 $= 3063.167 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102(1 \cdot 33.5 - 1 \cdot 29.4) - 2 \cdot 5.99$
 $(1 \cdot 33.5 - 1 \cdot 29.4) \cdot (1 - 0.86)$
 $= 408.621 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 45) \cdot (5.99 - 0.48) \cdot 0.86$

= 423.807 mm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (100) • 0.86 + (0) • 0.86 + 125² • 1
 = 210.063 mm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (200 - 114) • 30 • 1
 = 2571.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 1.9796 mm
 Wall Thickness per UG16(b), tr16b = 3.0000 mm
 Wall Thickness, shell/head, internal pressure trb1 = 34.2038 mm
 Wall Thickness tb1 = max(trb1, tr16b) = 34.2038 mm
 Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm
 Wall Thickness per table UG-45 tb3 = 6.7578 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[6.76, max(34.2, 3)]
 = 6.7578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.98, 6.76)
 = 6.7578 mm

Available Nozzle Neck Thickness = 7.4898 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	41.3,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	253.7 MPa	Passed
Occasional	:	4.4,	Allowable	:	156.9 MPa	Passed
Shear	:	27.8,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N104"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.9898	mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.1929 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	19.7960 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
 = max(0, (3063 - 409 + 2 • 5.99 • 0.86 •
 (1 • 33.5 - 29.4))138)
 = 372031.69 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
 = (424 + 2571 + 210 - 0 • 0.86) • 138
 = 442226.50 N

Weld Load [W2]:
 = (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv



$$= (424 + 0 + 85.5 + (343)) \cdot 138$$
$$= 117628.86 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$
$$= (424 + 0 + 210 + 2571 + (343)) \cdot 138$$
$$= 489577.22 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$
$$= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 118$$
$$= 103800. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$
$$= (3.14/2.0) \cdot 200 \cdot 28 \cdot 0.49 \cdot 138$$
$$= 594755. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$
$$= (3.14 \cdot 54.2) \cdot (7.49 - 1.5) \cdot 0.7 \cdot 118$$
$$= 84166. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$
$$= (3.14/2) \cdot 114 \cdot 30 \cdot 0.74 \cdot 138$$
$$= 549998. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$
$$= (3.14/2.0) \cdot 114 \cdot (35 - 1.5) \cdot 0.74 \cdot 138$$
$$= 614154. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (594755 + 84166) = 678922 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$
$$= (103800 + 549998 + 614154 + 0) = 1267952 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$
$$= (594755 + 614154 + 0) = 1208908 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 678921 N	, must exceed W = 372031 N	or W1 = 442226 N
Path 2-2 = 1267952 N	, must exceed W = 372031 N	or W2 = 117628 N
Path 3-3 = 1208908 N	, must exceed W = 372031 N	or W3 = 489577 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.13 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 12.0520 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 347.7885 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N104" :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	14760.000	mm
Vessel Thickness	Tv	35.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	99.320	mm
Nozzle Thickness	Tn	7.490	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	30.000	mm
Diameter of Reinforcing Pad	Dpad	200.000	mm
Design Internal Pressure	Dp	1.100	MPa
Include Pressure Thrust		No	

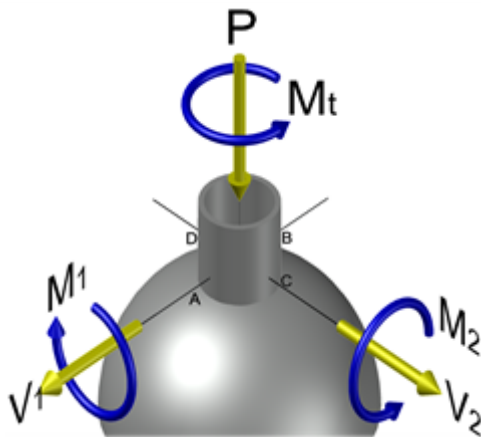
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-6399.7	N
Longitudinal Shear (SUS)	(V1) V1	5656.6	N
Circumferential Shear (SUS)	(Vc) V2	5656.6	N
Circumferential Moment (SUS)	(Mc) M1	1484877.4	N-mm
Longitudinal Moment (SUS)	(Ml) M2	1484877.4	N-mm
Torsional Moment (SUS)	Mt	1920683.5	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Error/Warning Messages for WRC 107/537 Analysis: 1
 Warning - The ratio R_o/R_m , 0.0077, must be > 0.0571 and < 0.571 .

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(7398.25 (35.0 - 1.5))}$
 $= 1757.161 \text{ mm}$

WRC 107 Stress Calculation for SUSained loads:

Radial Load		P	-6399.7	N
Circumferential Shear	(VC)	V2	5656.6	N
Longitudinal Shear	(VL)	V1	5656.6	N
Circumferential Moment	(MC)	M1	1484877.4	N-mm
Longitudinal Moment	(ML)	M2	1484877.4	N-mm
Torsional Moment		MT	1920683.5	N-mm

Dimensionless Param: $U = 0.08$ $\text{TAU} = 9.04$ $\text{RHO} = 4.00$ (10.60)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.10802	(A,B,C,D)
$M(x) / P$	SP 4	0.05377	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.60286	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.32142	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.60286	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.32142	(A,B,C,D)
$N(y) * T / P$	SP 4	0.32581	(A,B,C,D)
$M(y) / P$	SP 4	0.35935	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10183	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	2.38931	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10183	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	2.38931	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad.	Bend. P	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-1.0	1.0	1.0	-1.0



Rad. Memb. ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Rad. Bend. ML	-1.0	1.0	1.0	-1.0	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-0.7	0.4	2.0	-1.1	-0.7	0.4	2.0	-1.1
Tang. Memb. P	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Tang. Bend. P	3.4	-3.4	3.4	-3.4	3.4	-3.4	3.4	-3.4
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-7.7	7.7	7.7	-7.7
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-7.7	7.7	7.7	-7.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-3.8	4.7	11.7	-10.5	-3.8	4.7	11.7	-10.5
Shear VC	0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tot. Shear	2.0	2.0	1.0	1.0	1.0	1.0	2.0	2.0
Str. Int.	5.0	5.9	11.8	10.6	4.1	4.9	12.1	10.9

Unitless Prm: U = 0.20 TAU = 0.00 (16.19) RHO = 0.00 (5.59)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.22984	(A,B,C,D)
M(x) / P	SR 2	0.19635	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.15572	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.76902	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.15572	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.76902	(A,B,C,D)
N(y) * T / P	SR 2	0.06900	(A,B,C,D)
M(y) / P	SR 2	0.05775	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04770	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.22726	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04770	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.22726	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Rad. Bend. P		6.7	-6.7	6.7	-6.7	6.7	-6.7	6.7	-6.7
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Rad. Bend. MC		0.0	0.0	0.0	0.0	-12.3	12.3	12.3	-12.3
Rad. Memb. ML		-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Rad. Bend. ML		-12.3	12.3	12.3	-12.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-4.6	6.4	20.7	-17.3	-4.6	6.4	20.7	-17.3
Tang. Memb. P		0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Tang. Bend. P		2.0	-2.0	2.0	-2.0	2.0	-2.0	2.0	-2.0
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC		0.0	0.0	0.0	0.0	-3.6	3.6	3.6	-3.6
Tang. Memb. ML		-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML		-3.6	3.6	3.6	-3.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-1.4	1.9	6.1	-5.1	-1.4	1.9	6.1	-5.1
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5



Shear MT	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tot. Shear	1.4	1.4	0.4	0.4	0.4	0.4	1.4	1.4
Str. Int.	5.2	6.9	20.7	17.3	4.7	6.5	20.8	17.4

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Rad. Pl (SUS)		-0.2	-0.2	0.5	0.5	-0.2	-0.2	0.5	0.5
Rad. Q (SUS)		-0.5	0.5	1.5	-1.5	-0.5	0.5	1.5	-1.5
Long. Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Long. Pl (SUS)		0.5	0.5	0.6	0.6	0.5	0.5	0.6	0.6
Long. Q (SUS)		-4.3	4.3	11.1	-11.1	-4.3	4.3	11.1	-11.1
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Pm (SUS)		63.9	63.9	63.9	63.9	63.9	63.9	63.9	63.9
Pm+Pl (SUS)		64.7	64.7	65.0	65.0	64.7	64.7	65.0	65.0
Pm+Pl+Q (Total)		64.2	69.4	75.7	63.0	63.5	68.9	76.0	63.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	63.94	138.00	Passed
Pm+Pl (SUS)	64.97	207.00	Passed
Pm+Pl+Q (TOTAL)	76.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Rad. Pl (SUS)		0.9	0.9	1.7	1.7	0.9	0.9	1.7	1.7
Rad. Q (SUS)		-5.5	5.5	19.0	-19.0	-5.5	5.5	19.0	-19.0
Long. Pm (SUS)		121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Long. Pl (SUS)		0.3	0.3	0.5	0.5	0.3	0.3	0.5	0.5
Long. Q (SUS)		-1.6	1.6	5.6	-5.6	-1.6	1.6	5.6	-5.6
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Pm (SUS)		121.2	121.2	121.2	121.2	121.2	121.2	121.2	121.2
Pm+Pl (SUS)		122.4	122.4	123.1	123.1	122.4	122.4	123.1	123.1



Pm+Pl+Q (Total)	120.4	128.1	141.9	116.1	119.9	127.7	142.0	116.3
-----------------	-------	-------	-------	-------	-------	-------	-------	-------

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.19	138.00	Passed
Pm+Pl (SUS)	123.12	207.00	Passed
Pm+Pl+Q (TOTAL)	142.04	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.